Expected Analysis Note: H to ZZ to Four Leptons Signal-Strength Measurement with CMS 2017 Open Data

Analysis session odette_354d

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Abstract

This analysis note documents the Phase 4a expected-only measurement of the Higgs-boson signal strength in the four-lepton final state using CMS 2017 Open Data flat ntuples corresponding to a user-provided target luminosity of 10 fb^{-1} at $\sqrt{s} = 13$ TeV. The analysis follows the CMS H to ZZ to four-lepton reference methodology where the available ntuple content allows it, using a simultaneous binned template likelihood in final-state categories over $105 < m_{4\ell} < 140$ GeV. The expected Asimov signal-strength result is $\mu = 1.0^{+0.633}_{-0.517}$, with symmetric uncertainty 0.575. The final-state workspace is sparse, with 17 of 18 expected bins having $S + B < 5$, and the -20% corruption sensitivity is documented low-count infeasible after three attempts, not passed. No observed-data signal-strength result is reported in this document; 10% and full-data values are reserved for later phases.

Contents

Change Log

Phase 4a v1, 2026-05-30

- Initial AN version for expected-only inference.
- Established the full document structure, staged all 49 available upstream figure PDFs, and recorded the current analysis state: final-state S1 categories nominal; VBF, classifier categories, and neural-network categories not promoted.
- Added the expected signal-strength result, systematic model, validation tests, comparison context, and limitations needed for Doc 4b and Doc 4c updates.

1 Introduction

The $H \to ZZ^{(*)} \to 4\ell$ channel, central to the original LHC Higgs-boson observations [1, 2], is a high-resolution final state for Higgs-boson measurements because the four charged leptons provide a narrow reconstructed invariant-mass peak on top of a continuum ZZ background. The goal of this open-data analysis is narrower than the official CMS program: it measures an expected detector-level signal strength μ in the four-lepton mass spectrum and documents which elements of the CMS reference analysis can be reproduced with the available flat ntuples.

The user request asked for a CMS-HIG-16-041-like analysis, including the $105 < m_{4\ell} < 140$ GeV fit window, categorization, a VBF category if possible, and an angular neural-network discriminator. The analysis implements the shared signal-strength parameter and simultaneous final-state fit, but it does not claim official equivalence. The flat ntuples contain no recoverable jet or VBF discriminant information, and the classifier/neural-network route did not pass promotion gates. These outcomes are part of the result because they define what the open-data ntuple can support.

The principal comparison target is CMS-HIG-16-041, published as JHEP 11 (2017) 047, which reports $\mu = 1.05^{+0.19}_{-0.17}$ at $m_H = 125.09$ GeV, a fiducial cross section of $2.92^{+0.48}_{-0.44}(\text{stat})^{+0.28}_{-0.24}(\text{syst})$ fb, $m_H = 125.26 \pm 0.21$ GeV, and $\Gamma_H < 1.10$ GeV at 95% CL [3]. A later CMS Run-2 result gives $\mu = 0.94 \pm 0.07(\text{stat})^{+0.09}_{-0.08}(\text{syst})$ and an inclusive fiducial cross section of $2.84^{+0.23}_{-0.22}(\text{stat})^{+0.26}_{-0.21}(\text{syst})$ fb [4]. PDG world-average values are used only as external context and validation references, not as fit inputs [5].

2 Data Samples

The analysis uses flat ntuples derived from CMS open-data-style event content and simulation workflows [6–10]. It uses the prompt-specified primary paths, not the local worktree copies, because Phase 1 found non-identical file sizes, metadata rows, and event entries between the two locations. The open-data file contains a single h4lTree with 854 candidate entries and metadata luminosity information. All simulated samples are normalized with the prompt-provided effective cross sections and generated-event denominators from the Metadata trees.

The normalization rule is

$$w_s = \frac{\sigma_s^{\text{eff}}\,\mathcal{L}}{N_s^{\text{Metadata}}}, \tag{1}$$

where $\mathcal{L} = 10000\,\text{pb}^{-1}$ and s denotes an MC sample. The data weight is one. This rule is used for fit inputs and stacked yield plots; no MC component is normalized to the observed data integral.

Table 1: Data sample summary. The luminosity is the user-provided target luminosity from the founding prompt and Phase 1 inventory.

Period	$\sqrt{s}$ [TeV]	Preselection entries	$\mathcal{L}$ [fb^{-1}]
2017 CMS Open Data	13	854	10

Table 2: MC normalization inputs from machine-readable Phase 3 normalization records.

Sample	Group	σ_{eff} [pb]	N_{gen}	Weight
DYJetsToLL	background reducible	5.4e+03	8.24e+07	0.6545
GGZZ2E2Mu	background ggZZ	0.003185	4.99e+05	6.38e-05

Sample	Group	σ_{eff} [pb]	N_{gen}	Weight
GGZZ4E	background ggZZ	0.001619	9.93e+05	1.63e-05
GGZZ4Mu	background ggZZ	0.001575	9.97e+05	1.58e-05
GluGluToHToZZ	signal ggH	0.006024	9.84e+05	6.12e-05
TTBar	background top	52.7	1.48e+07	0.03566
VBF_HToZZ	signal VBF	0.000488	4.98e+05	9.8e-06
WMHToZZ	signal VH	6.71e-05	1.93e+05	3.47e-06
WPHToZZ	signal VH	0.000107	2.95e+05	3.64e-06
ZHToZZ	signal VH	9.84e-05	4.86e+05	2.02e-06
ZZTo4L	background ZZ	1.325	5.21e+07	0.000254

3 Event Selection

The retained ntuples already contain a best four-lepton candidate. The selection therefore applies event-level checks that are auditable from retained branches: finite kinematic values, a nonzero trigger bitmask, valid final-state composition, flavor-matched lepton identification, Z-pair sanity, and the mass-window requirements. The broad validation and display range is $70 < m_{4\ell} < 170$ GeV; the inference range is $105 < m_{4\ell} < 140$ GeV.

The final nominal categories are the reconstructed final states 4μ, $4e$, and $2e2\mu$. No lepton-only category is labeled VBF because Phase 3 found no jet branches, no VBF discriminants, no allowed upstream join source, and no safe event-key recovery path. Classifier and neural-network categories were attempted with standard boosted-tree and machine-learning tools [11, 12] after angular closure, but failed promotion because they did not improve the expected μ uncertainty by the required amount and did not satisfy all input/score/low-stat gates.

Table 3: Selection cutflow summary. Data counts are raw events and MC yields are prompt-normalized weighted yields.

Step	Data events	MC weighted yield
all	854	842
trigger bitmask nonzero	798	790
flavor matched lepton ID	467	466
broad validation 70-170	203	210
fit window 105-140	69	56.6

Table 4: Input-variable modeling gate for classifier attempts. Only variables passing all D7 requirements were eligible, and the final classifier categories still failed promotion.

Variable	χ^2/ndf	p-value	max ratio dev.	D7 pass
cos_theta1	1.6	0.1562	0.5568	no
cos_theta2	1.289	0.2655	0.4591	no
cos_theta_star	1.007	0.4115	0.3637	no
eta4l	0.4039	0.8464	0.2204	no
lead_abs_eta	0.4734	0.7553	0.1931	yes
lead_lepton_pt	4.056	0.002731	1	no
mZ1	122.1	6.22e-155	1	no

Variable	χ^2/ndf	p-value	max ratio dev.	D7 pass
mZ2	32.08	7.49e-39	1	no
phi	0.7802	0.5637	0.2509	no
phi1	0.1552	0.9785	0.08262	yes
pt4l	0.3328	0.8934	0.3478	no
sublead_abs_eta	1.076	0.3664	0.2433	no
sublead_lepton_pt	849.5	0	1	no
y4l	996.7	0	1	no

4 Corrections and Model Construction

This measurement is detector-level. There is no particle-level unfolding because the flat ntuples do not retain generator-level truth branches or a response matrix. The relevant correction step is MC normalization and template construction: selected MC events are weighted with Eq. 1, grouped into signal and background templates, and binned in $m_{4\ell}$ by final-state category.

The expected bin content for process group g, category c, and mass bin b is

$$\nu_{gcb}(\boldsymbol{\theta}) = \sum_{s\in g} \sum_{i\in(s,c,b)} w_s\, a_{gcb}(\boldsymbol{\theta}), \tag{2}$$

where $a_{gcb}(\boldsymbol{\theta})$ is the nuisance-dependent rate or shape modifier. The total expected content is

$$\lambda_{cb}(\mu, \boldsymbol{\theta}) = \mu \sum_{g\in H} \nu_{gcb}(\boldsymbol{\theta}) + \sum_{g\in B} \nu_{gcb}(\boldsymbol{\theta}). \tag{3}$$

The Phase 4a observation is Asimov pseudo-data equal to the nominal expectation, not observed Open Data counts.

Shape and rate variations are propagated by producing up/down shifted templates. For a rate-only systematic with relative size δ_k, the variation is

$$\nu^{\pm}_{gcb,k} = \nu^{0}_{gcb}(1 \pm \delta_k), \tag{4}$$

while the $m_{4\ell}$ scale systematic is implemented as a histosys-like shifted-template variation. The grouped MC-stat approximation uses process/category normalization nuisances derived from Phase 3 sum of squared weights:

$$\delta^{\mathrm{MCstat}}_{gc} = \frac{\sqrt{\sum_b \mathrm{sumw2}_{gcb}}}{\sum_b \nu_{gcb}}. \tag{5}$$

This is formally downscoped relative to full per-bin HistFactory staterror profiling [13]; alternative-binning stability is used to check that it does not dominate the expected conclusion.

The template construction keeps two normalizations separate. Shape-comparison plots used during input validation may rescale MC to the data integral to expose mismodeling in candidate classifier variables, but those diagnostic scales never enter Eq. 2. Fit templates use only the luminosity and prompt-effective cross-section weights in Eq. 1. This separation is important because otherwise the signal-strength fit would become partly circular: a background or signal normalization inferred from observed data in the signal window would reduce the very rate information used to estimate μ.

The nominal grouping also preserves the distinction between physics components even when they are added in the final likelihood. The Higgs signal group contains ggH, VBF, WH, WplusH, WminusH,

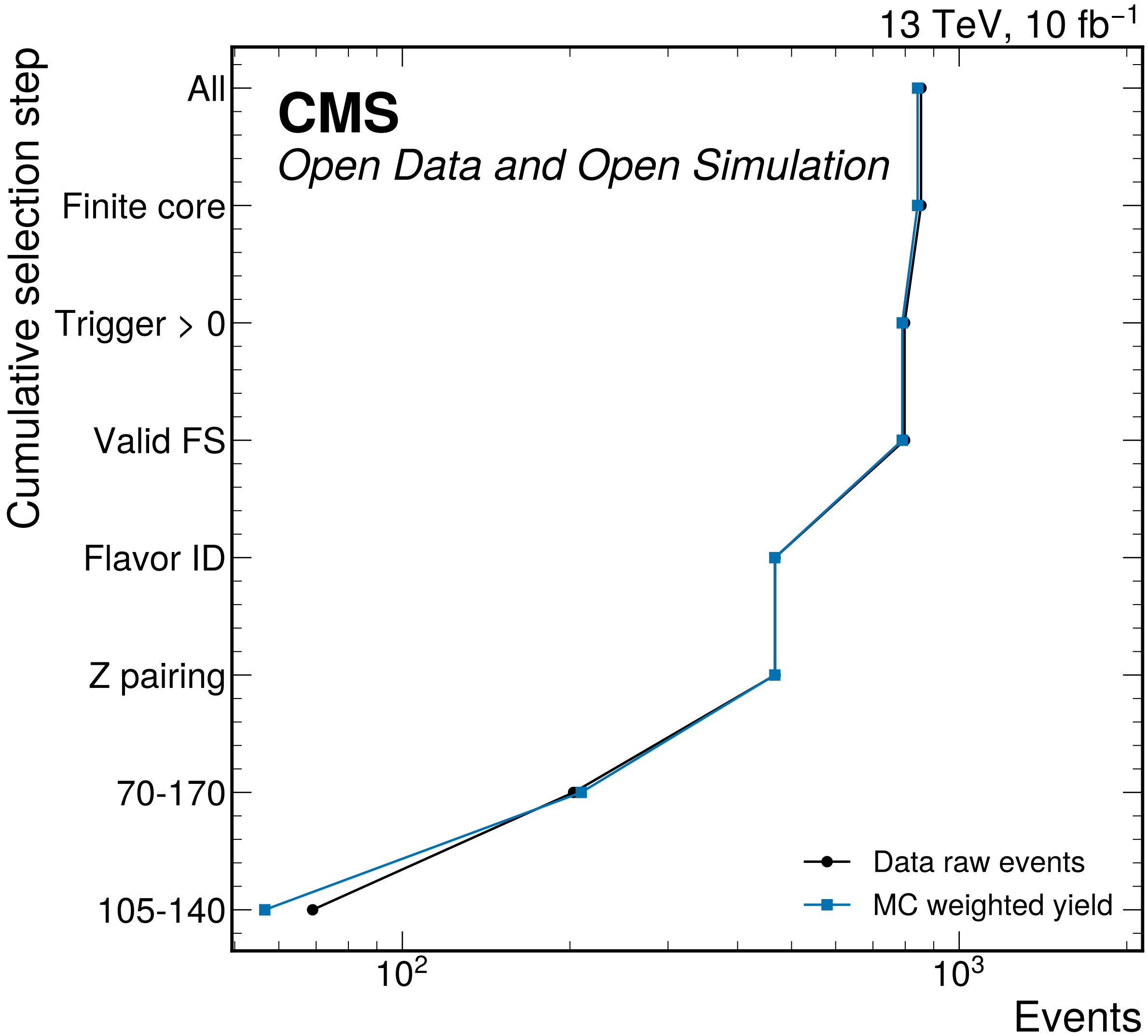

Figure 1: Cumulative Phase 3 cutflow for data raw counts and prompt-normalized MC weighted yields. The rendered step labels are abbreviated for readability; 70-170 is the broad validation mass window and 105-140 is the fit window. Every sample-level cutflow is monotonic, and the fit-window endpoint is 69 data events with 56.6098 expected MC events.

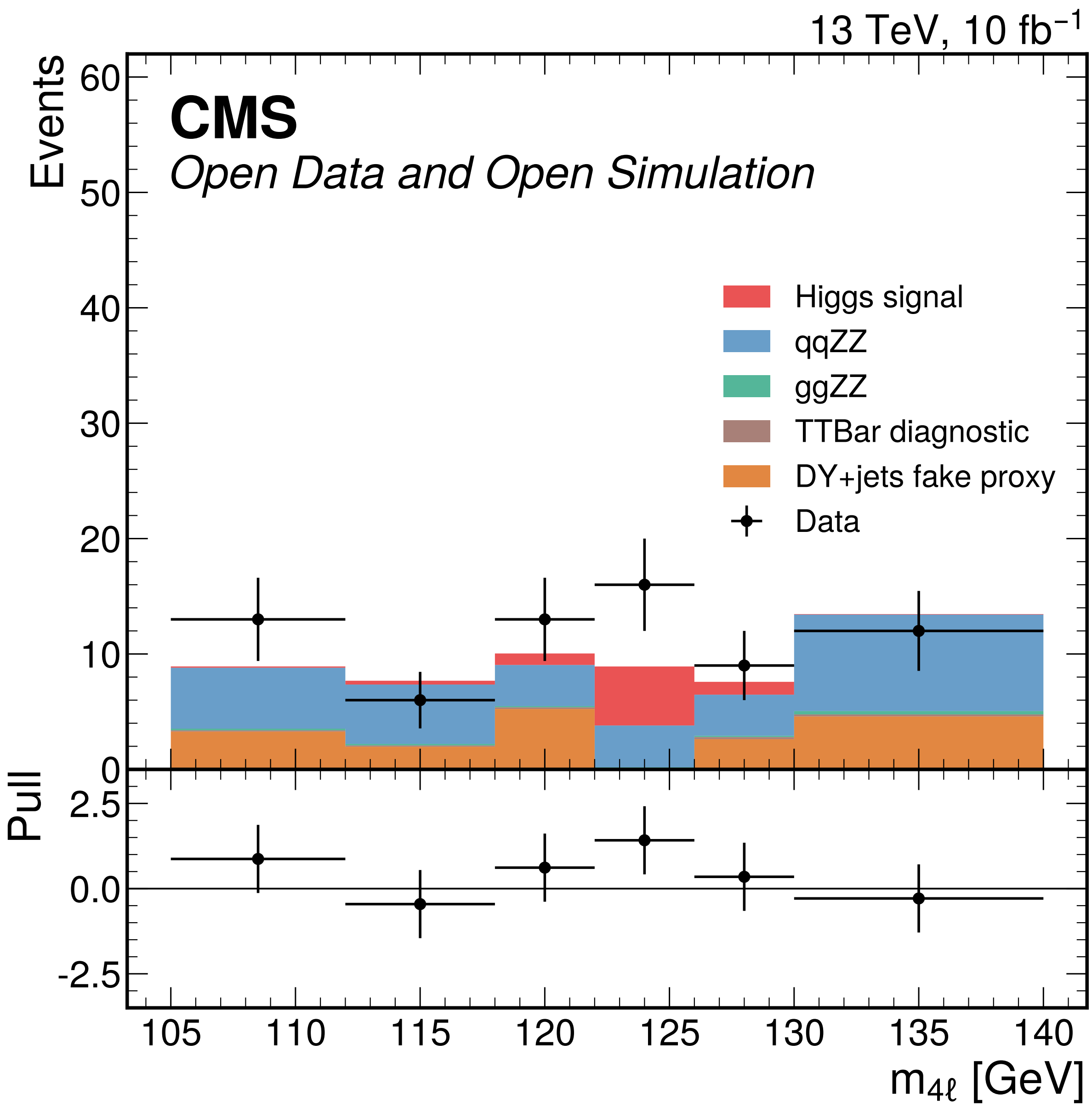

Figure 2: Inclusive four-lepton mass distribution in the fit window. MC is normalized with prompt effective cross sections and Metadata denominators, while DY+jets remains the nominal fake proxy. The lower panel shows pulls and the fit-window version is the handoff input for downstream template fits.

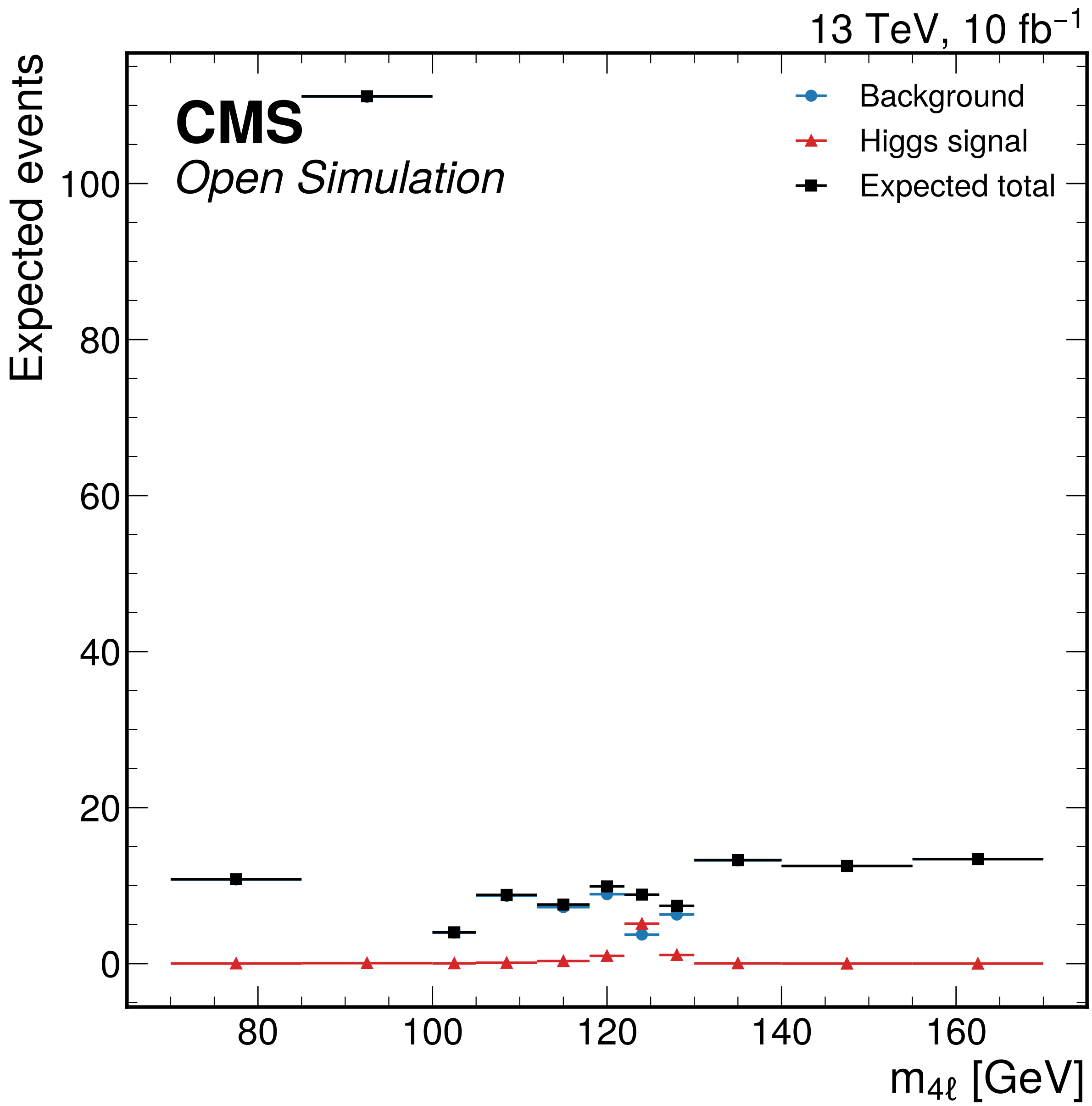

Figure 3: Expected inclusive four-lepton mass distribution in the broad validation range 70 < m4l < 170 GeV. The broad display is used for validation/sideband context; the signal-strength fit window remains 105 < m4l < 140 GeV.

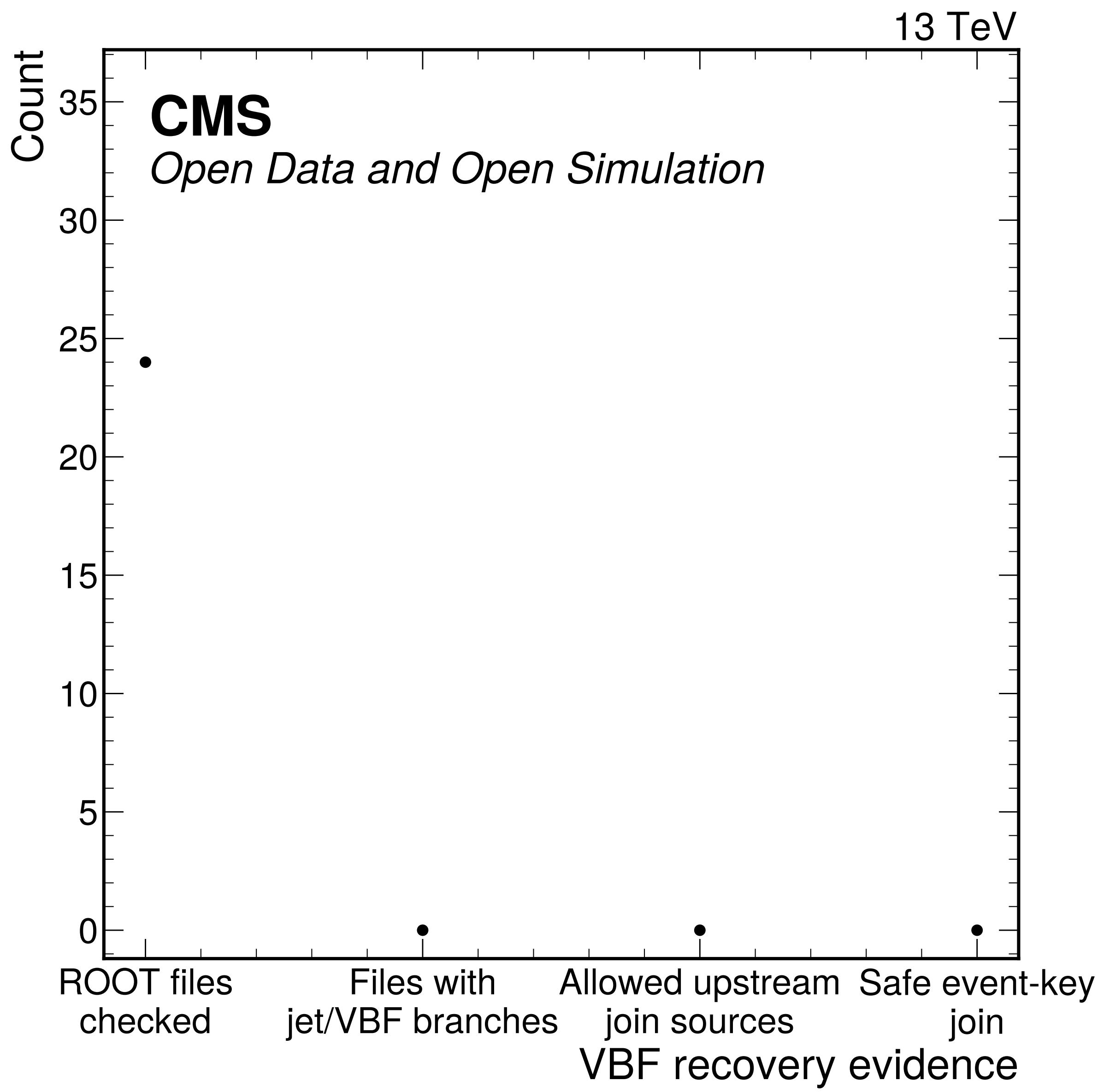

Figure 4: VBF recovery and downscope evidence from branch inventories and allowed join sources. No checked flat ntuple contains real jet or VBF discriminator branches, no allowed upstream join source exists, and no lepton-only category is labeled VBF.

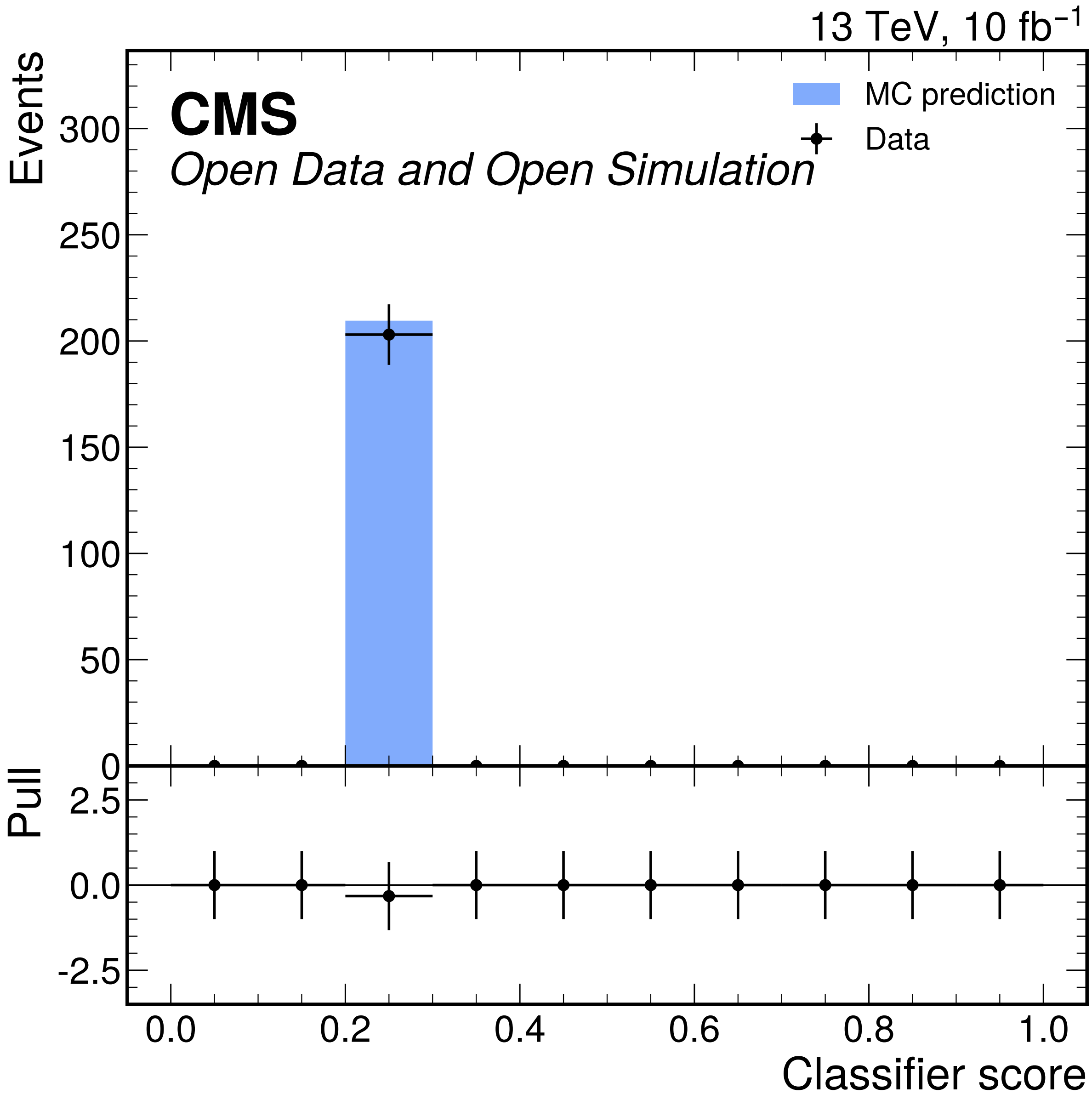

Figure 5: Best S2 classifier score data/MC comparison in the broad validation window. The score-shape gate and low-stat category viability failed, so this diagnostic is preserved as rejected-approach evidence rather than used in the nominal fit.

and ZH templates; the single POI multiplies their sum. The irreducible background is split into qqZZ and ggZZ groups because they have different production mechanisms and nuisance sizes. The reducible background remains DY+jets, with TTBar retained only as an omission diagnostic. This grouping is the minimum structure needed to compare to CMS H to four-lepton analyses without claiming production-mode measurements that the available categories cannot support.

The downscope decisions enter the model construction directly. Since no VBF category is recoverable, VBF events contribute only to the inclusive signal template and no jet-energy or VBF-migration nuisance is created. Since the classifier categories are rejected, angular variables and MVA scores do not partition the likelihood. Since no particle-level response exists, the result is not unfolded and no acceptance correction is applied. These absences are encoded in the systematic-source table rather than hidden in prose.

The correction/model section is intentionally explicit because it replaces an unfolding section for this detector-level analysis. The procedure is reproducible from the sample inventory, selection definitions, binning, and JSON template payloads. It also defines the main limitation: the reported μ is a signal-strength fit relative to the prompt-effective MC expectation and not a fiducial cross-section extraction with independent acceptance, branching-fraction, or particle-level corrections.

5 Systematic Uncertainties

The systematic model follows the reviewed Phase 4a expected workspace. Active sources are propagated as normalization or shape modifiers in the binned likelihood; downscoped or inapplicable sources are still documented because they affect interpretation and comparison to CMS reference analyses.

The covariance decomposition for the single POI is

$$V_{\mathrm{tot}} = V_{\mathrm{stat}} + V_{\mathrm{syst}}, \tag{6}$$

with $V_{\mathrm{stat}} = 0.3063689401723719$, $V_{\mathrm{syst}} = 0.023955108369782596$, and $V_{\mathrm{tot}} = 0.33032404854215447$ for μ. The MC-stat component inside the systematic budget contributes variance 0.0029560843175981955.

Table 5: Systematic source inventory from systematics_sources.json.

Label	Key	Source	Size	Status
SP1	lumi	Integrated luminosity	0.008072	implemented
SP4	lepton eff	Lepton reco/ID/trigger	0.03	implemented
SP7	signal theory	Signal production	0.05	fallback prior
SP9	ZZ norm	qqZZ background normalization	0.1	fallback prior
SP9	ggZZ norm	ggZZ background normalization	0.2	fallback prior
SP10	DY norm	DY+jets fake proxy	0.5	implemented
SP11	ttbar omission	TTBar omission diagnostic	0.0444	implemented
SP5	m4l scale	Lepton m4l scale	0.001	implemented
SP3	mc stat	MC statistical uncertainty	n/a	grouped MC-stat downscope
SP2	prompt xsecs	Prompt xsecs	per-sample prompt xsecs	implemented prompt fallback
SP6	pileup/PV	Pileup/PV	n/a	not propagated
SP8	Higgs BR	Higgs BR	n/a	not applicable
SP12	classifier	Classifier migration	n/a	not applicable

Label	Key	Source	Size	Status
SP13	angular	Angular reco	n/a	not propagated

5.1 Integrated Luminosity

Integrated luminosity changes the normalization of all simulated templates. It is evaluated as a log-normal normalization nuisance using the public CMS 2017 luminosity uncertainty as a scale reference for the user-provided 10 fb^{-1} subset. The relative variation is 0.008072174738841406, producing a small expected impact on μ because the fit uncertainty is dominated by statistical power and background modeling. This source would become more important in an observed cross-section extraction, but in Phase 4a it is propagated only as a rate uncertainty.

5.2 Prompt Effective Cross Sections

Prompt cross sections define nominal MC weights and are not independently tuned to data. Phase 4a mitigates this by carrying process normalization nuisances for signal and background groups and by recording the prompt source trail. The result should be interpreted as a signal strength relative to these prompt-effective sample normalizations, not as a standalone fiducial cross section.

5.3 MC Statistical Uncertainty

MC statistical uncertainty is propagated through grouped process/category normalization nuisances derived from Phase 3 sumw2. This is a formal downscope from full bin-by-bin HistFactory staterror profiling, which was not computationally stable in the sandbox. The grouped component contributes variance 0.0029560843175981955 and is checked through alternative binnings. Downstream observed-data phases must repeat the stability checks.

5.4 Lepton Efficiency

Lepton reconstruction, identification, and trigger efficiency affect every signal and background template because all selected objects are electrons or muons. The Phase 4a implementation uses a 0.03 rate envelope derived from Phase 3 trigger and flavor-matched ID data/MC step-efficiency closure. This is a detector-level envelope rather than an official CMS scale-factor chain. The expected μ impact is visible in the nuisance impact table and is subdominant to DY fake-proxy and ZZ normalization uncertainties.

5.5 Lepton Momentum Scale and Resolution

Lepton momentum scale and resolution influence the reconstructed four-lepton mass shape. Phase 4a propagates this as an $m_{4\ell}$ template shift corresponding to a 0.001 variation. This is not an official CMS lepton calibration; it is an external performance-envelope approximation applied to the available detector-level templates. The low-count corruption sensitivity study shows that a -20% mass-response corruption is not rejected in the final-state workspace, so this source is also tied to a documented sensitivity limitation.

5.6 Pileup and Primary Vertex Modeling

Pileup and primary-vertex modeling are documented but not propagated because PV variables are not used in the nominal selection categories or fit templates. Phase 3 excluded pvNdof and related variables under the validation gate. This is an explicit not-propagated status, not a silent omission.

5.7 Signal Production Composition

Signal production and composition uncertainty accounts for the relative mixture of ggH, VBF, WH, WminusH, WplusH, and ZH signal templates. The flat ntuples do not provide all official generator-composition inputs, so Phase 4a uses a 0.05 fallback prior and marks it as an expected-phase approximation. It is propagated separately over signal production groups while one global μ scales the total Higgs yield. Because no VBF category is available, this source mostly changes the inclusive signal normalization rather than category migration.

5.8 Higgs Branching Fraction

The Higgs BR is not used in Phase 4a because no fiducial or inclusive cross-section conversion is performed. The reported quantity is detector-level μ relative to the nominal Higgs MC templates. If a later phase converts to a cross section, the branching-fraction source must be activated and cited.

5.9 qqZZ Background Normalization

The dominant irreducible continuum background is qqZZ. Its normalization is varied by 0.1 as a fallback prior due to unavailable campaign-specific validation of the prompt effective cross section. The variation is applied to the background ZZ template in all final states. It is one of the largest systematic variance components, which is expected because the signal region contains substantial ZZ continuum under the Higgs peak.

5.10 ggZZ Background Normalization

The loop-induced ggZZ background is represented by the three GGZZ final-state samples. A 0.2 fallback normalization prior is used because the small effective cross sections and prompt-provided sample definitions cannot be independently validated to official precision in this sandbox. Its expected impact on μ is small compared with qqZZ and DY because the selected ggZZ yield is small.

5.11 DY plus Jets Fake Proxy

DY+jets is the nominal reducible fake-background proxy by user instruction. The official CMS analyses use data-driven Z+X estimates, so this source is a comparability limitation. Phase 4a assigns a 0.5 normalization variation based on low sideband statistics and fake-model limitations. It is the largest single nuisance impact on expected μ, which correctly reflects the weak constraint provided by the available DY selected sample.

5.12 TTBar Omission Diagnostic

TTBar is not promoted to a nominal separate fake component because its weighted yield is below the Phase 2 thresholds relative to DY+jets in the signal and sideband windows. The omission diagnostic is retained as a 0.0444 envelope on the reducible background. Its numerical impact is small, but documenting it prevents the DY-only fake proxy from being interpreted as a complete reducible-background estimate.

5.13 Classifier Migration

Classifier migration uncertainty is not applicable in the nominal fit because S2 classifier categories, including the neural-network attempt, were rejected. The diagnostics are retained in the appendix to show why the route was not promoted. If a later phase revisits classifier categories, this source must be reactivated with score-boundary migration variations.

5.14 Angular Reconstruction

Angular reco passed closure, with median mass differences far below 0.1 GeV and no out-of-range angles in checked selected events. It is not propagated as a nominal fit systematic because angular variables are not used after S2 rejection. The closure remains important evidence that the NN attempt was made on physically sensible inputs.

5.15 VBF and Jet Systematics

Jet/VBF category systematics are formally downscoped because there is no recoverable jet or VBF information in the allowed ntuples. No VBF category exists in the nominal model, and no lepton-only category is labeled VBF. This makes official VBF-category comparisons unavailable rather than weakly measured.

Table 6: Largest nuisance impacts on expected μ. Shifts are fixed-variation expected impacts, not observed-data pulls.

Nuisance	Down shift	Up shift	Max impact
dy_fake_norm	0.1427	-0.06093	0.1427
qqZZ_norm	0.08392	-0.08207	0.08392
lepton eff	0.05804	-0.05382	0.05804
m4l scale_shape	-0.009583	-0.05273	0.05273
signal_ggH_theory	0.0462	-0.04237	0.0462
lumi	0.01497	-0.01476	0.01497
ggZZ_norm	0.004351	-0.004283	0.004351
signal_VBF_theory	0.004308	-0.004223	0.004308
ttbar omission	0.003187	-0.002649	0.003187
signal_VH_theory	0.001326	-0.001318	0.001326

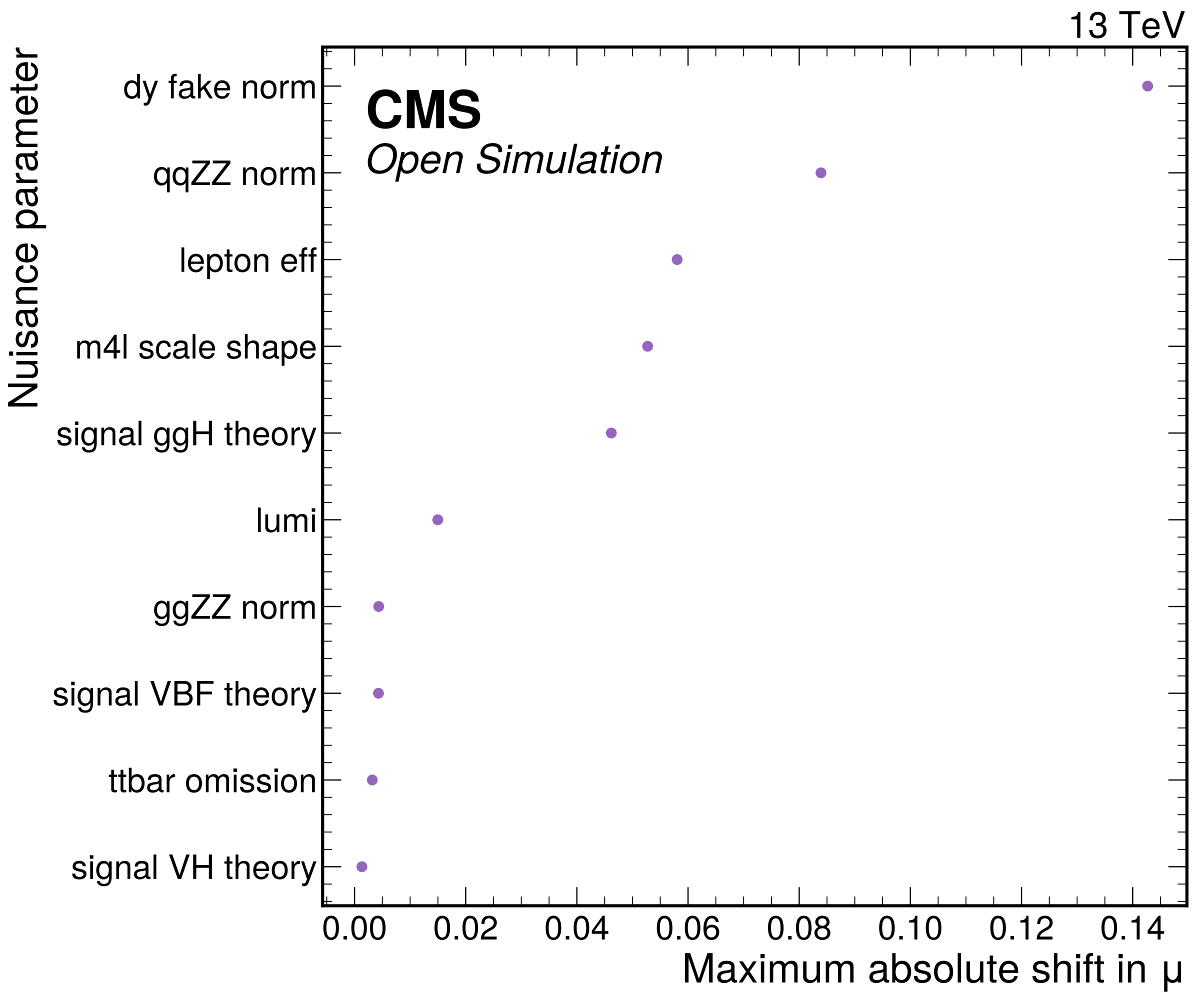

Figure 6: Expected nuisance impact ranking on the global signal strength, evaluated by fixing each nuisance at plus or minus one standard deviation and refitting the Asimov model.

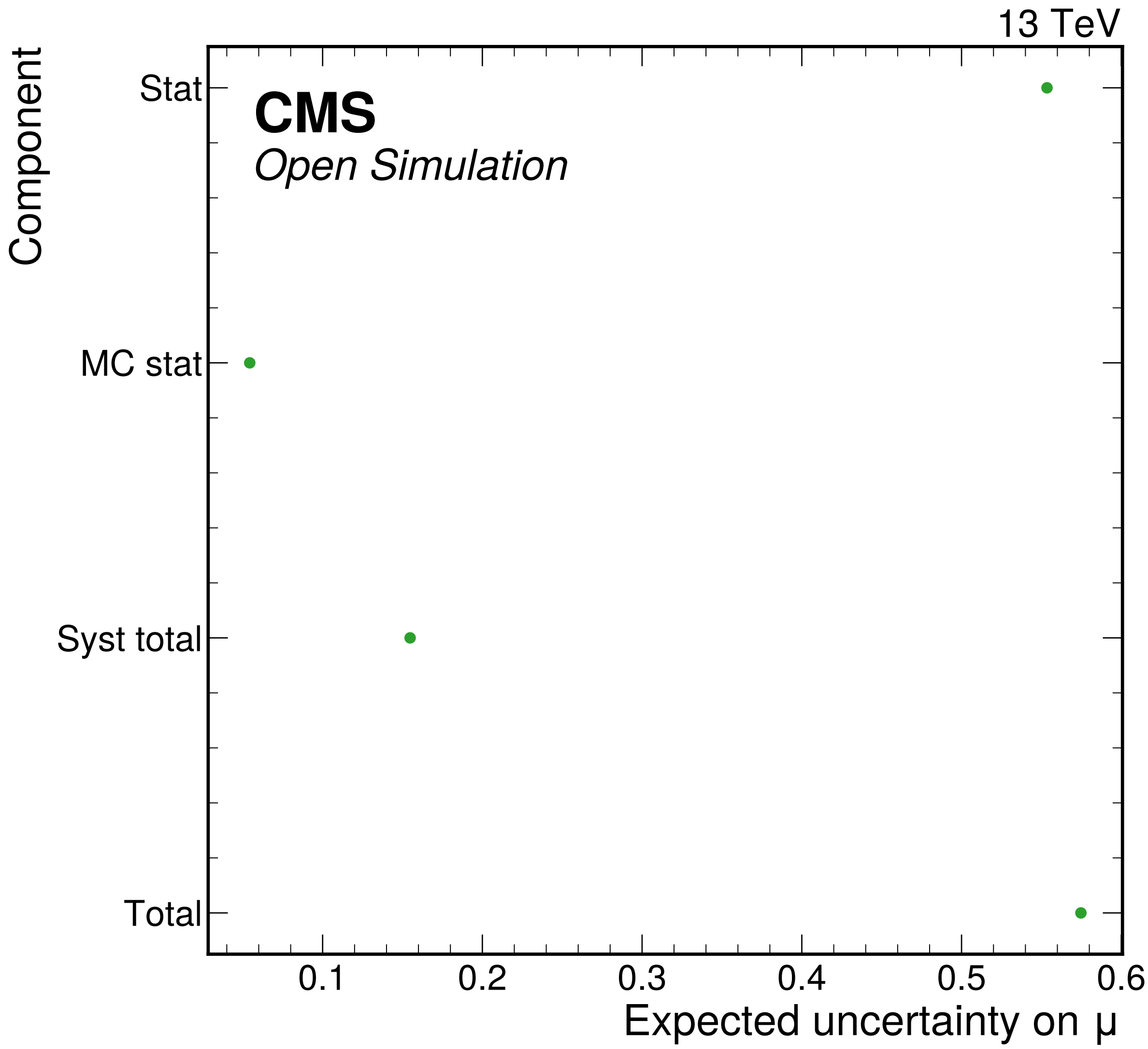

Figure 7: Expected uncertainty breakdown for the Phase 4a Asimov signal-strength fit, derived from the stat-only, MC-stat, and full nuisance fits.

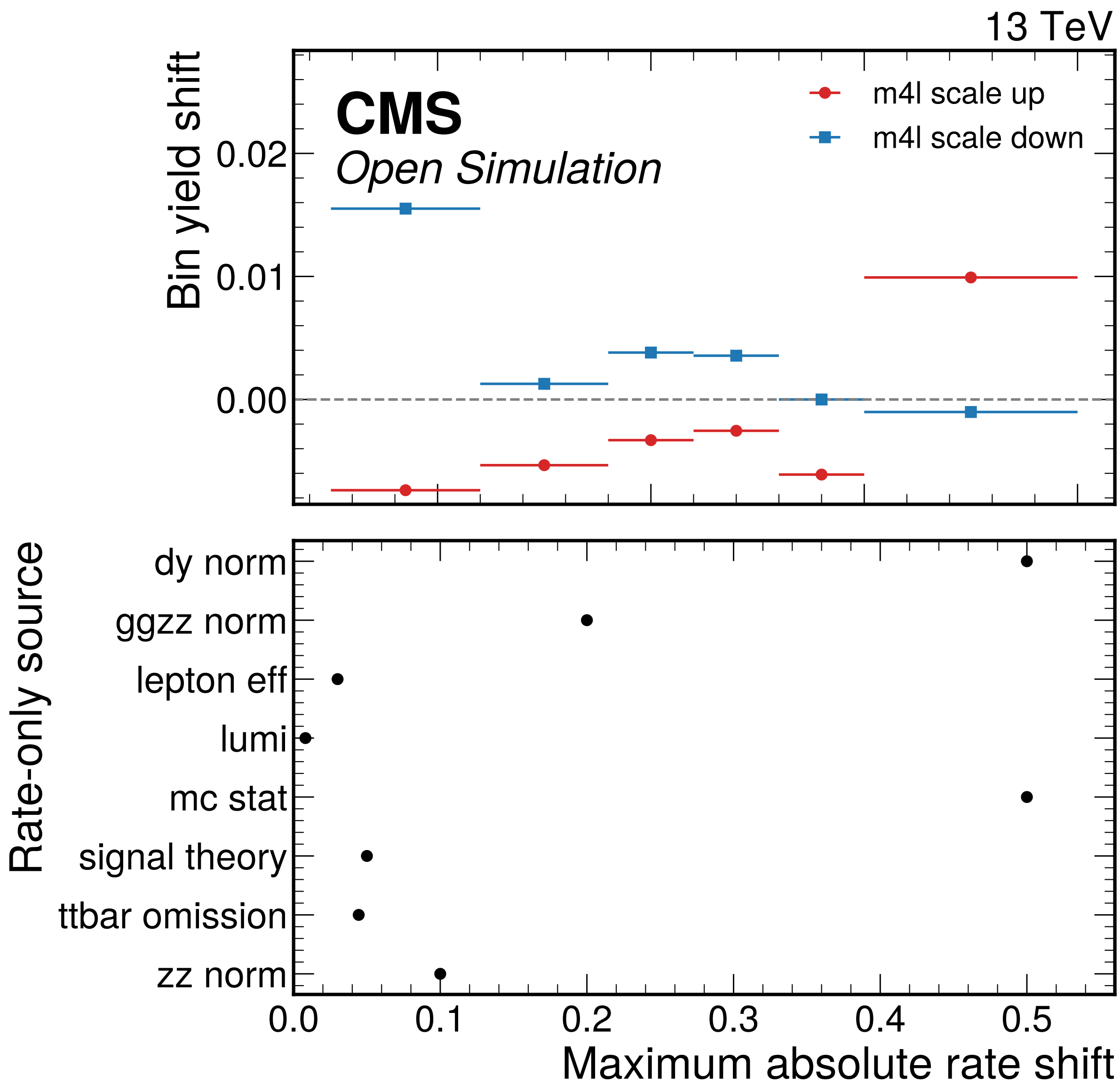

Figure 8: Per-systematic shifted-bin summary. The upper panel shows the actual m4l-scale shape-source bin shifts for one representative process/channel; the lower panel shows maximum integral effects for rate-only sources without inventing fake shape dependence.

6 Cross-checks

The cross-checks are designed to answer whether the low-count final-state model can be used honestly for an expected-only result. The Asimov goodness-of-fit is exactly zero by construction and is not treated as independent validation. Independent checks are low-count toys, signal injection and recovery, alternative-binning stability, channel compatibility, and corrupted-model sensitivity.

For a comparison vector $\mathbf{d} - \mathbf{m}$ with covariance C, the generic validation statistic is

$$\chi^2 = (\mathbf{d} - \mathbf{m})^T C^{-1} (\mathbf{d} - \mathbf{m}). \tag{7}$$

For low-count Poisson bins the preferred diagnostic is the saturated-model deviance,

$$D = 2 \sum_b \left[n_b \log \frac{n_b}{\lambda_b} - (n_b - \lambda_b) \right], \tag{8}$$

with the usual zero-count convention. In Phase 4a the pseudo-data are generated by the nominal model, so observed-data GoF interpretation is deferred.

Table 7: Signal injection and recovery. All injected Asimov configurations pass the 20% relative-bias gate.

Injected μ	Fitted μ	Bias	Relative bias	Pass
0	5.39e-21	5.39e-21	5.39e-21	yes
1	1	0	0	yes
2	2	-4.47e-06	2.24e-06	yes
5	5	-7.58e-05	1.52e-05	yes

Table 8: Alternative-binning stability for the expected signal-strength fit.

Config.	Channels	σ_μ	bins < 5	p-value
final_state_nominal	4mu, 4e, 2e2mu	0.5724	17	1
inclusive_nominal	inclusive	0.5755	0	1
inclusive_coarse	inclusive	0.5868	0	1
inclusive_peak_side	inclusive	0.5895	0	1

Table 9: Documented low-count corruption-sensitivity limitation for the -20% mass-response test. None of the three final-state-aligned profiled tests rejects at p<0.05; this criterion is documented low-count infeasible, not passed.

Attempt	Test	Statistic	ndf	p-value	Rejects
1	profiled Poisson saturated-likelihood deviance in the final-state simultaneous workspace	16.92	17	0.4595	no
2	profiled per-channel shape-only Poisson deviance	12.97	15	0.6049	no
3	profiled Pearson chi2 on the final-state simultaneous workspace	22.72	17	0.1584	no

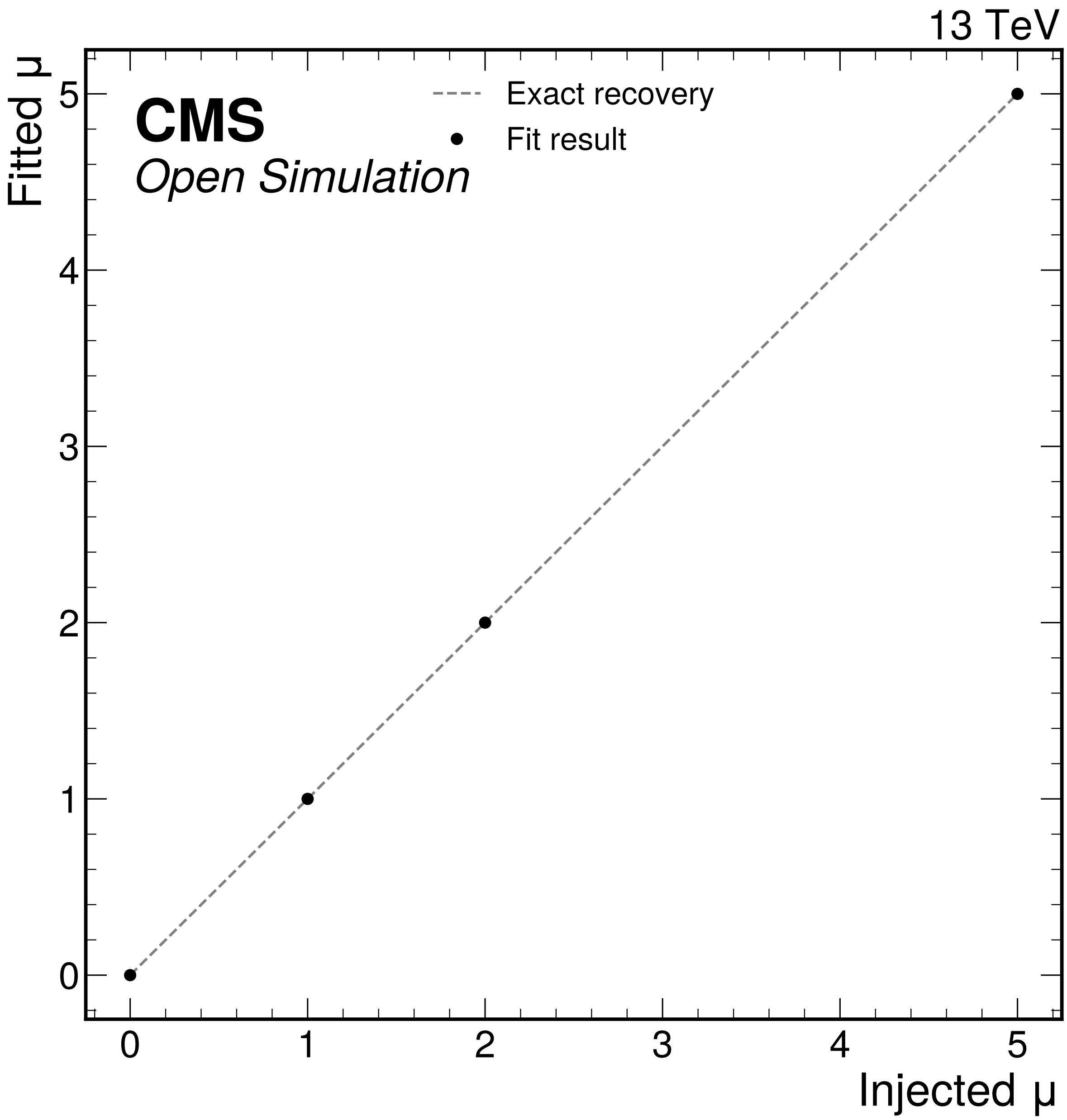

Figure 9: Signal injection and recovery test at mu = 0, 1, 2, and 5. All injected Asimov configurations are refit with the same nuisance model and checked against the 20 percent bias gate.

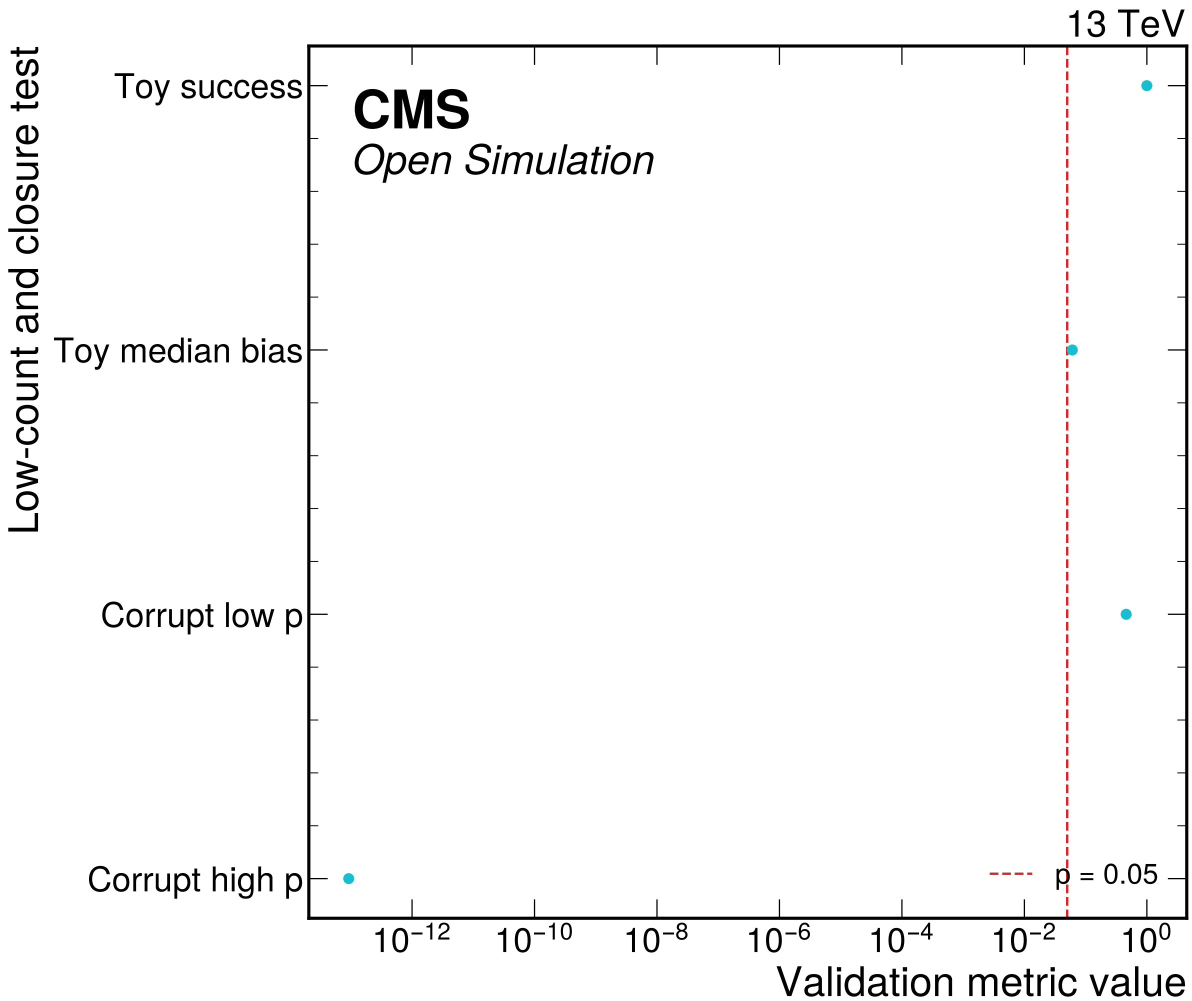

Figure 10: Low-count toy and closure-sensitivity validation summary. The final-state simultaneous corruption test rejects the +20 percent mass-response case; the -20 percent case is retained as a documented low-count sensitivity limitation in the JSON.

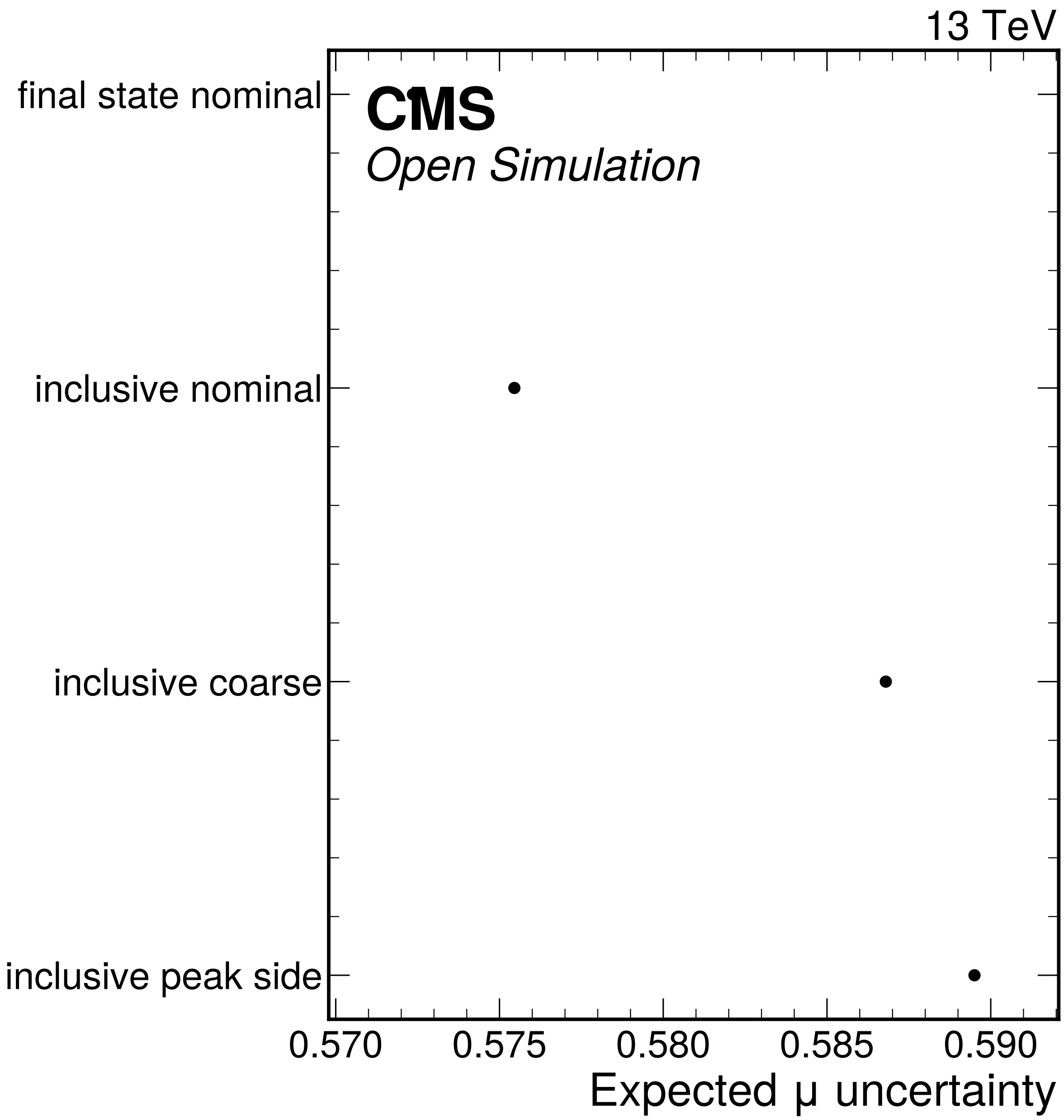

Figure 11: Alternative-binning stability comparison for the expected signal-strength fit. The final-state nominal configuration is retained for the Asimov result after low-count toy validation, while inclusive/coarse variants provide stability cross-checks.

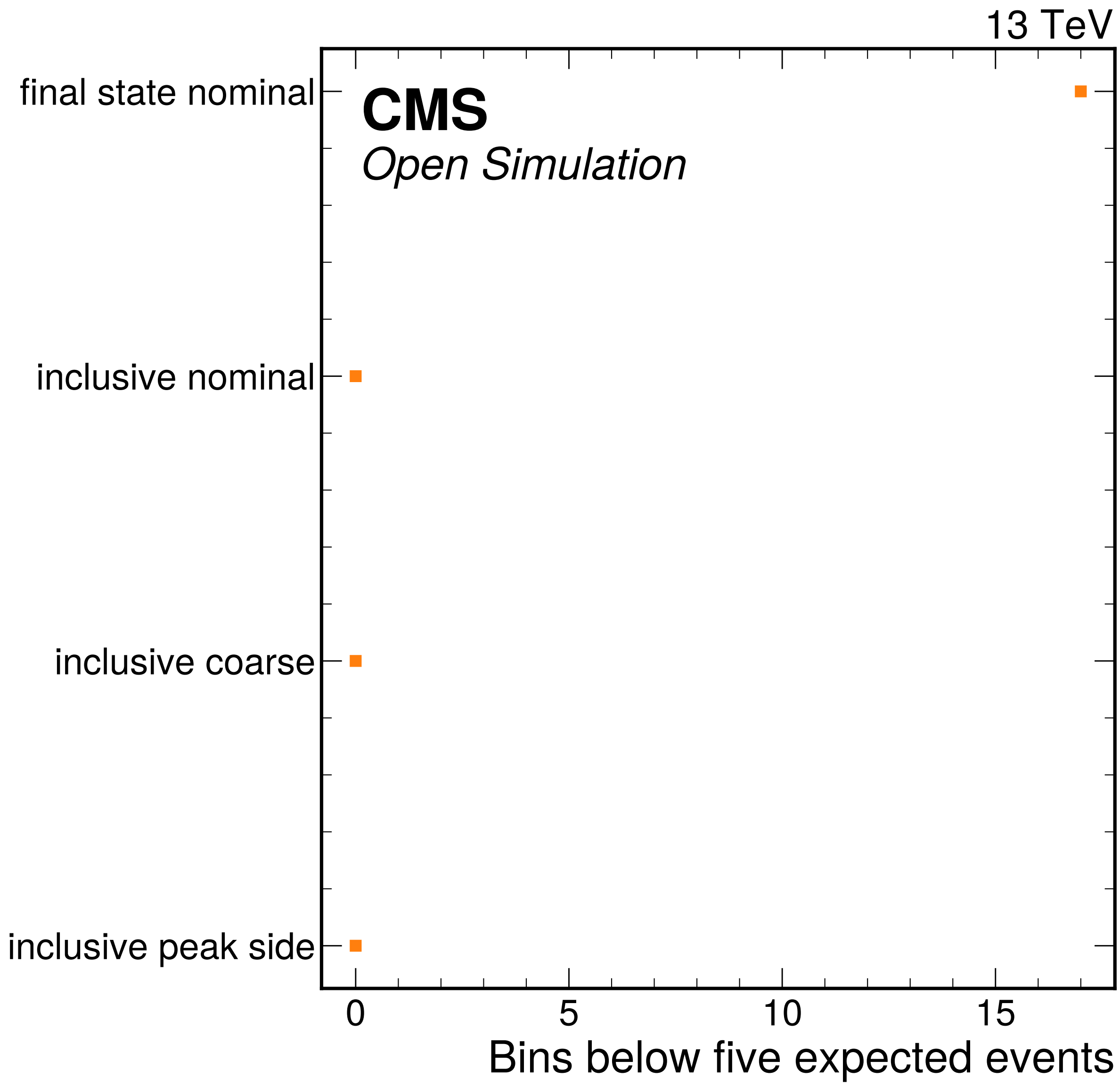

Figure 12: Low-count bin summary for the alternative-binning stability comparison. The nominal final-state model retains low expected-count bins only after the dedicated Poisson toy validation passes.

7 Statistical Method

The nominal inference uses a binned simultaneous template likelihood implemented with pyhf [13, 14]. One global parameter of interest μ scales all Higgs production templates together. The active channels are 4μ, $4e$, and $2e2\mu$, with mass-bin edges $[105, 112, 118, 122, 126, 130, 140]$ GeV.

The likelihood is

$$L(\mu, \boldsymbol{\theta}) = \prod_{c,b} \mathrm{Pois}\left(n_{cb} \mid \lambda_{cb}(\mu, \boldsymbol{\theta})\right) \prod_{k} \pi_k(\theta_k), \tag{9}$$

where π_k are Gaussian or log-normal constraints depending on the modifier. The profile-likelihood test statistic is

$$q(\mu) = -2 \log \frac{L(\mu, \hat{\hat{\boldsymbol{\theta}}}_\mu)}{L(\hat{\mu}, \hat{\boldsymbol{\theta}})}. \tag{10}$$

The one-sigma interval is read from $\Delta q(\mu) = 1$ on the expected profile scan. Asymptotic profile-likelihood methods and numerical minimization tools are standard for HEP measurements [15, 16], but the low expected bin counts are why the note also reports toy and binning-stability diagnostics [15].

The parameter convention is deliberately simple. A single global signal-strength POI multiplies all Higgs production components, so the fit measures the total Higgs yield relative to the nominal prompt-normalized MC expectation. The production-mode labels are retained inside the template model only so that signal-composition uncertainties can be propagated. They are not independent POIs, and the note does not quote ggH, VBF, or VH signal strengths. This avoids over-interpreting the open-data sample and the lepton-only categories.

The nuisance parameters are interpreted differently in expected and observed phases. In Phase 4a the Asimov pseudo-data are generated from the same nominal model that is fitted, so nuisance best fits at their nominal values are expected and are not evidence that the data constrain those systematics. The nuisance-impact table instead answers a different question: if a source is shifted by its prescribed prior variation and the model is refit, how much can the expected signal-strength estimate move? That is why the impact ranking is meaningful in Phase 4a even though the nuisance pulls themselves are not observed-data measurements.

The goodness-of-fit numbers are also staged quantities. The nominal Asimov deviance and chi-squared are zero by construction when the fitted expectation equals the generated expectation, so a reported $p = 1$ is only a bookkeeping check. The validation evidence is therefore the ensemble of non-tautological tests: injected signal strengths at $\mu = 0$, 1, 2, and 5; Poisson toy behavior in the sparse final-state bins; alternative inclusive and coarser binnings; channel compatibility; and intentionally corrupted mass-response models. The corrupted-model test is especially important because it probes whether the low-count likelihood can reject a wrong shape rather than merely recover an injected normalization.

The mass-profile scan uses the same likelihood structure but replaces the signal template at each grid mass with a shifted detector-level M125-derived template and profiles μ at each grid point. The scan therefore tests method parity with the CMS-style simultaneous category mass extraction, but it is not promoted to a calibrated mass measurement. The statistical reason is that the template family does not include independent simulated mass hypotheses or official per-lepton calibration and resolution inputs. The reported closure table shows that the implemented scan can recover injected grid masses under its own assumptions; it does not establish the full experimental mass scale.

The expected interval is asymmetric because it is read from the profile scan rather than from a fixed quadratic approximation. The symmetric uncertainty reported in summary tables is the arithmetic average of the upward and downward one-sigma distances and is used only for compact comparison metrics such as the precision ratio to CMS-HIG-16-041. Whenever the interval is the result, the note quotes the asymmetric pair from expected_parameters.json.

8 Expected Results

The Phase 4a expected-only result uses Asimov pseudo-data equal to the nominal model expectation. The fitted value is therefore centered at $\mu = 1.0$ by construction; the physics content of Phase 4a is the expected uncertainty, the validation of the low-count workspace, and the complete documentation of approximations.

Table 10: Expected signal-strength result from expected_parameters.json.

Quantity	Value	Minus uncertainty	Plus uncertainty
μ	1	0.5167	0.6327
Symmetric uncertainty	0.5747	–	–

Table 11: Variance decomposition for μ.

Component	Variance
Statistical	0.3064
MC statistical approximation	0.002956
Systematic including MC stat	0.02396
Total	0.3303

8.1 Mass-Profile Attempt

The simultaneous mass-profile attempt uses shifted detector-level M125 templates in the same final-state categories as the signal-strength workspace, with μ profiled at each mass grid point. The scan range is 110–140 GeV. This is method-parity evidence and closure validation, not an official Higgs mass measurement, because independent mass-hypothesis MC and official lepton calibration/morphing inputs are unavailable.

Table 12: Mass-template closure. The closure gate passes for injected masses, but the result is not promoted to an official mass measurement.

Injected mass [GeV]	Recovered grid [GeV]	Bias [GeV]	Pass
115	115	0	yes
125	125	0	yes
135	135	0	yes

9 Comparison to Prior Results and Theory

The expected μ result is contextualized against CMS-HIG-16-041 and CMS-HIG-19-001, but those values are not inputs to the fit and the analysis is not tuned to match them. With the CMS-HIG-16-041

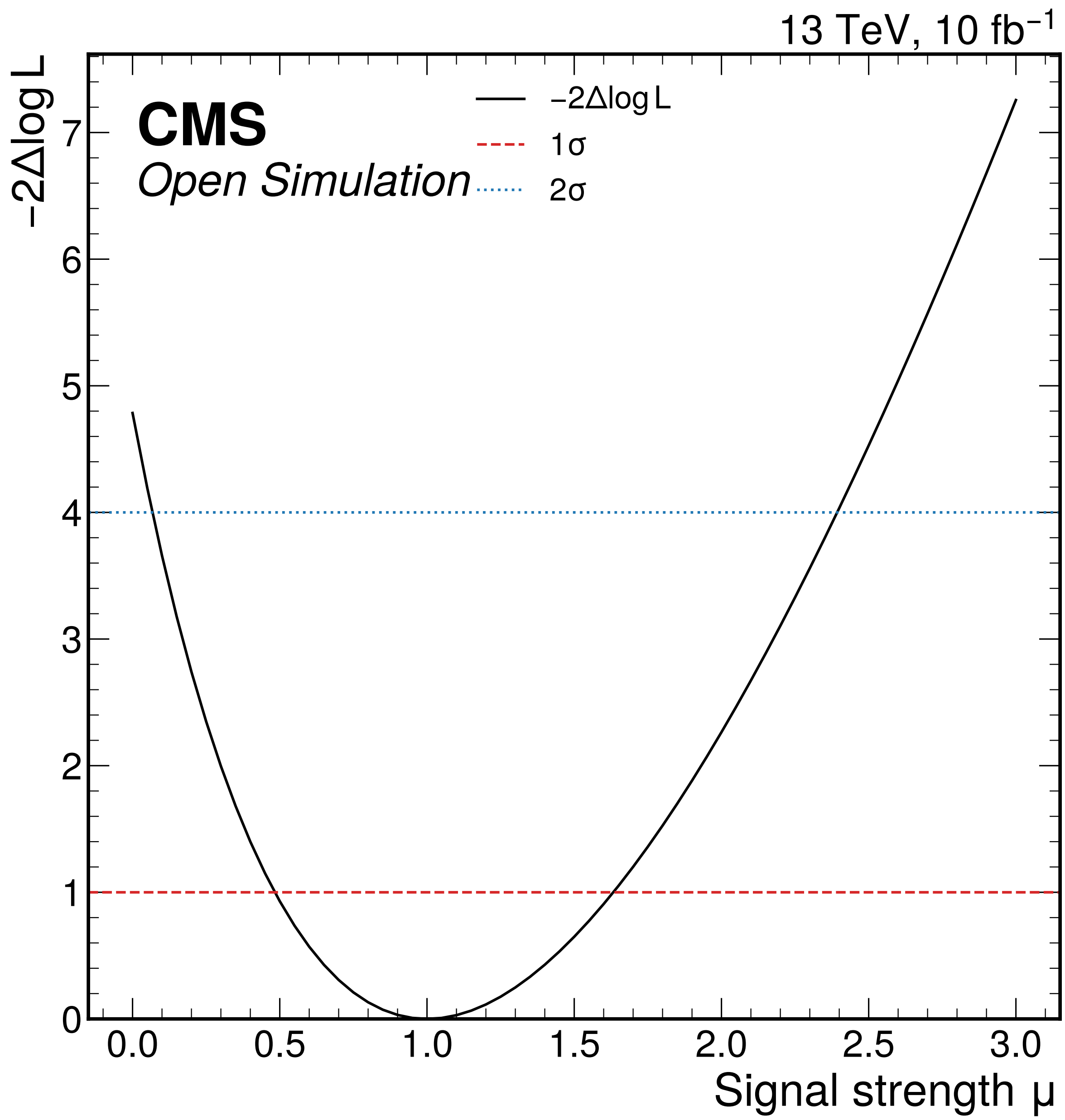

Figure 13: Expected profile-likelihood scan for the global Higgs signal-strength parameter. The Asimov best fit is mu = 1 with symmetric expected uncertainty 0.575.

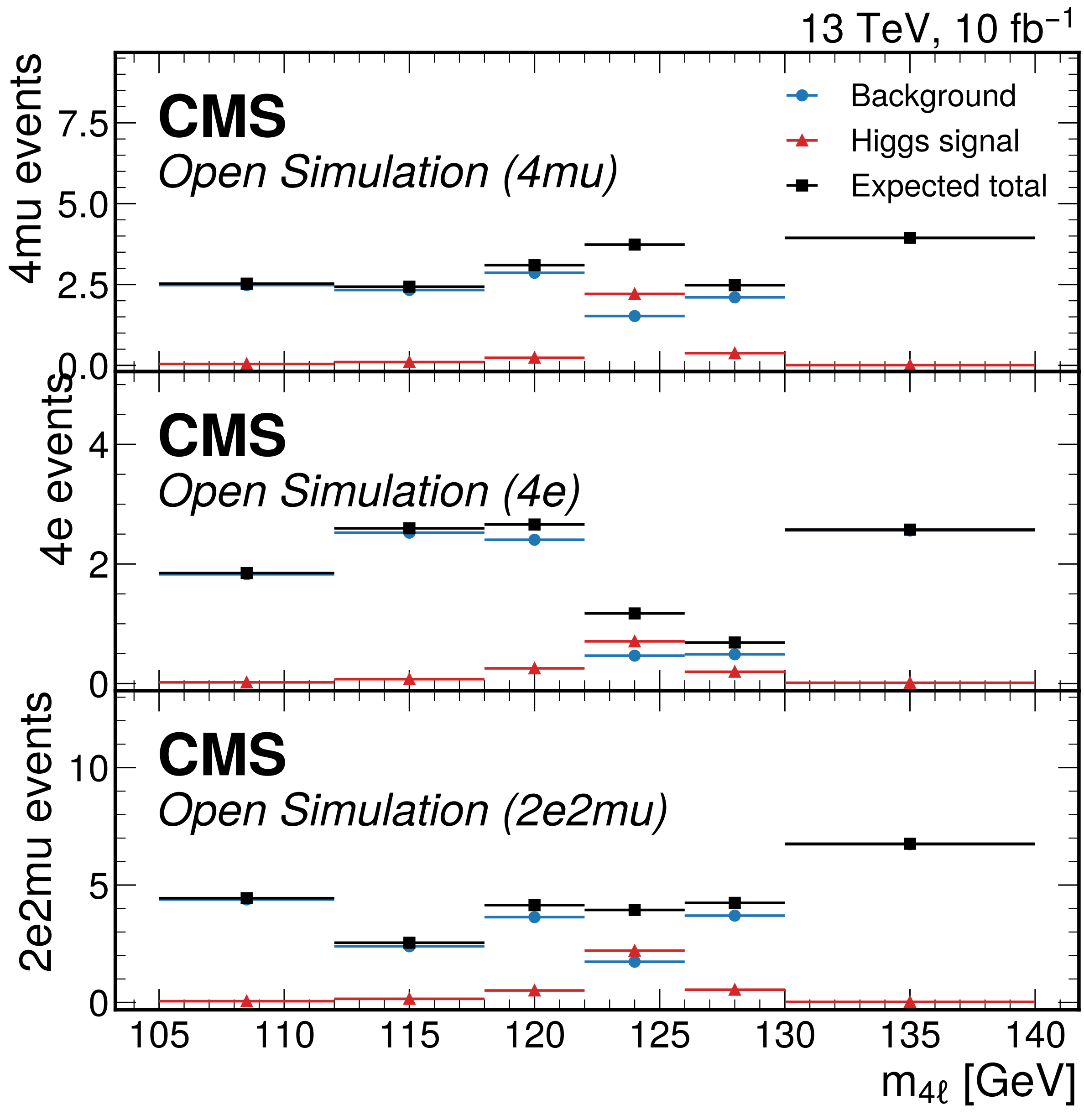

Figure 14: Expected Asimov four-lepton mass templates in the final-state categories used by the Phase 4a simultaneous fit. Points show the background expectation, Higgs signal expectation, and their total; no observed Open Data counts are used as pseudo-data.

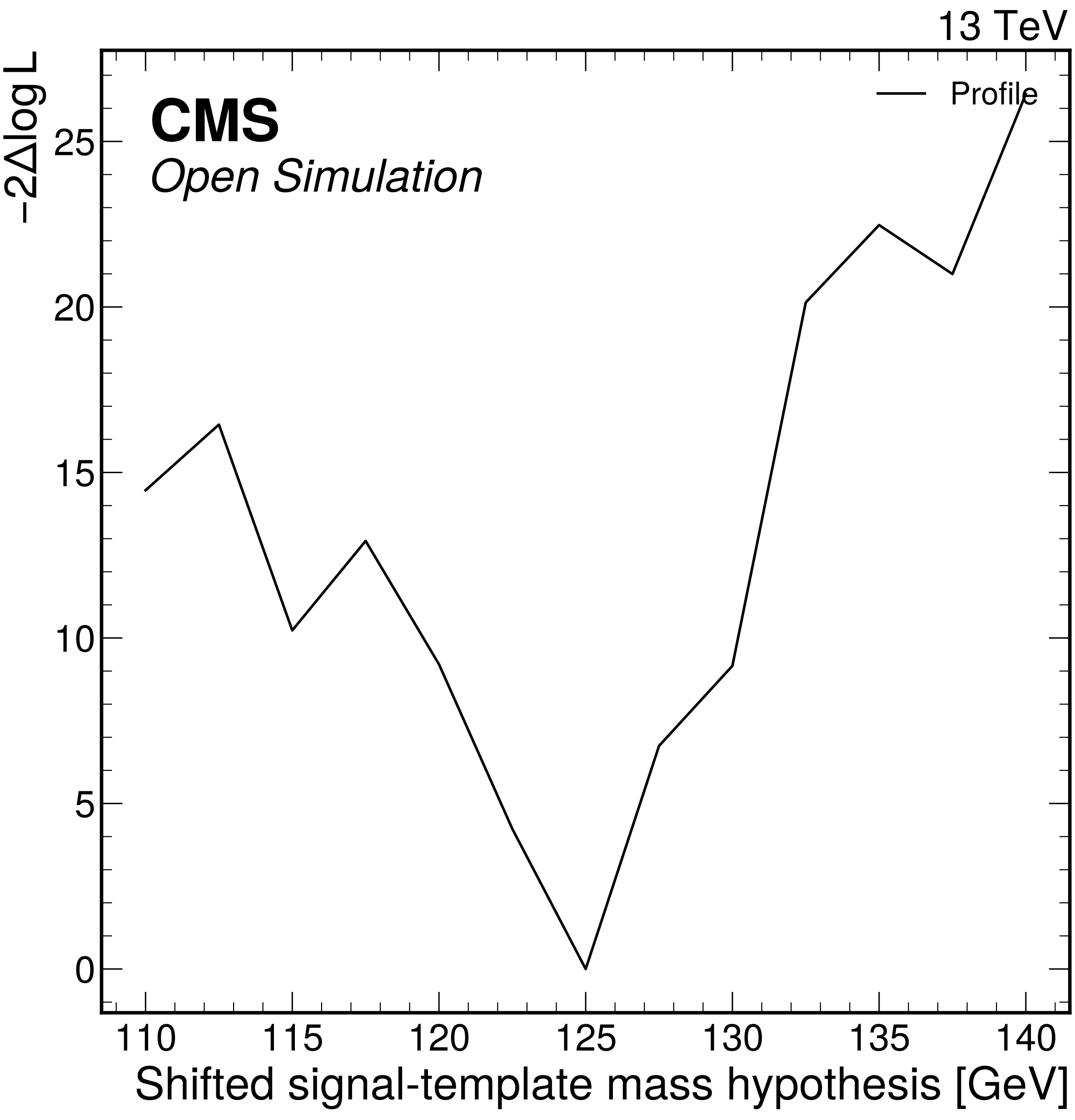

Figure 15: Detector-level shifted-template mass-profile attempt with mu profiled. This is retained as method-parity evidence; it is not promoted to an official-quality mass measurement because independent mass-hypothesis MC and official calibration inputs are unavailable.

symmetric uncertainty approximated as $(0.19+0.17)/2 = 0.18$, the Phase 4a expected uncertainty ratio is 0.575 / 0.18 = 3.19. This comparison is limited to precision and analysis scope because the Phase 4a value is an Asimov expectation, not an observed measurement.

The mass-profile closure best grid point is 125 GeV for the nominal injected template. This is retained only as an internal closure check of the shifted-template scan under its own assumptions. It is not compared as a residual against CMS or PDG Higgs-mass values because the Phase 4a mass scan is a shifted-template detector-level closure exercise rather than a calibrated mass measurement.

Table 13: Comparison matrix. Non-comparable entries are explicitly labeled, and Asimov expected results are compared only by uncertainty and scope.

Quantity	This Phase 4a note	Reference	Comparability
Inclusive μ	$1.0^{+0.633}_{-0.517}$	CMS-HIG-16-041: $1.05^{+0.19}_{-0.17}$	Contextual precision comparison; expected-only result
Inclusive μ	$1.0^{+0.633}_{-0.517}$	CMS-HIG-19-001: $0.94 \pm 0.07(\text{stat})^{+0.09}_{-0.08}(\text{syst})$	Contextual; different dataset/scope
Mass	125 GeV grid closure	CMS-HIG-16-041 mass result	Internal closure only; not an official-quality measurement
Fiducial cross section	not extracted	CMS-HIG-16-041: $2.92^{+0.48}_{-0.44}(\text{stat})^{+0.28}_{-0.24}(\text{syst})$ fb	unavailable in Phase 4a
Width	not extracted	CMS-HIG-16-041: $\Gamma_H < 1.10$ GeV	unavailable
VBF categories	downscoped	CMS seven production-sensitive categories	unavailable without jets
Reducible background	DY+jets MC proxy	CMS data-driven Z+X	approximated

10 Conclusions

The expected-only Phase 4a signal-strength measurement is complete enough to support downstream 10% and full-data updates. The expected result is $\mu = 1.0^{+0.633}_{-0.517}$ with symmetric uncertainty 0.575 in the $105 < m_{4\ell} < 140$ GeV fit window. The precision is about 3.19 times weaker than the symmetrized CMS-HIG-16-041 signal-strength uncertainty, which is consistent with the smaller 10 fb^{-1} open-data sample, detector-level simplifications, missing official calibrations, no VBF category, and DY+jets fake-proxy treatment.

The analysis satisfies the expected inference gate with documented limitations. The most important caveats are the grouped MC-stat approximation, the low-count final-state bins, the non-rejected -20% mass-response corruption test after three final-state-aligned attempts, and the fact that the mass-profile result is closure evidence rather than an official mass measurement. The final-state workspace is sparse, with 17 of 18 expected bins having $S + B < 5$, and the -20% corruption sensitivity is documented low-count infeasible after three attempts, not passed.

11 Future Directions

The Phase 4b update should repeat low-count toy validation on the fixed-seed 10% observed data subset, compare the observed partial-data result with the expected interval, and preserve the same fit window

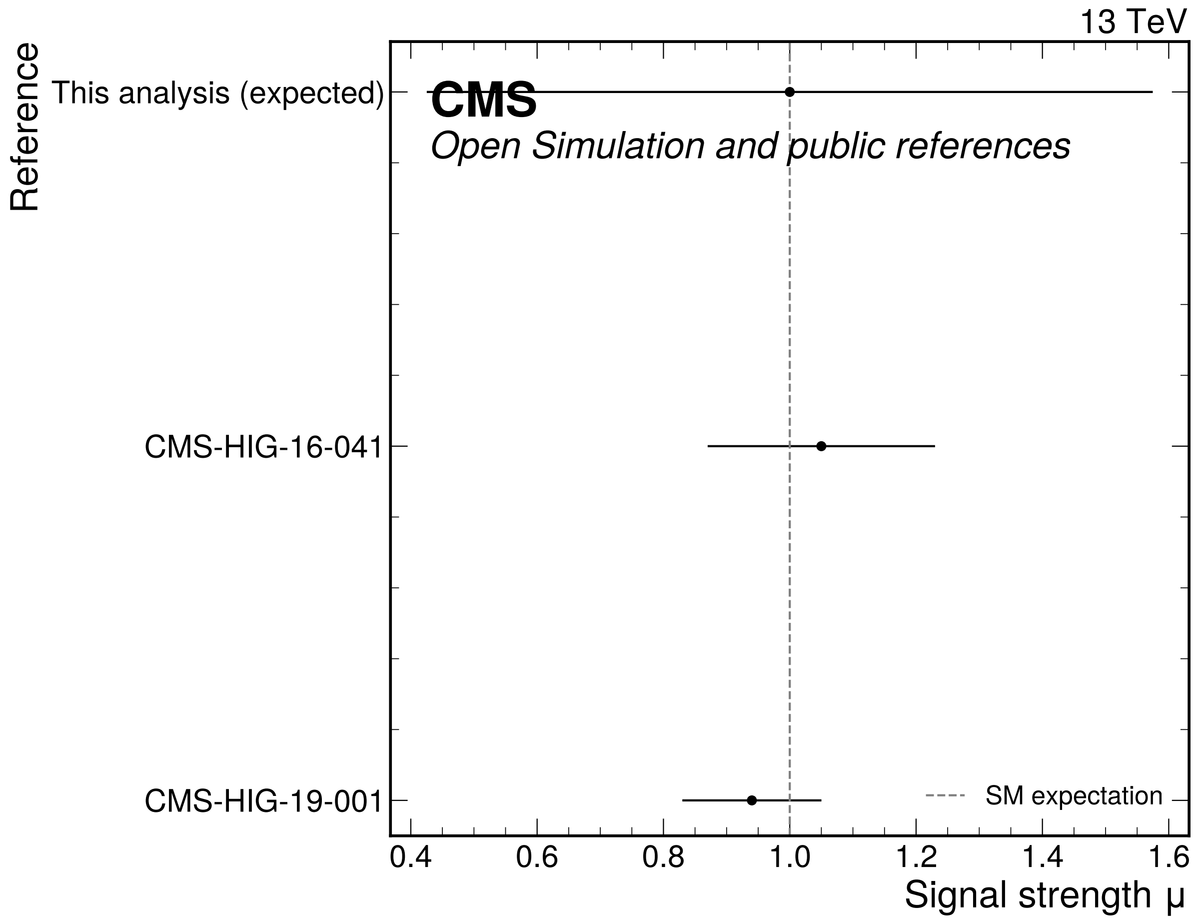

Figure 16: Expected Phase 4a signal-strength precision compared with public CMS H to ZZ to four-lepton references. The expected uncertainty/reference ratio 3.19 is computed relative to CMS-HIG-16-041 only; CMS-HIG-19-001 is shown as an additional public comparison.

and category definitions unless a reviewed regression changes them. The Phase 4c update should repeat the fit on the full observed data and revisit alternative binning if observed GoF or nuisance behavior changes. A true VBF category requires upstream jet information or a reviewed expansion of allowed inputs; it cannot be recovered from the current flat ntuples. A more official fake-background treatment would require a data-driven Z+X estimate, which is intentionally absent from the current scope.

12 Known Limitations and Open Questions

First, the analysis is detector-level and uses prompt-effective MC normalization. It does not unfold to a fiducial phase space, and it does not produce a fiducial cross section. The practical impact is that comparisons to CMS fiducial cross sections are unavailable, while μ comparisons remain contextual.

Second, the reducible background is a DY+jets MC proxy. This was requested by the user and propagated with a broad nuisance, but it is not equivalent to the official CMS data-driven Z+X method. Its dominant impact on μ is an honest statement of weak fake-background control.

Third, the final-state simultaneous workspace is sparse: Phase 3 had 17 of 18 bins with expected $S + B < 5$. Phase 4a toys and alternative binnings support expected-only use, but the -20% mass-response corruption test remains documented low-count infeasible after three attempts. This limitation must be rechecked with observed data.

Fourth, the VBF and neural-network goals were attempted but not promoted. VBF is unavailable because jet/VBF information is absent; the NN does not pass the promotion gates and is not used nominally. These are downscopes, not hidden analysis choices.

A Figure Inventory and Auxiliary Plots

All 49 upstream figure PDFs are staged under analysis_note/figures and included in this note. The following appendix figures carry the detailed validation and reconnaissance trail so that the body can focus on the physics narrative.

B Limitation Index

Table 14: Constraint, decision, and limitation index carried into Doc 4a.

Label	Status	Impact
A1/D1	Resolved	Primary prompt paths used; local copies not mixed.
A2/SP2	Implemented fallback	Prompt cross sections used with process nuisances.
A3/D4	Formally downscoped	No VBF category or jet systematics.
A4/D8	Resolved for attempt	Angular variables recomputed and validated before NN attempt.
A5	Limitation	No truth-level object closure claims.
A6/SP6	Documented not propagated	PV variables excluded; no PV nuisance in nominal fit.
L1	Active limitation	Detector-level open-data analysis, not official CMS calibration.
L2/SP10	Active limitation	DY+jets MC fake proxy replaces data-driven Z+X.
L3/D9	Active limitation	Mass scan is closure evidence, not official mass measurement.
SP3	Formal downscope	Grouped MC-stat approximation replaces full bin-by-bin staterror.

Label	Status	Impact
VT12	Passed with caveat	Mass-template injected closure passes; result not promoted to official measurement.
Corruption -20%	Documented infeasible	Three final-state-aligned tests do not reject in low-count workspace.

C Reproduction Contract

A fresh analyst should reproduce the current expected result with the following commands from the analysis root:

1. 'pixi install' to materialize the analysis environment.
2. 'pixi run all' to run the chain through Phase 4a with the current task definition.
3. 'pixi run p3-all' to rebuild selection tables, VBF downscope evidence, angular closure, input validation, MVA diagnostics, and selection figures.
4. 'pixi run p4a-all' to rebuild the expected pyhf workspace outputs, result JSON files, figures, and commitments.
5. 'cd analysis_note && pixi run build-pdf ANALYSIS_NOTE_doc4a_v1.tex' to compile this note.

The execution DAG is: prompt and ROOT ntuples → Phase 1 metadata/literature inventory → Phase 2 strategy commitments → Phase 3 selection and categories → Phase 4a expected workspace/results JSON → Doc 4a LaTeX/PDF. Manual steps are limited to writing this note and reviewing rendered output.

D Machine-Readable Result Files

The numerical source of truth for Doc 4a is the JSON collection in the results directory. It contains the expected parameters, covariance, validation, mass scan, systematic summary, systematic shifts, and systematic-source records. Upstream selection JSON provides normalization, cutflow, input-validation, MVA metadata, approach-comparison, and selected-configuration records.

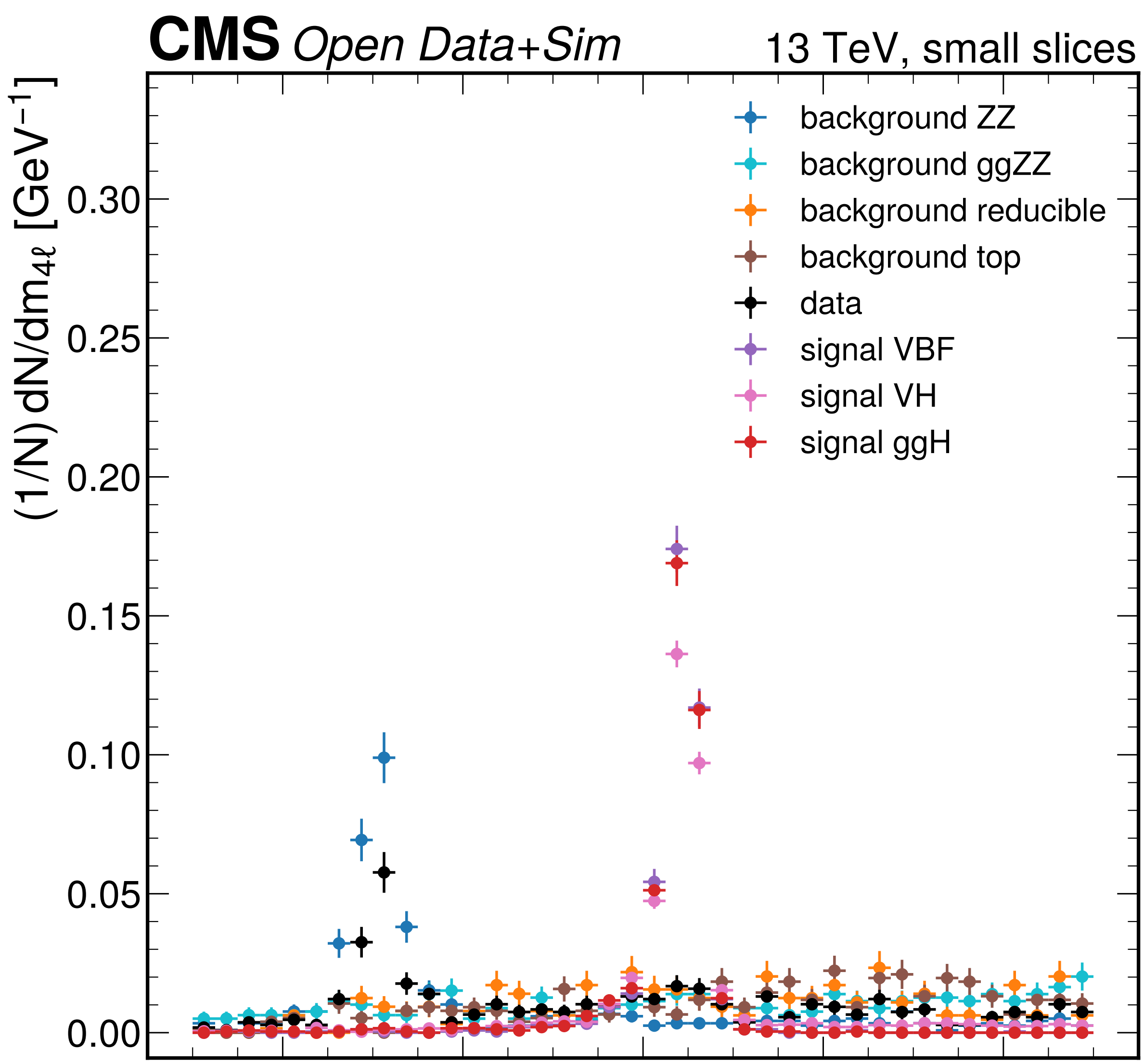

Figure 17: m_{4ell} [GeV] small-slice shape comparison. The distributions use at most 1000 entries per primary ROOT file and are area-normalized only for Phase 1 shape reconnaissance, not for yield validation. Large differences here flag candidate variables for Phase 2/3 data-MC quality checks rather than final selection conclusions.

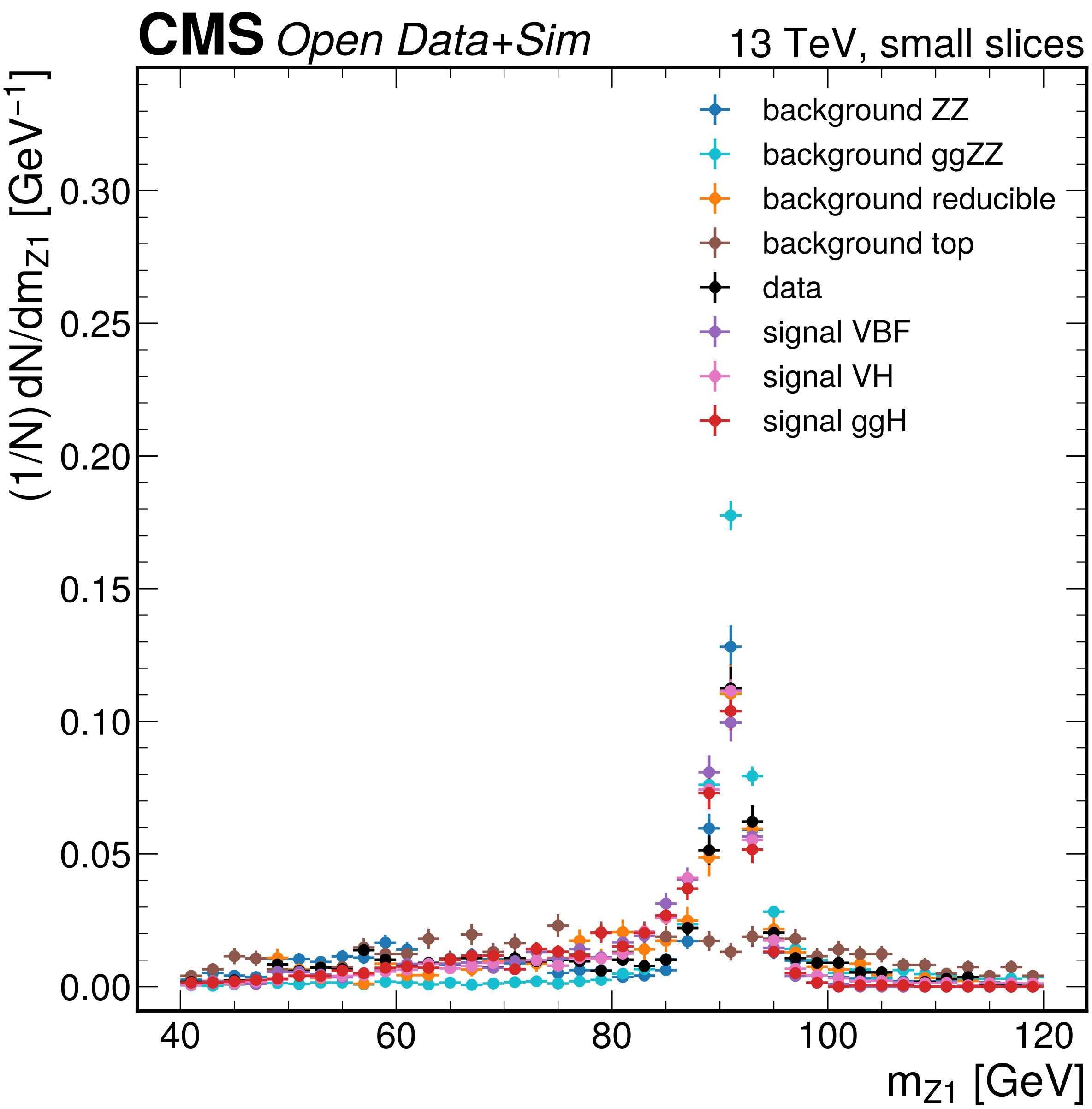

Figure 18: m_{Z1} [GeV] small-slice shape comparison. The distributions use at most 1000 entries per primary ROOT file and are area-normalized only for Phase 1 shape reconnaissance, not for yield validation. Large differences here flag candidate variables for Phase 2/3 data-MC quality checks rather than final selection conclusions.

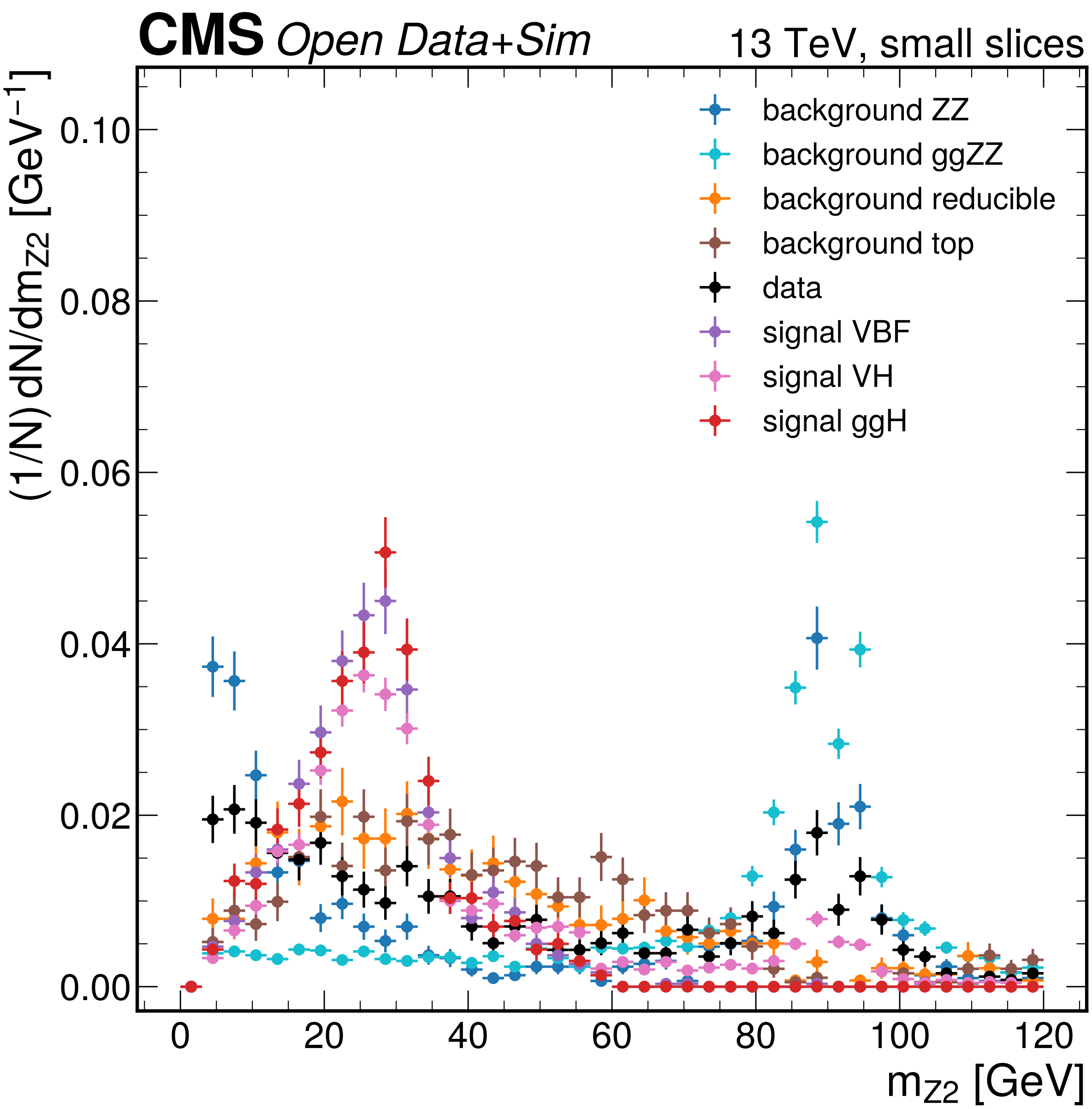

Figure 19: m_{Z2} [GeV] small-slice shape comparison. The distributions use at most 1000 entries per primary ROOT file and are area-normalized only for Phase 1 shape reconnaissance, not for yield validation. Large differences here flag candidate variables for Phase 2/3 data-MC quality checks rather than final selection conclusions.

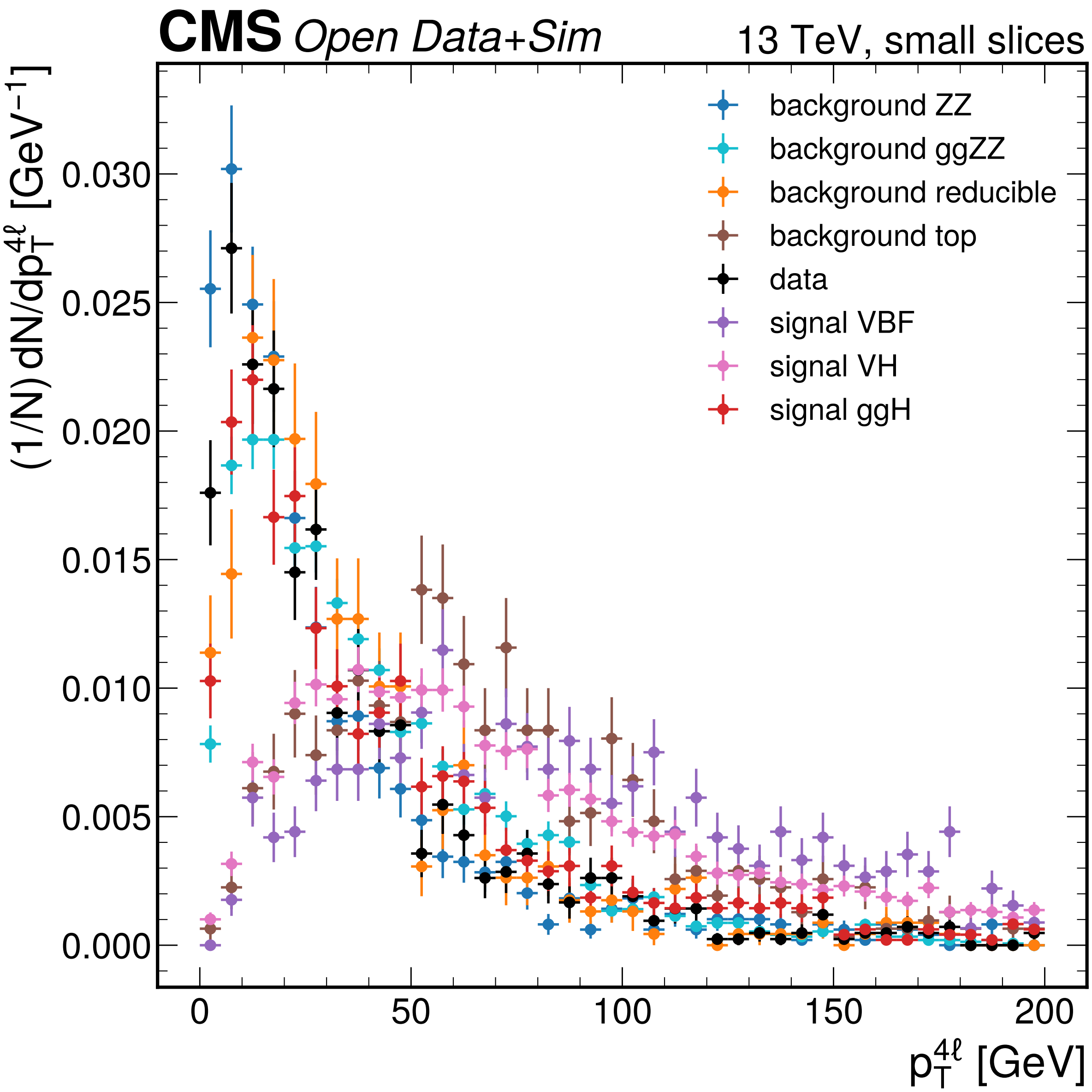

Figure 20: p_T^{4ell} [GeV] small-slice shape comparison. The distributions use at most 1000 entries per primary ROOT file and are area-normalized only for Phase 1 shape reconnaissance, not for yield validation. Large differences here flag candidate variables for Phase 2/3 data-MC quality checks rather than final selection conclusions.

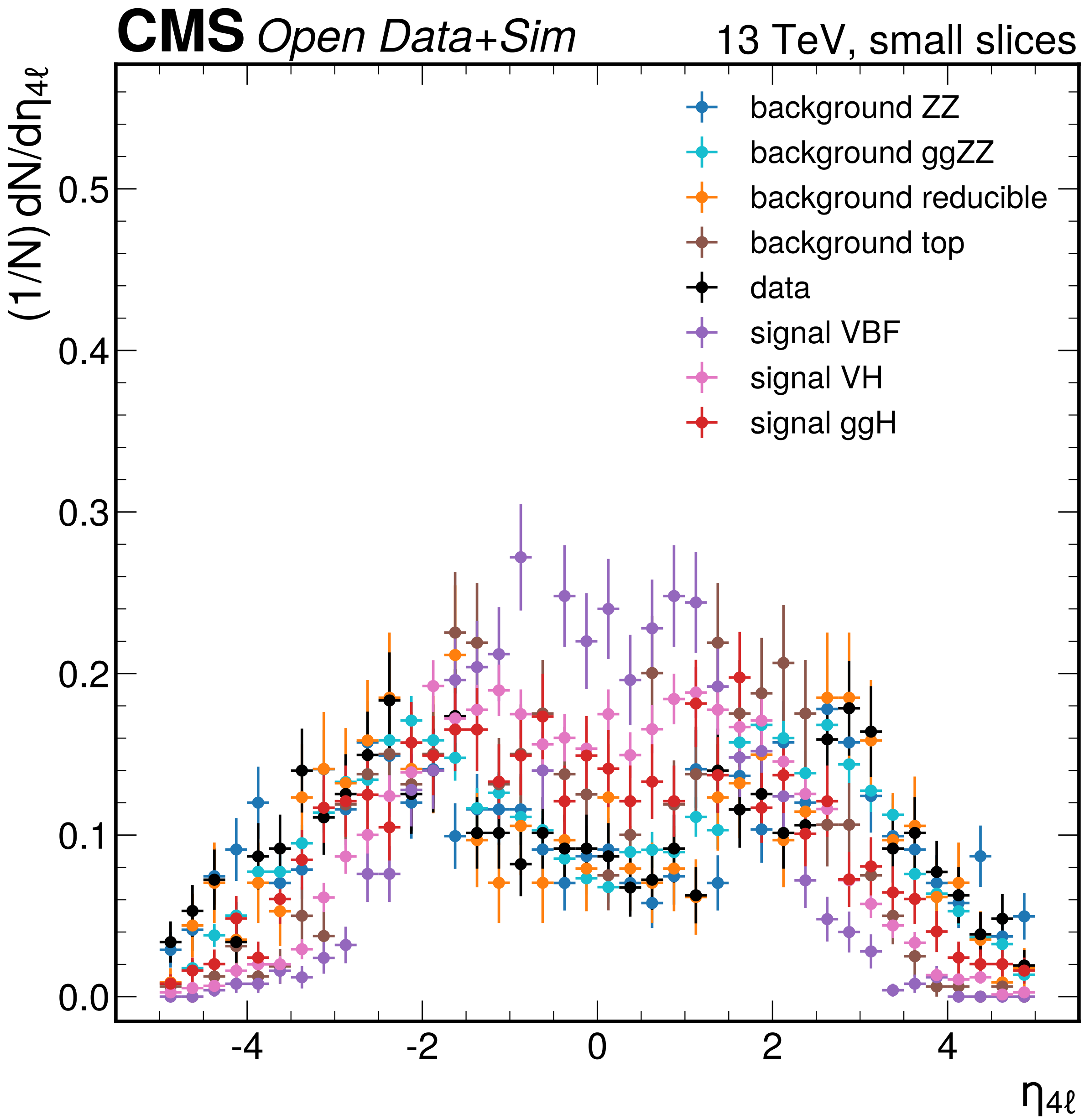

Figure 21: eta_{4ell} small-slice shape comparison. The distributions use at most 1000 entries per primary ROOT file and are area-normalized only for Phase 1 shape reconnaissance, not for yield validation. Large differences here flag candidate variables for Phase 2/3 data-MC quality checks rather than final selection conclusions.

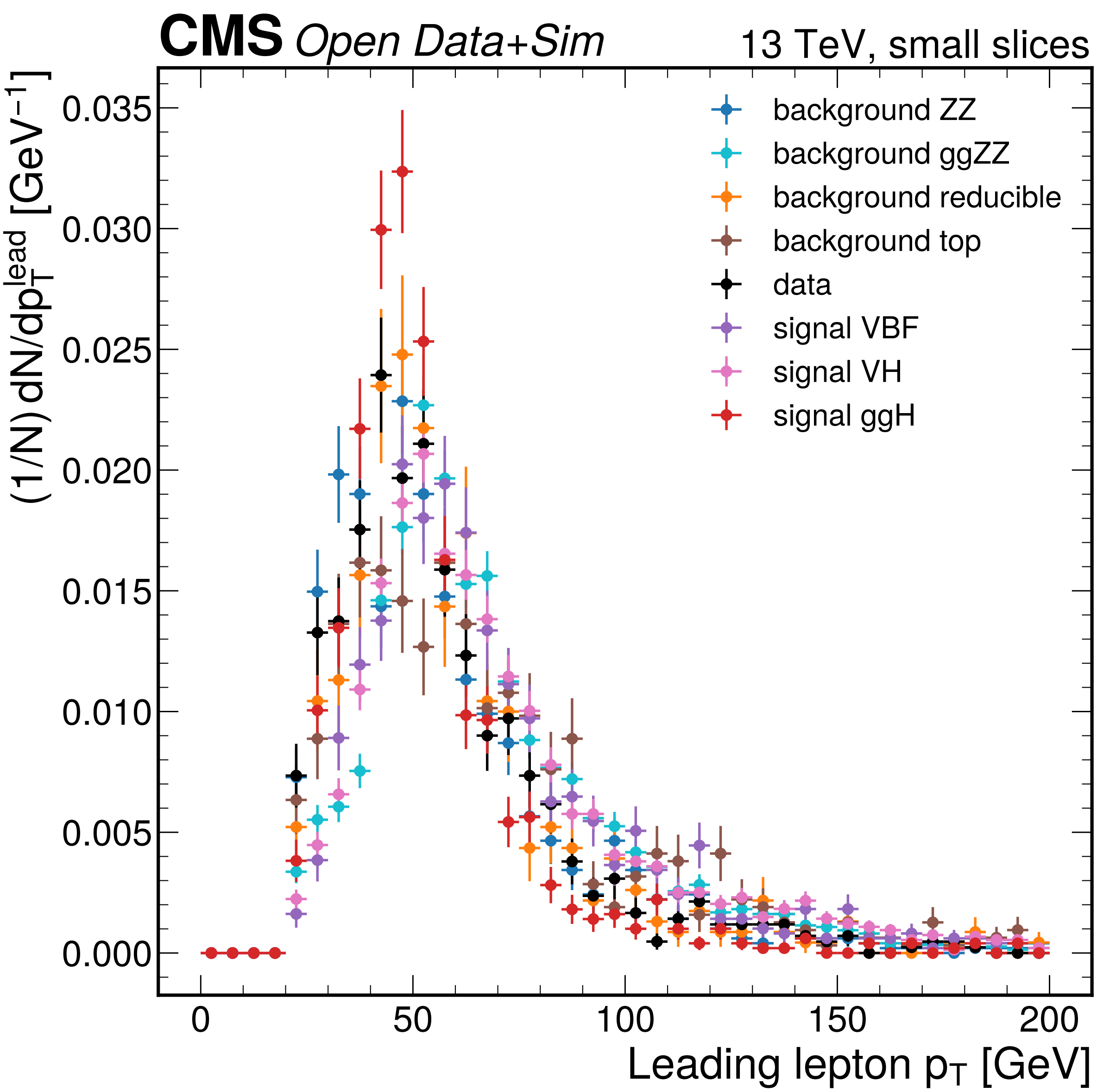

Figure 22: Leading lepton p_T [GeV] small-slice shape comparison. The distributions use at most 1000 entries per primary ROOT file and are area-normalized only for Phase 1 shape reconnaissance, not for yield validation. Large differences here flag candidate variables for Phase 2/3 data-MC quality checks rather than final selection conclusions.

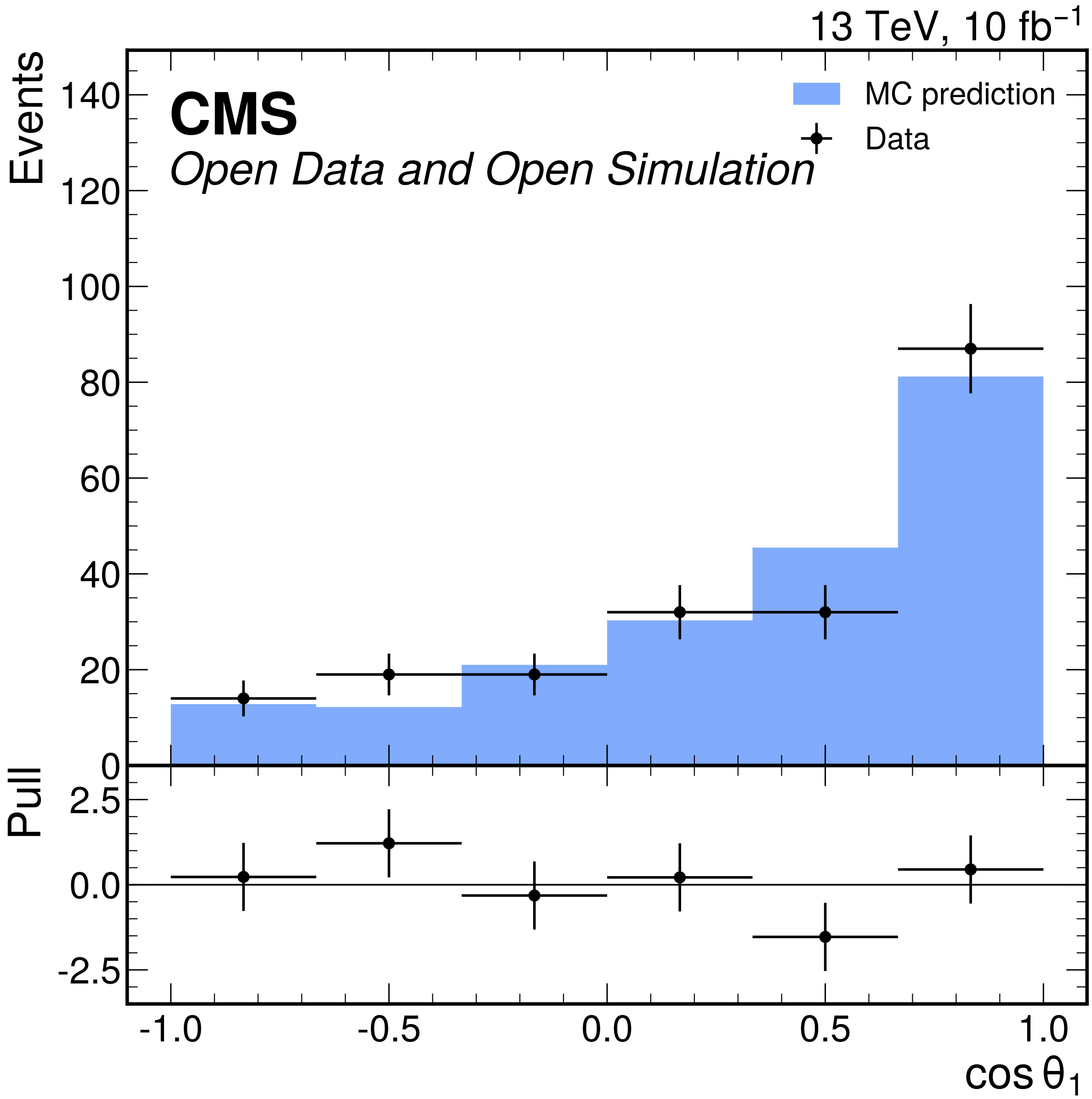

Figure 23: costheta_1 input-validation distribution in the broad $70 \leq m_{4\ell} \leq 170$ GeV window. The D7 shape gate fails with chi2/ndf = 1.60 and p = 0.156; this comparison is used only to decide classifier-input eligibility, not to normalize fit templates.

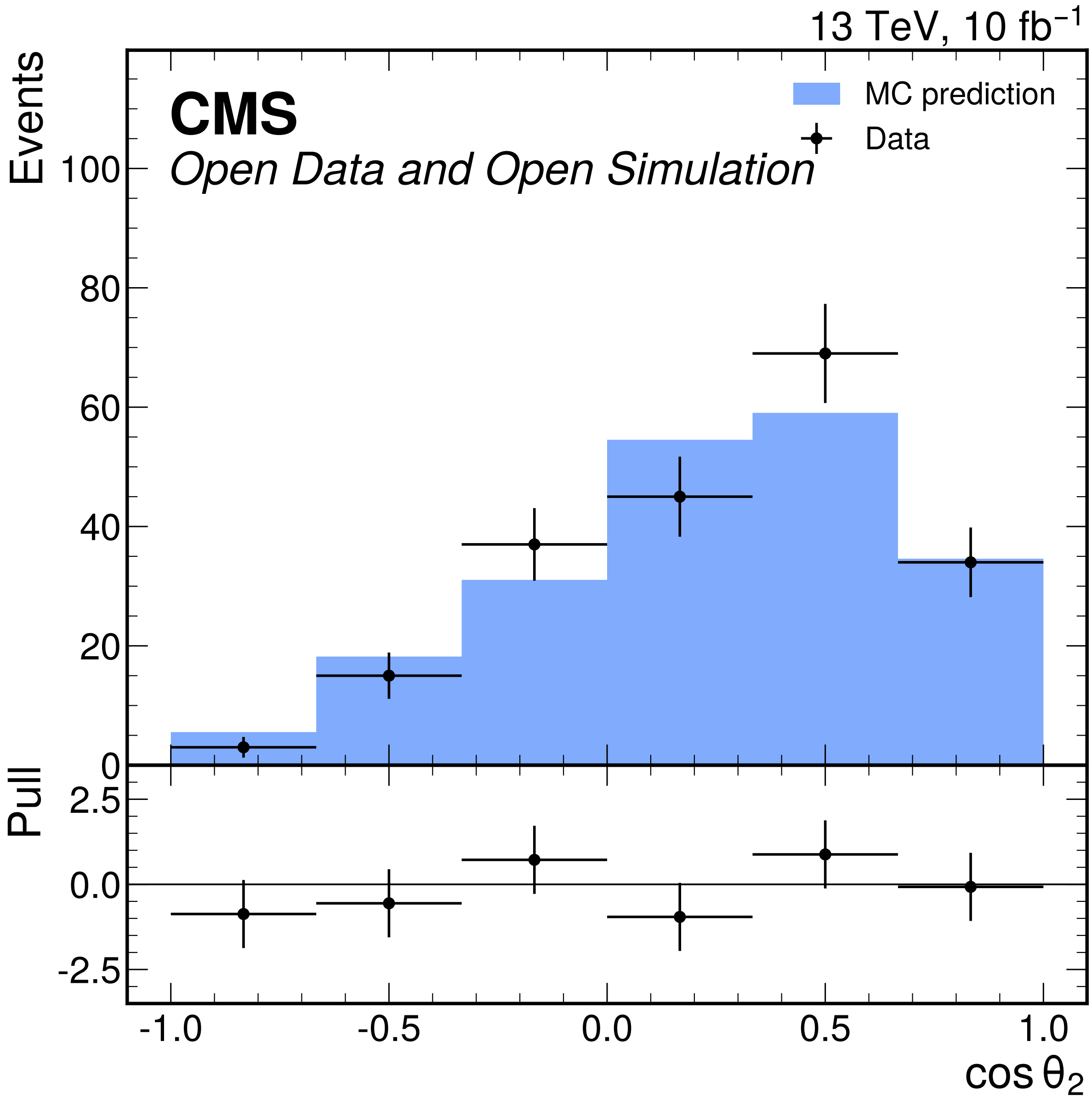

Figure 24: costheta_2 input-validation distribution in the broad $70 \leq m_{4\ell} \leq 170$ GeV window. The D7 shape gate fails with chi2/ndf = 1.29 and p = 0.265; this comparison is used only to decide classifier-input eligibility, not to normalize fit templates.

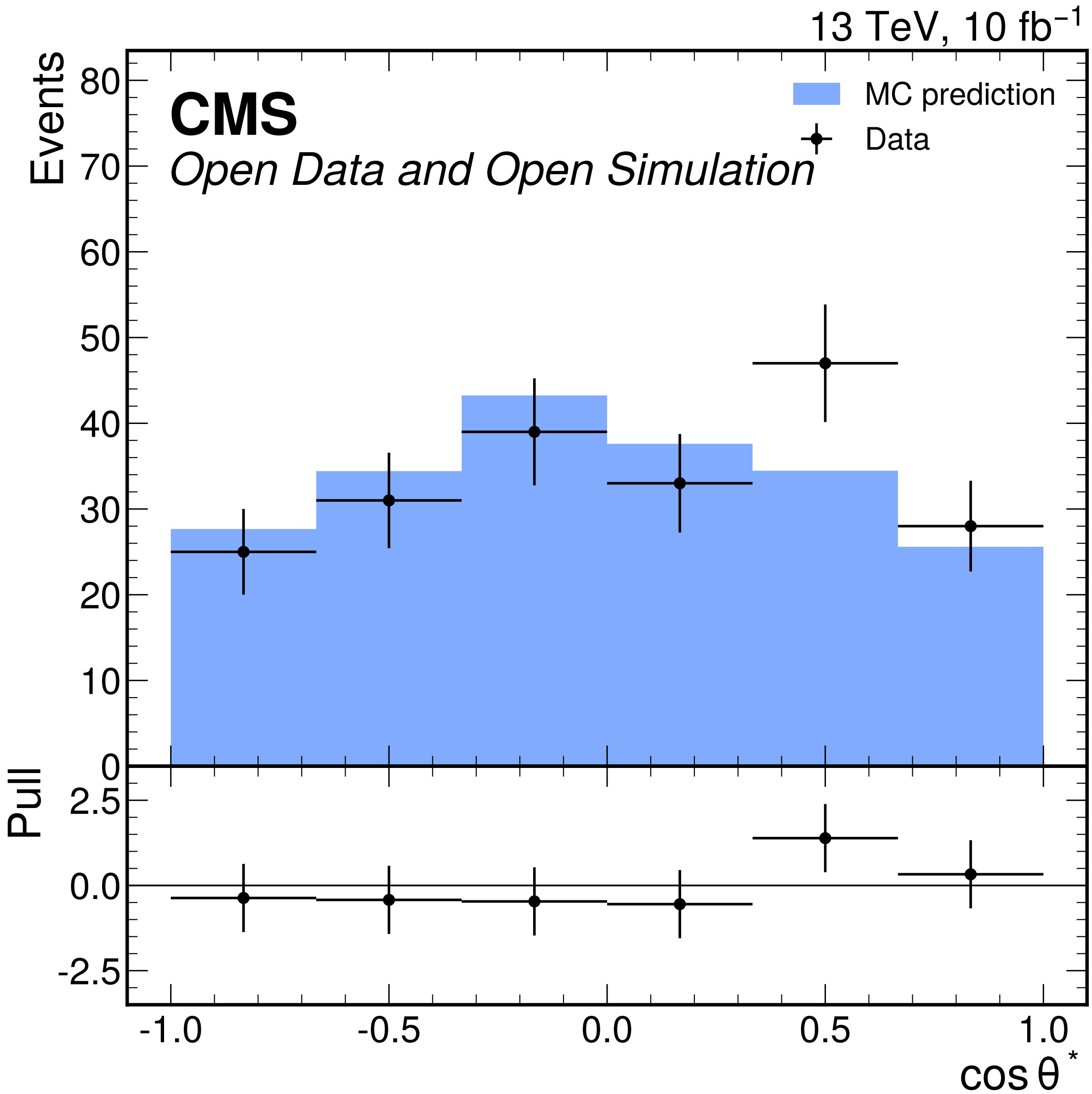

Figure 25: costheta^* input-validation distribution in the broad $70 \leq m_{4\ell} \leq 170$ GeV window. The D7 shape gate fails with chi2/ndf = 1.01 and p = 0.412; this comparison is used only to decide classifier-input eligibility, not to normalize fit templates.

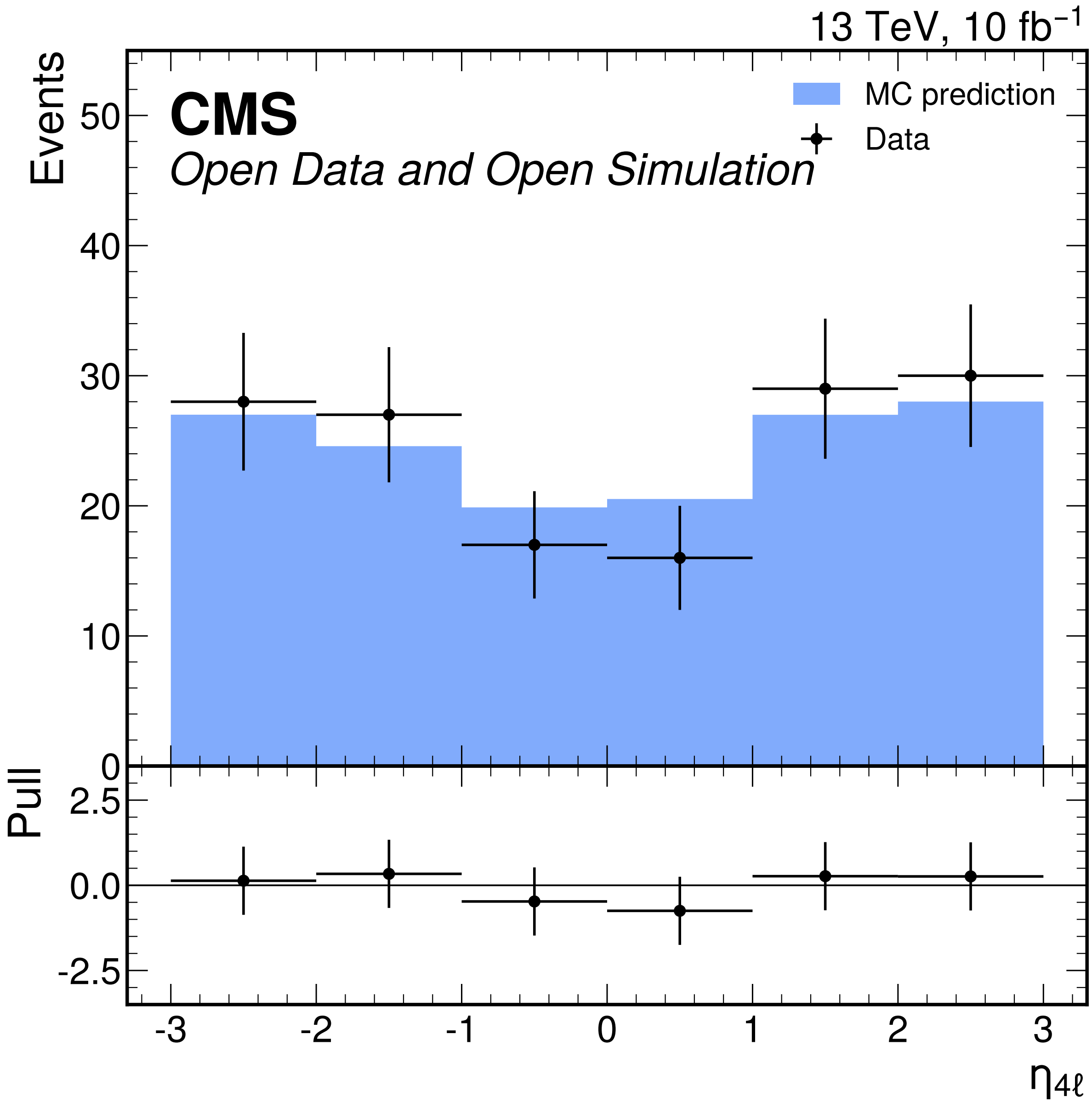

Figure 26: eta_{4ell} input-validation distribution in the broad $70 \leq m_{4\ell} \leq 170$ GeV window. The D7 shape gate fails with chi2/ndf = 0.404 and p = 0.846; this comparison is used only to decide classifier-input eligibility, not to normalize fit templates.

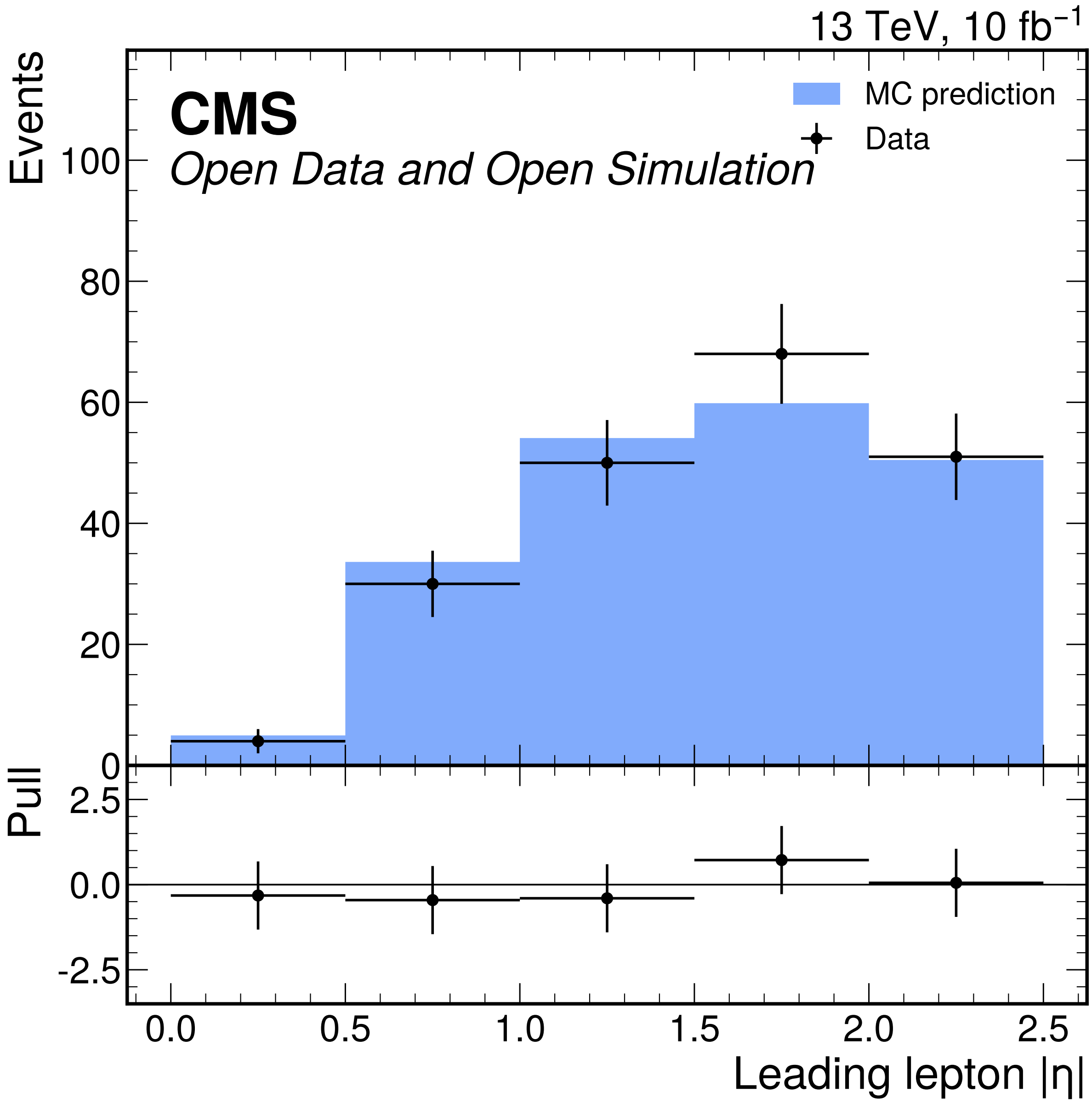

Figure 27: Leading lepton |eta| input-validation distribution in the broad $70 \leq m_{4\ell} \leq 170$ GeV window. The D7 shape gate passes with chi2/ndf = 0.473 and p = 0.755; this comparison is used only to decide classifier-input eligibility, not to normalize fit templates.

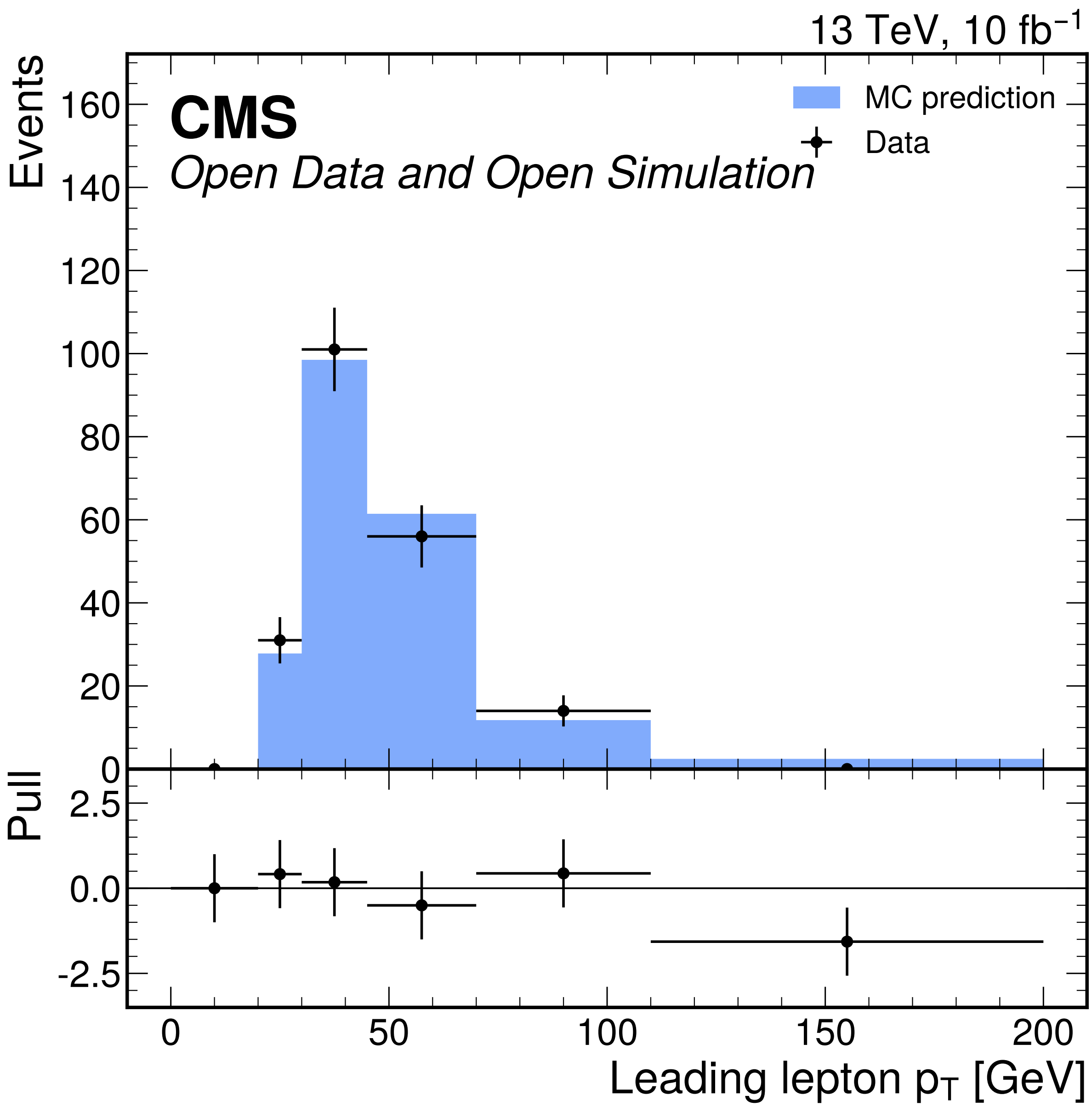

Figure 28: Leading lepton p_T [GeV] input-validation distribution in the broad $70 \leq m_{4\ell} \leq 170$ GeV window. The D7 shape gate fails with chi2/ndf = 4.06 and p = 0.00273; this comparison is used only to decide classifier-input eligibility, not to normalize fit templates.

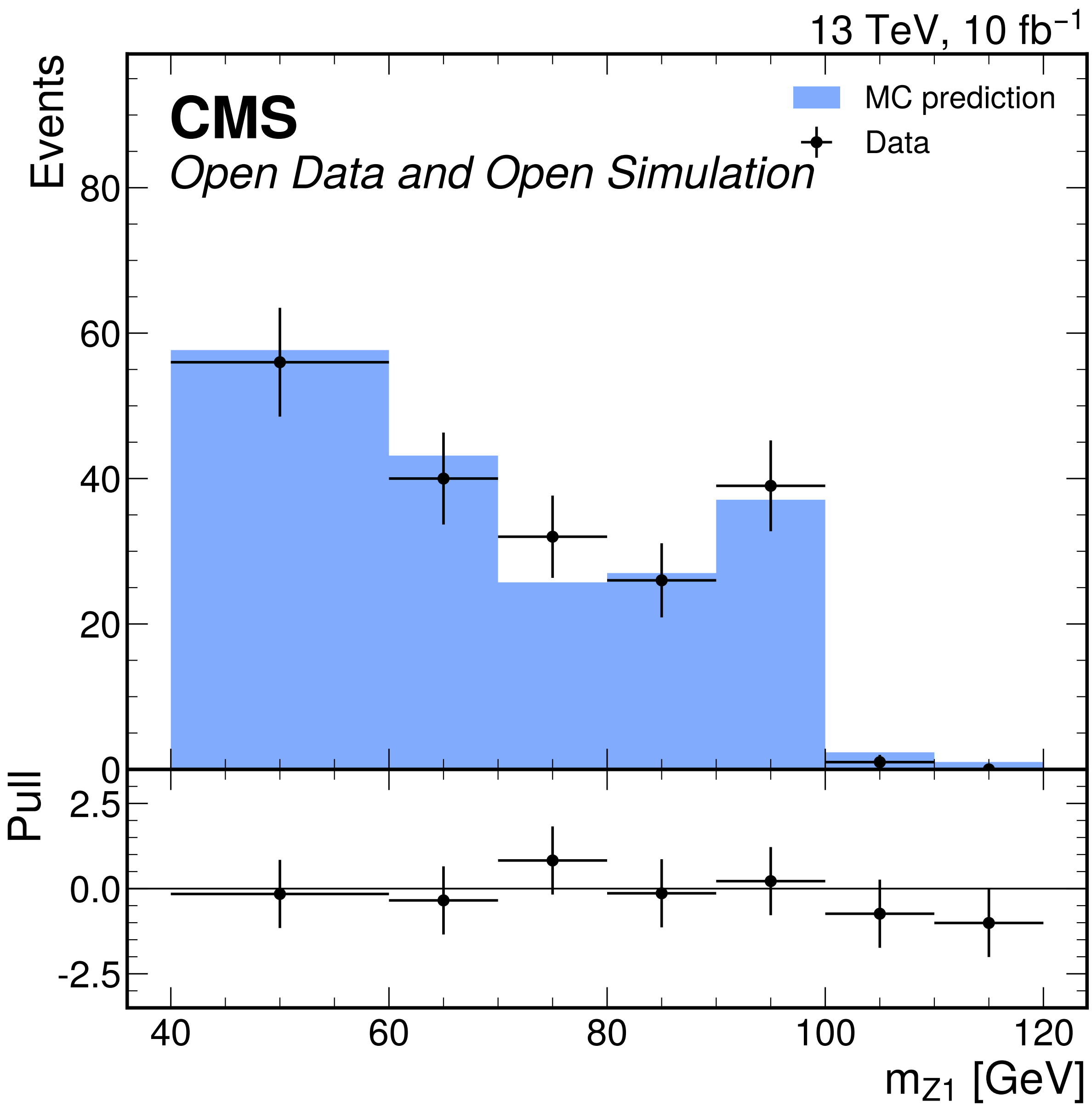

Figure 29: m_{Z1} [GeV] input-validation distribution in the broad $70 \leq m_{4\ell} \leq 170$ GeV window. The D7 shape gate fails with chi2/ndf = 122.06 and p = 6.22×10^{-155}; this comparison is used only to decide classifier-input eligibility, not to normalize fit templates.

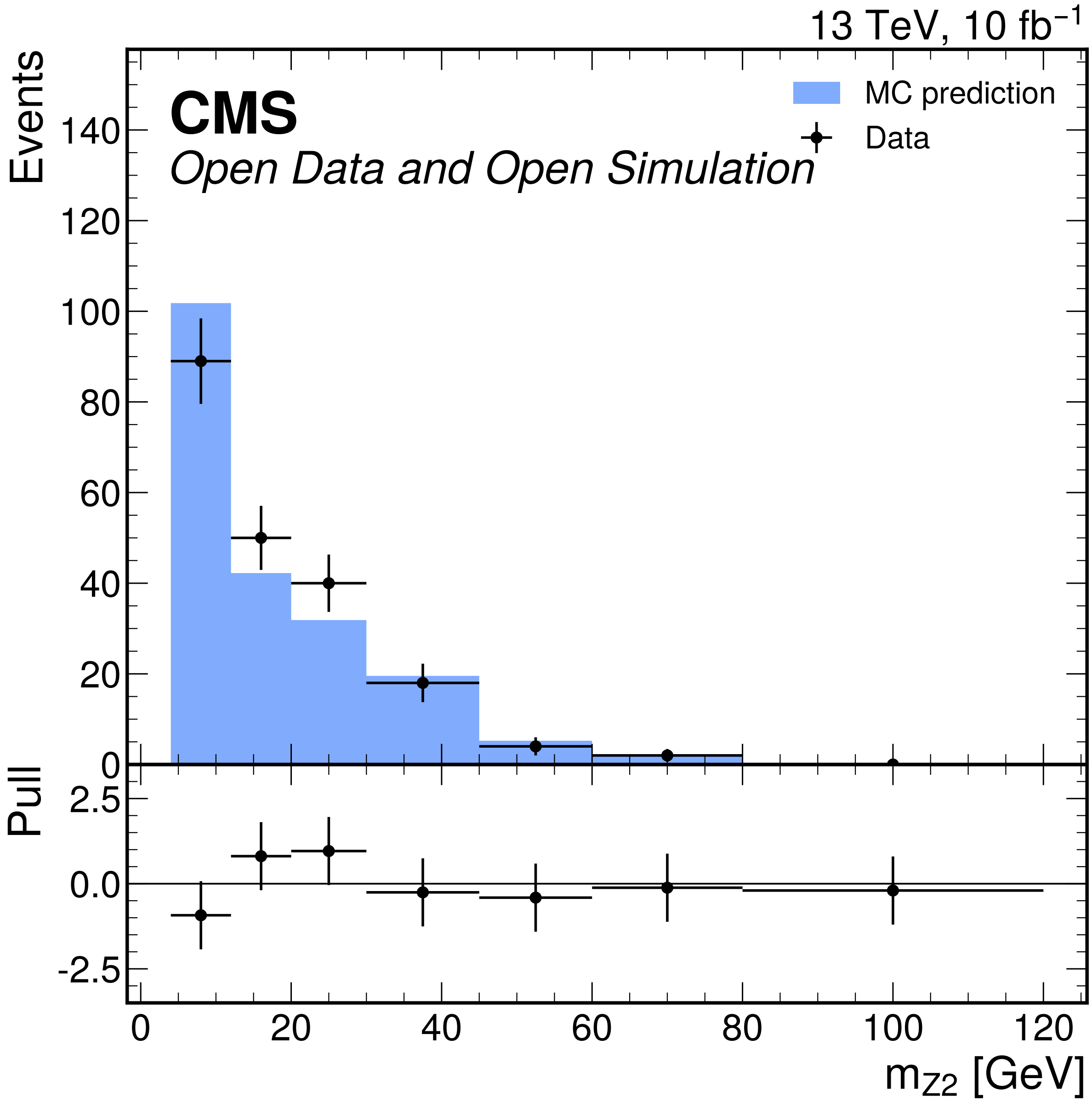

Figure 30: m_{Z2} [GeV] input-validation distribution in the broad $70 \leq m_{4\ell} \leq 170$ GeV window. The D7 shape gate fails with chi2/ndf = 32.08 and p = 7.49×10^{-39}; this comparison is used only to decide classifier-input eligibility, not to normalize fit templates.

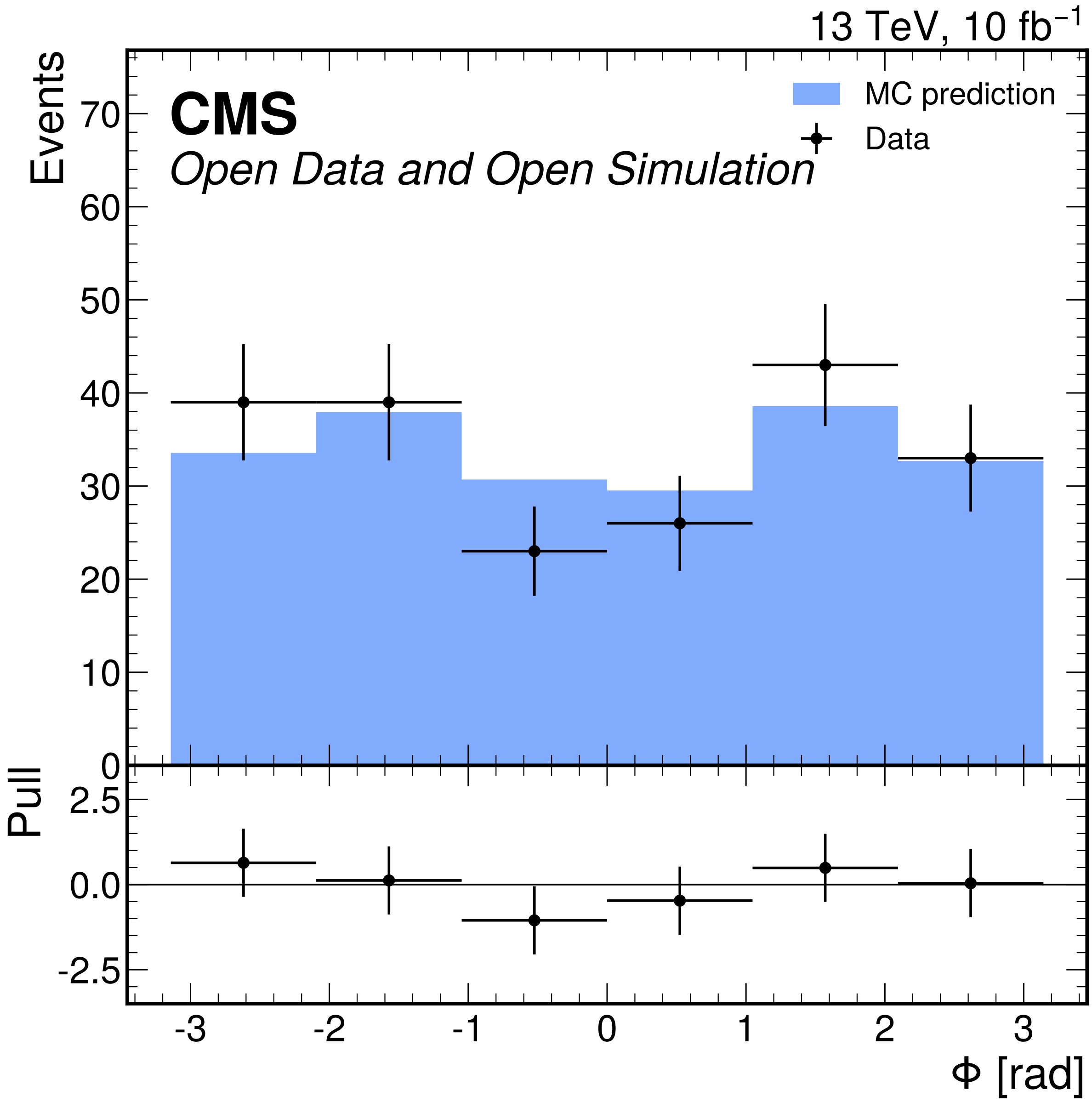

Figure 31: Phi [rad] input-validation distribution in the broad $70 \leq m_{4\ell} \leq 170$ GeV window. The D7 shape gate fails with chi2/ndf = 0.780 and p = 0.564; this comparison is used only to decide classifier-input eligibility, not to normalize fit templates.

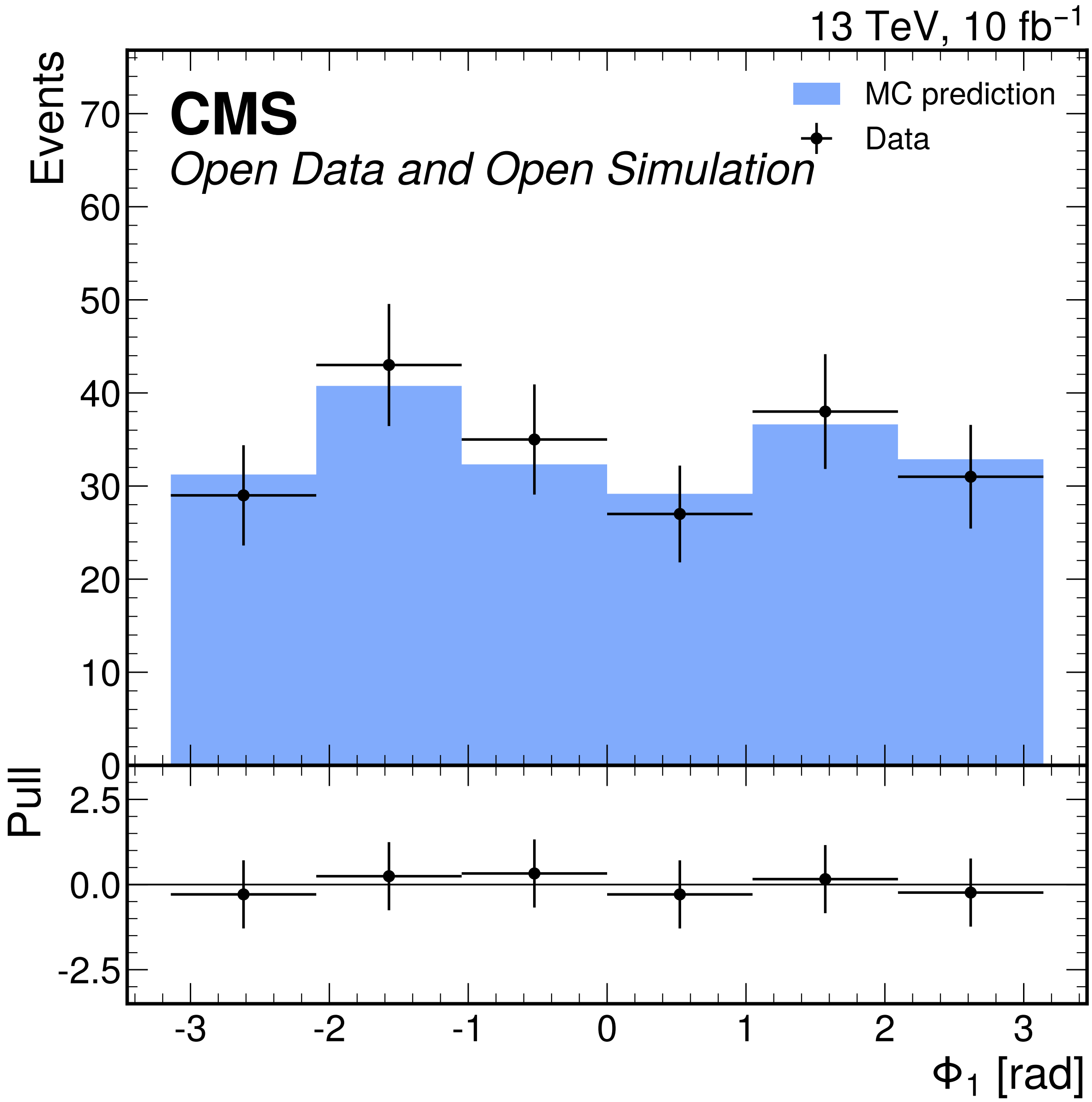

Figure 32: Phi_1 [rad] input-validation distribution in the broad $70 \leq m_{4\ell} \leq 170$ GeV window. The D7 shape gate passes with chi2/ndf = 0.155 and p = 0.979; this comparison is used only to decide classifier-input eligibility, not to normalize fit templates.

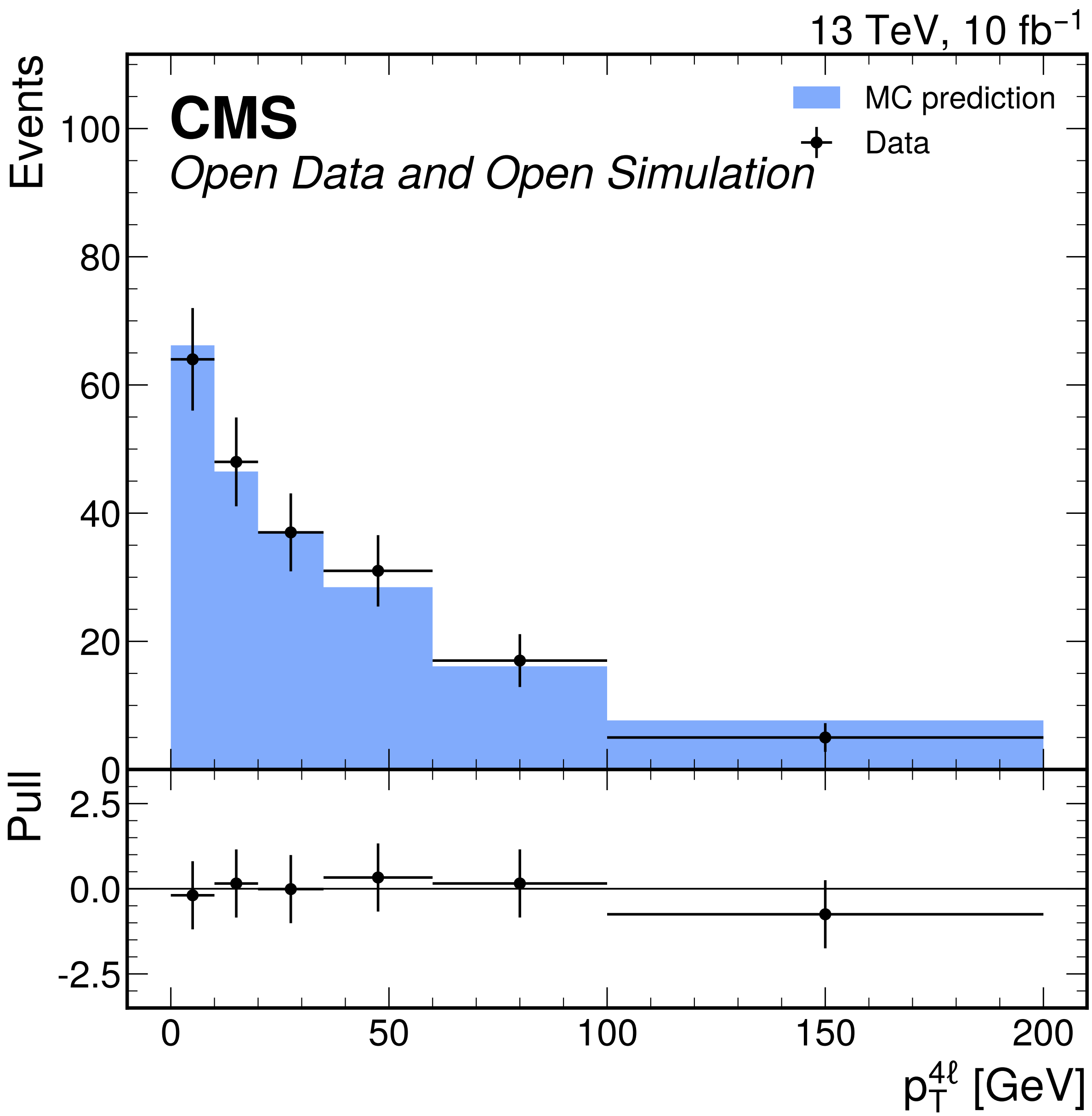

Figure 33: p_T^{4ell} [GeV] input-validation distribution in the broad $70 \leq m_{4\ell} \leq 170$ GeV window. The D7 shape gate fails with chi2/ndf = 0.333 and p = 0.893; this comparison is used only to decide classifier-input eligibility, not to normalize fit templates.

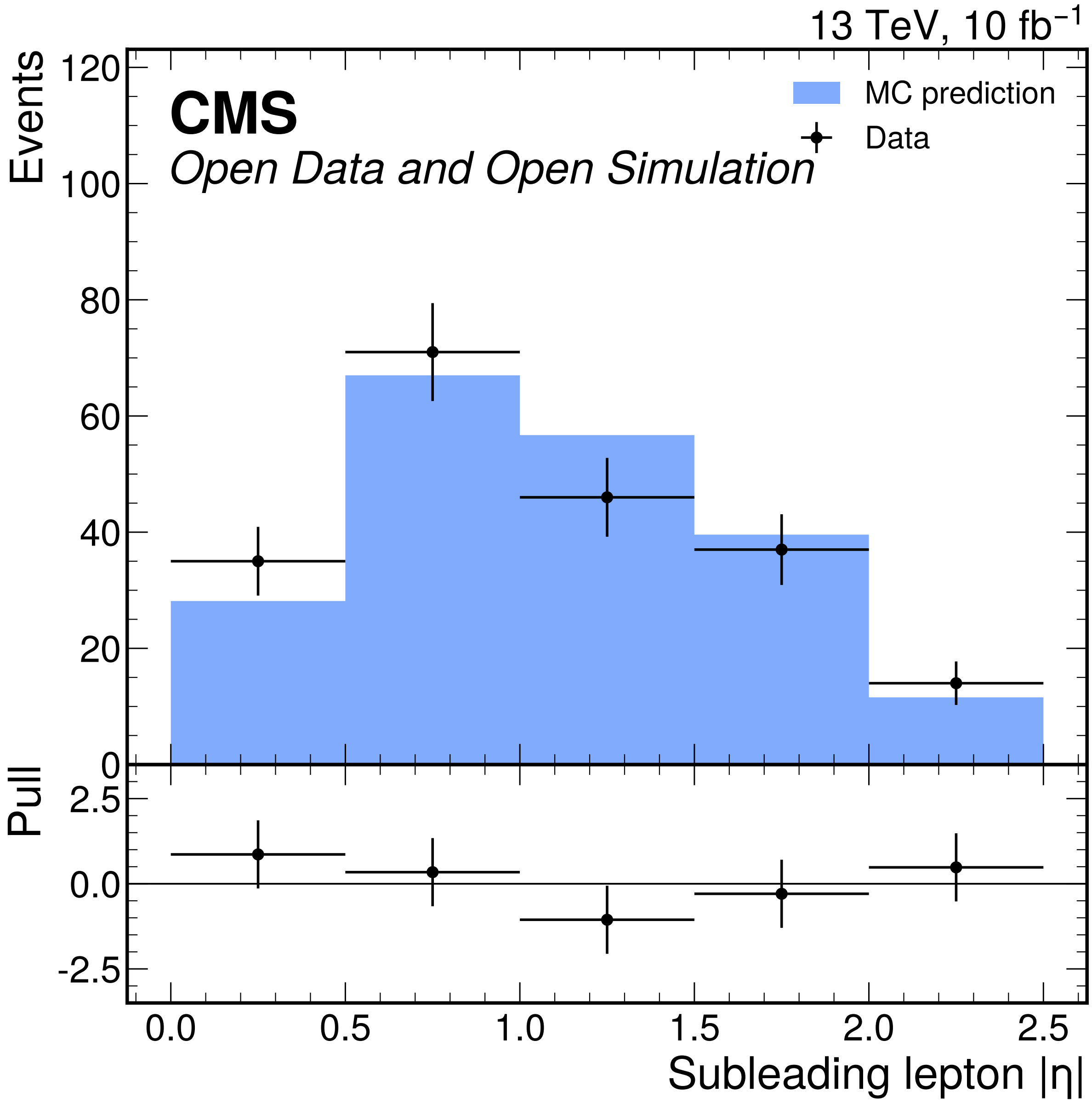

Figure 34: Subleading lepton |eta| input-validation distribution in the broad $70 \leq m_{4\ell} \leq 170$ GeV window. The D7 shape gate fails with chi2/ndf = 1.08 and p = 0.366; this comparison is used only to decide classifier-input eligibility, not to normalize fit templates.

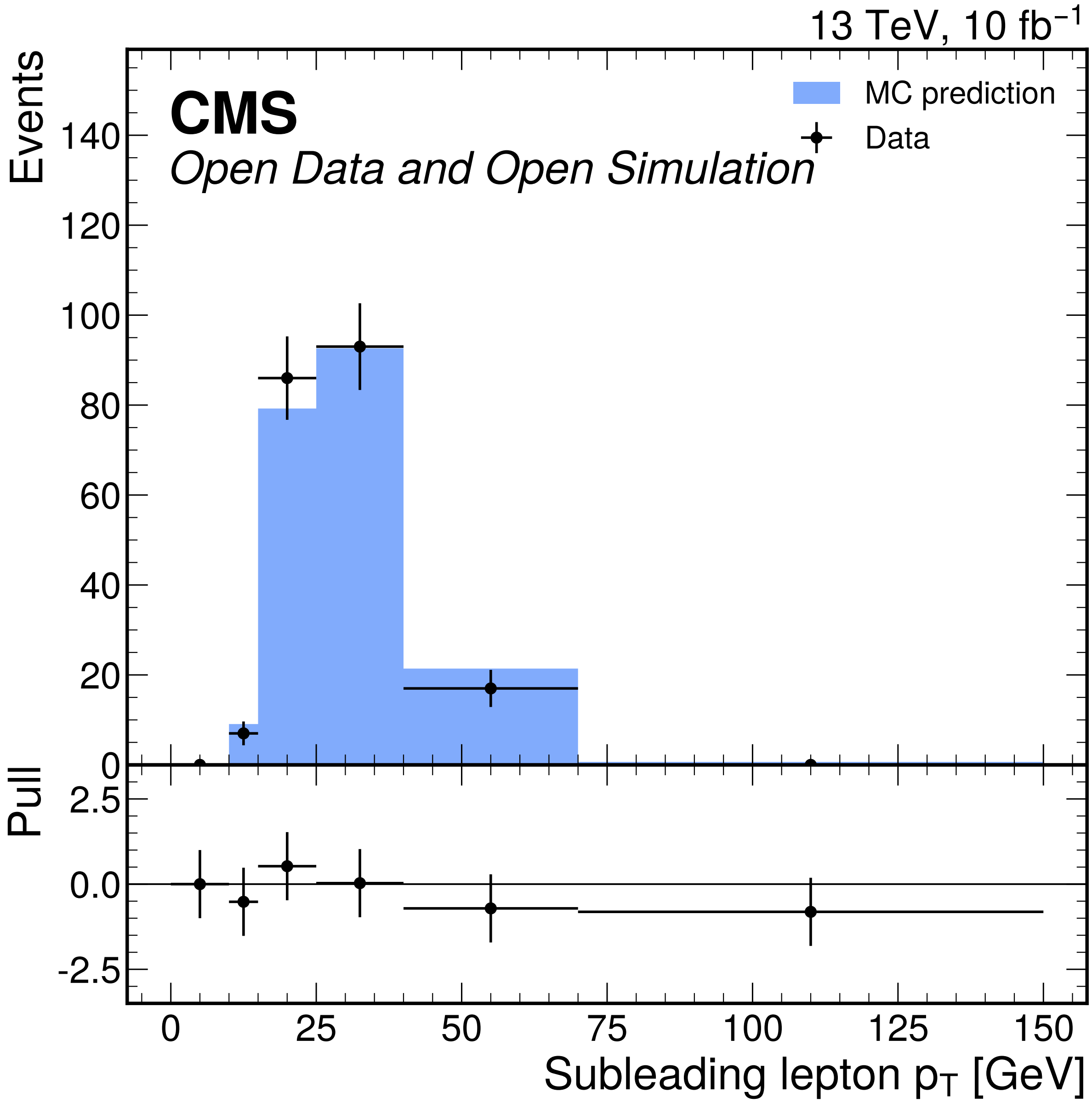

Figure 35: Subleading lepton p_T [GeV] input-validation distribution in the broad $70 \leq m_{4\ell} \leq 170$ GeV window. The D7 shape gate fails with chi2/ndf = 849.48 and p = 0.0; this comparison is used only to decide classifier-input eligibility, not to normalize fit templates.

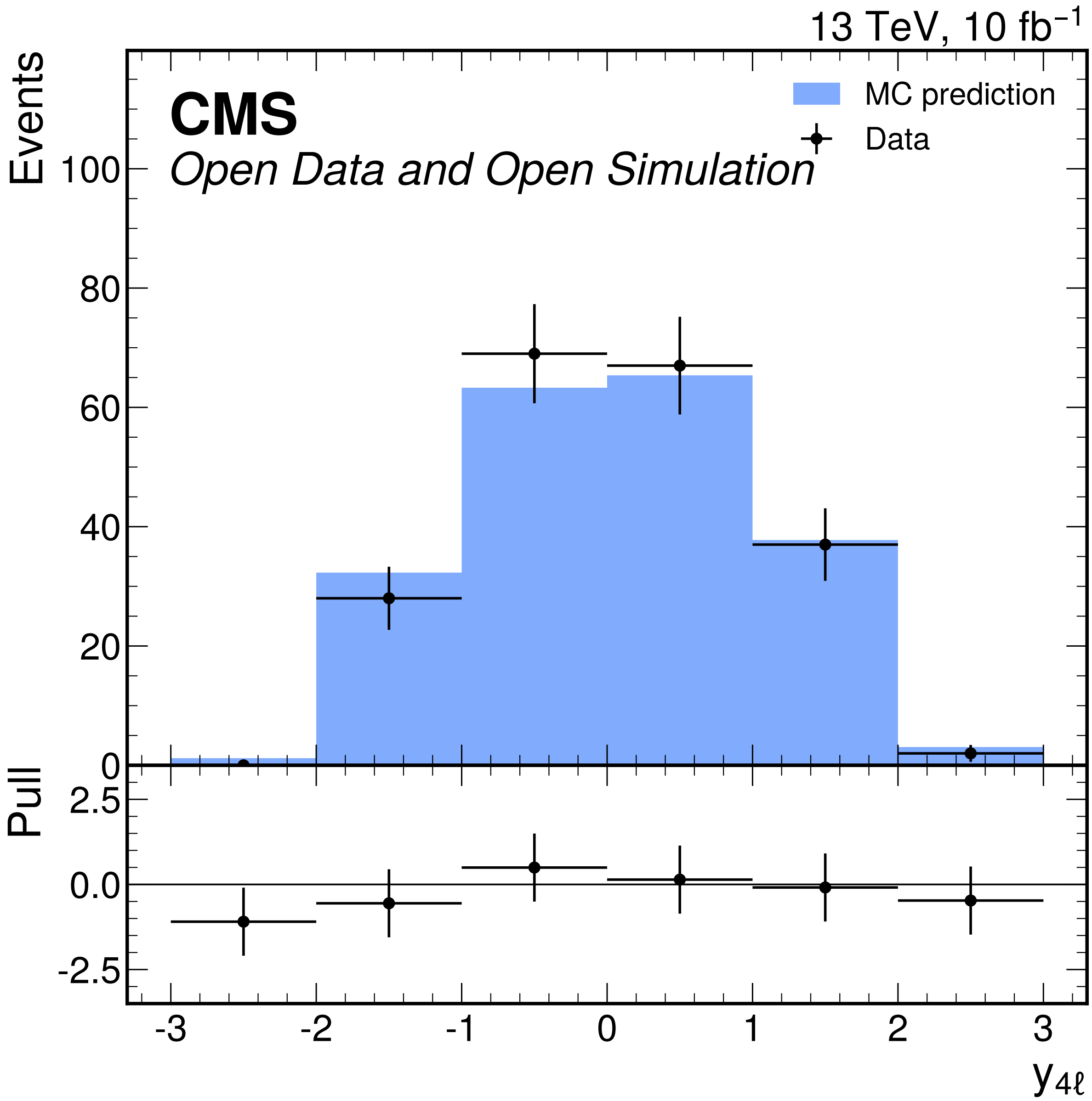

Figure 36: y_{4ell} input-validation distribution in the broad $70 \leq m_{4\ell} \leq 170$ GeV window. The D7 shape gate fails with chi2/ndf = 996.68 and p = 0.0; this comparison is used only to decide classifier-input eligibility, not to normalize fit templates.

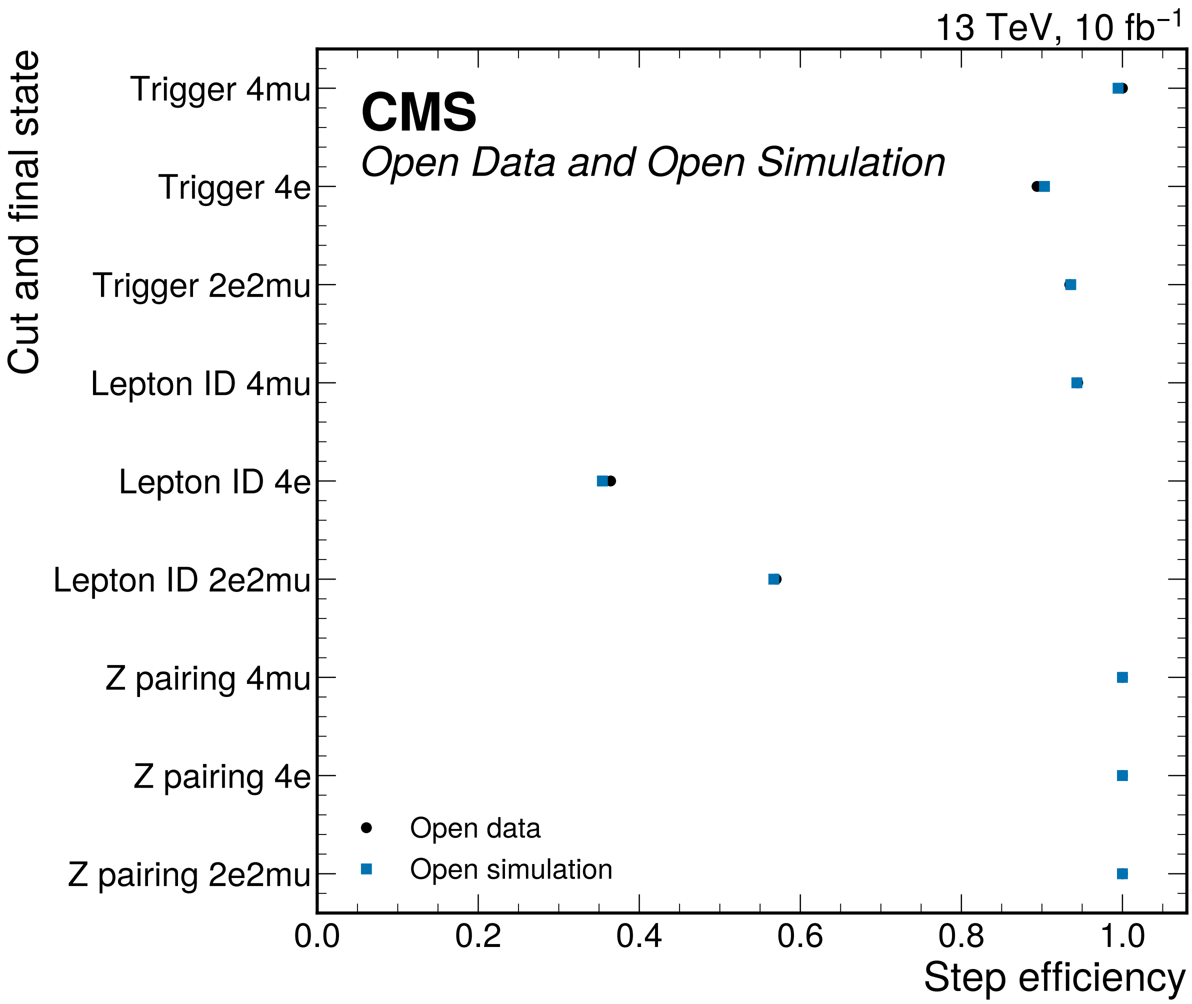

Figure 37: Trigger, flavor-matched lepton-ID, and Z-pair sanity efficiencies by final state. Efficiencies are ratios to the previous predeclared selection step; data uses raw events and MC uses prompt-normalized weighted yields.

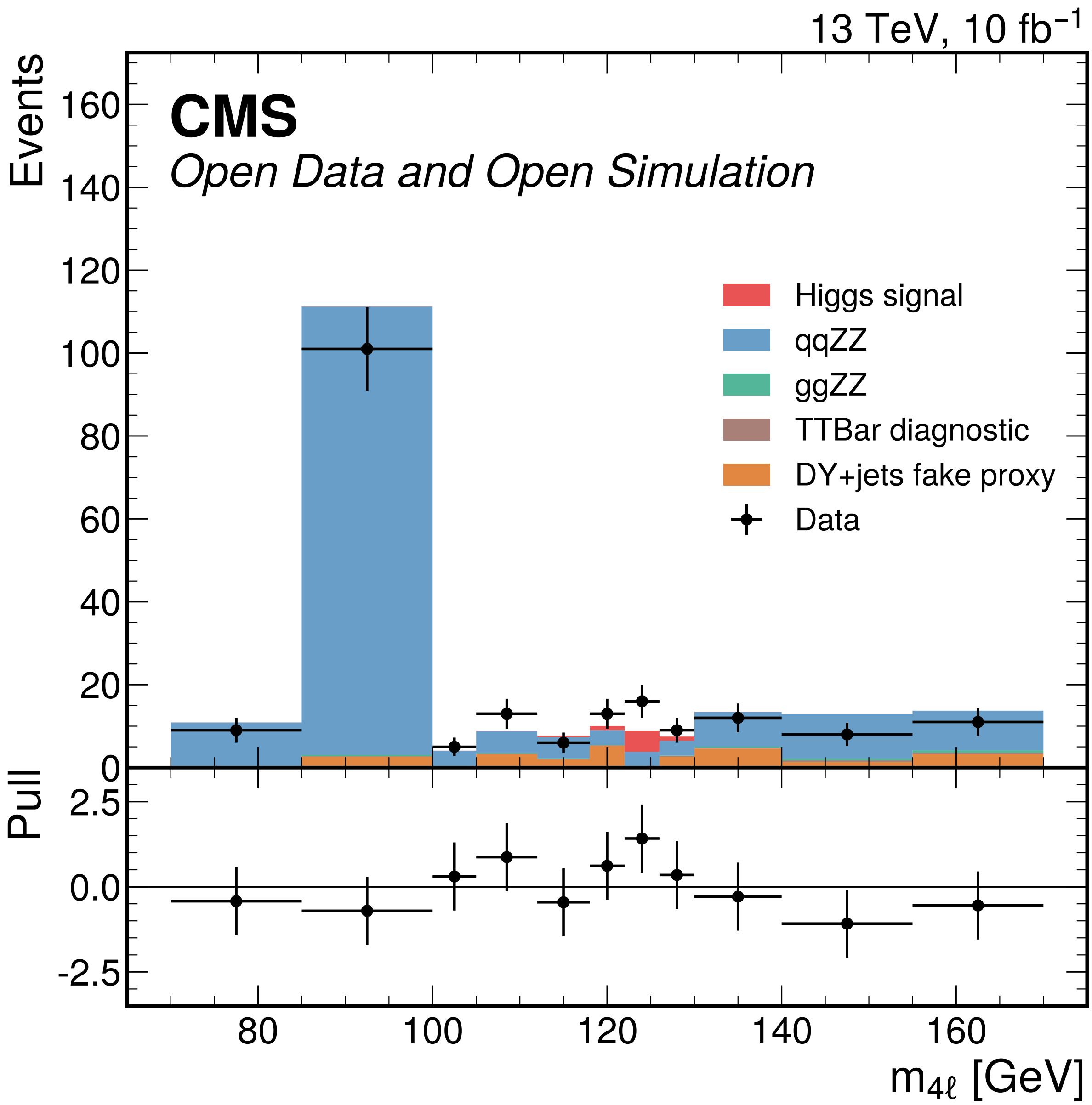

Figure 38: Inclusive four-lepton mass distribution in the broad validation window. MC is normalized with prompt effective cross sections and Metadata denominators, while DY+jets remains the nominal fake proxy. This display is sideband and validation context only; the separate 105–140 GeV fit-window templates define the downstream inference input.

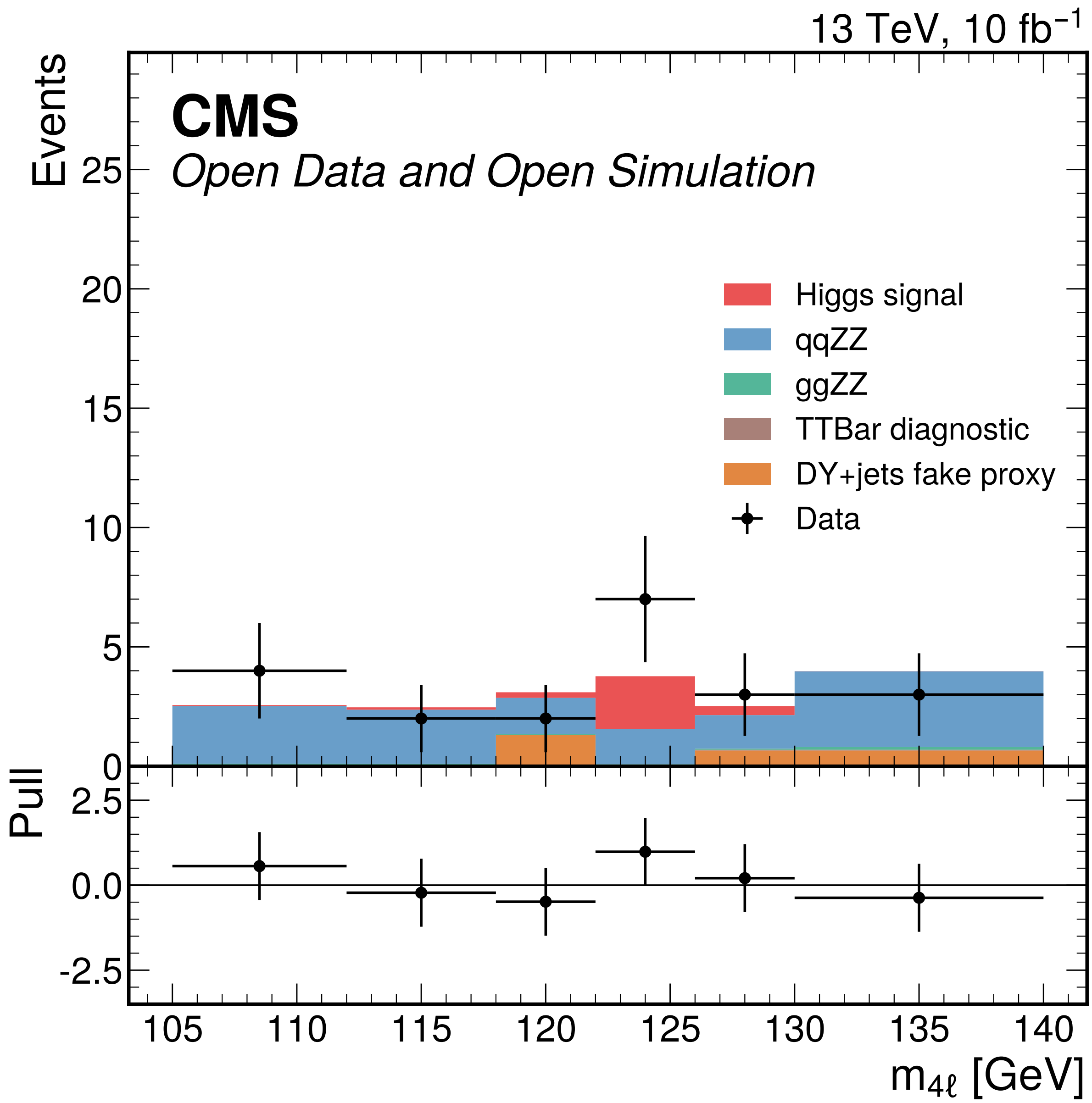

Figure 39: Four-lepton mass distribution for the 4mu final-state category in 105 < m4l < 140 GeV. These final-state categories are the conditional Phase 4 simultaneous-fit handoff because the classifier split failed the promotion gates and no real VBF category is available; Phase 4 must validate low-count Poisson/toy behavior and MC-stat stability before reporting fit results.

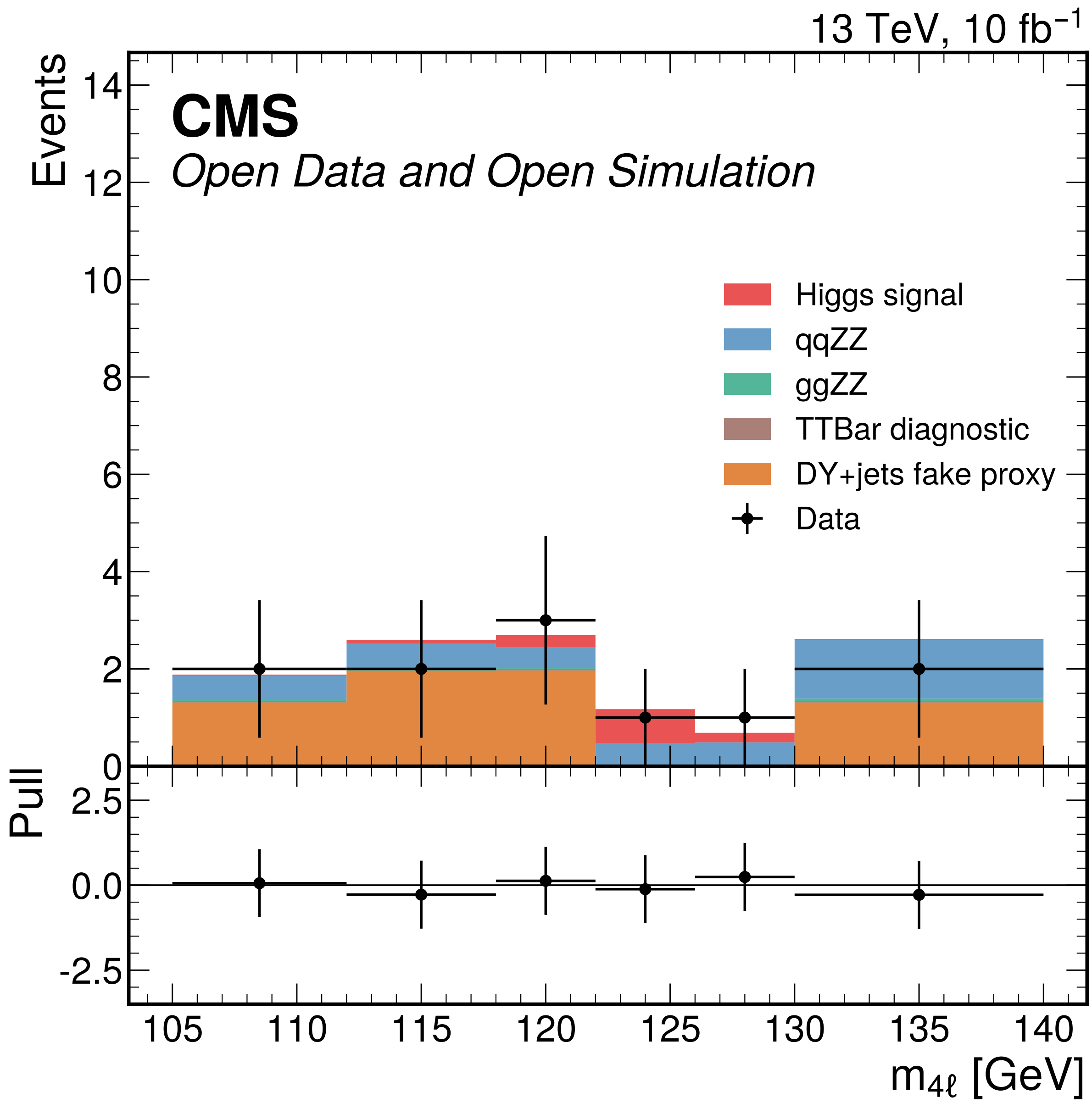

Figure 40: Four-lepton mass distribution for the 4e final-state category in 105 < m4l < 140 GeV. These final-state categories are the conditional Phase 4 simultaneous-fit handoff because the classifier split failed the promotion gates and no real VBF category is available; Phase 4 must validate low-count Poisson/toy behavior and MC-stat stability before reporting fit results.

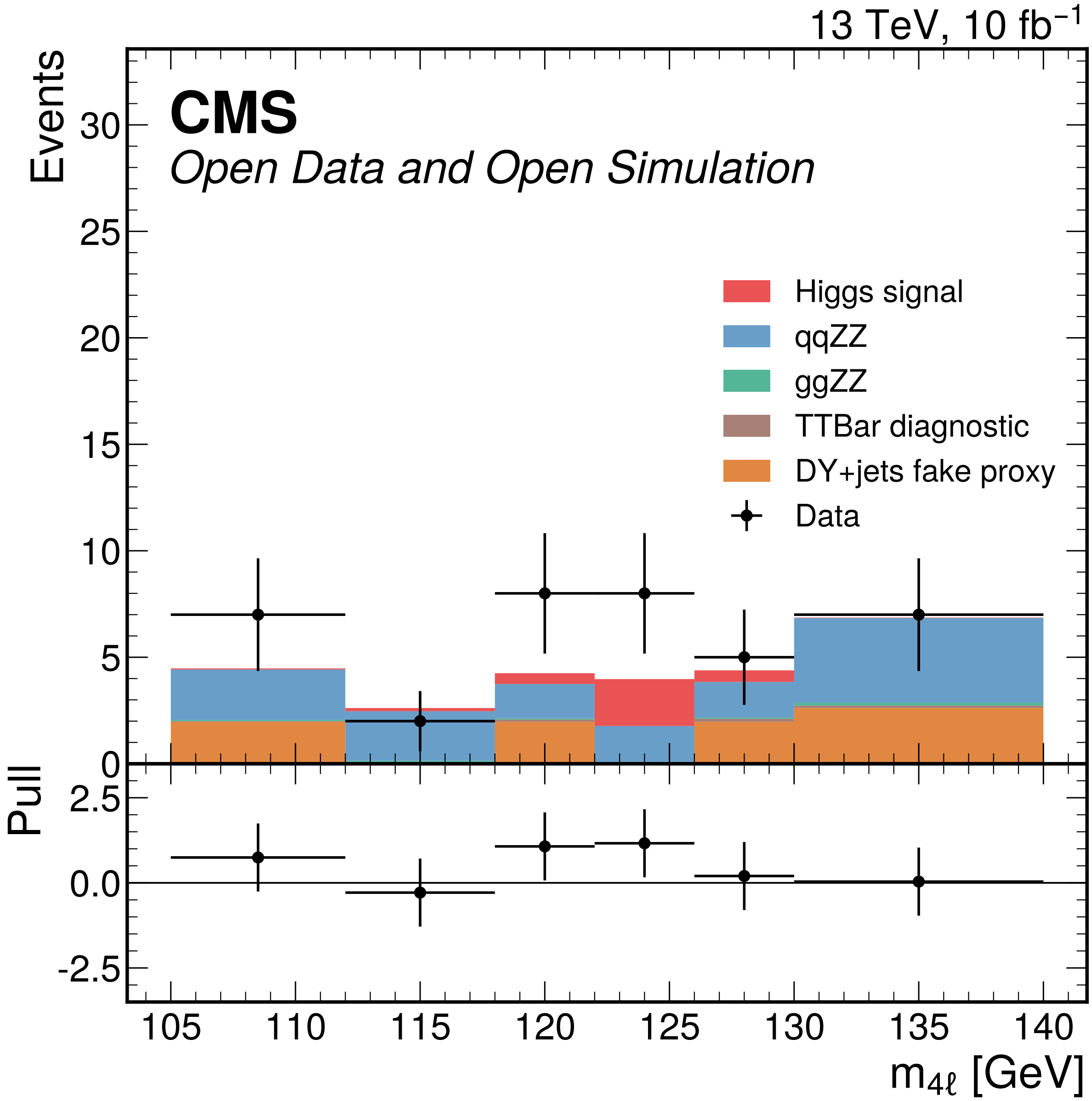

Figure 41: Four-lepton mass distribution for the 2e2mu final-state category in $105 < m4l < 140$ GeV. These final-state categories are the conditional Phase 4 simultaneous-fit handoff because the classifier split failed the promotion gates and no real VBF category is available; Phase 4 must validate low-count Poisson/toy behavior and MC-stat stability before reporting fit results.

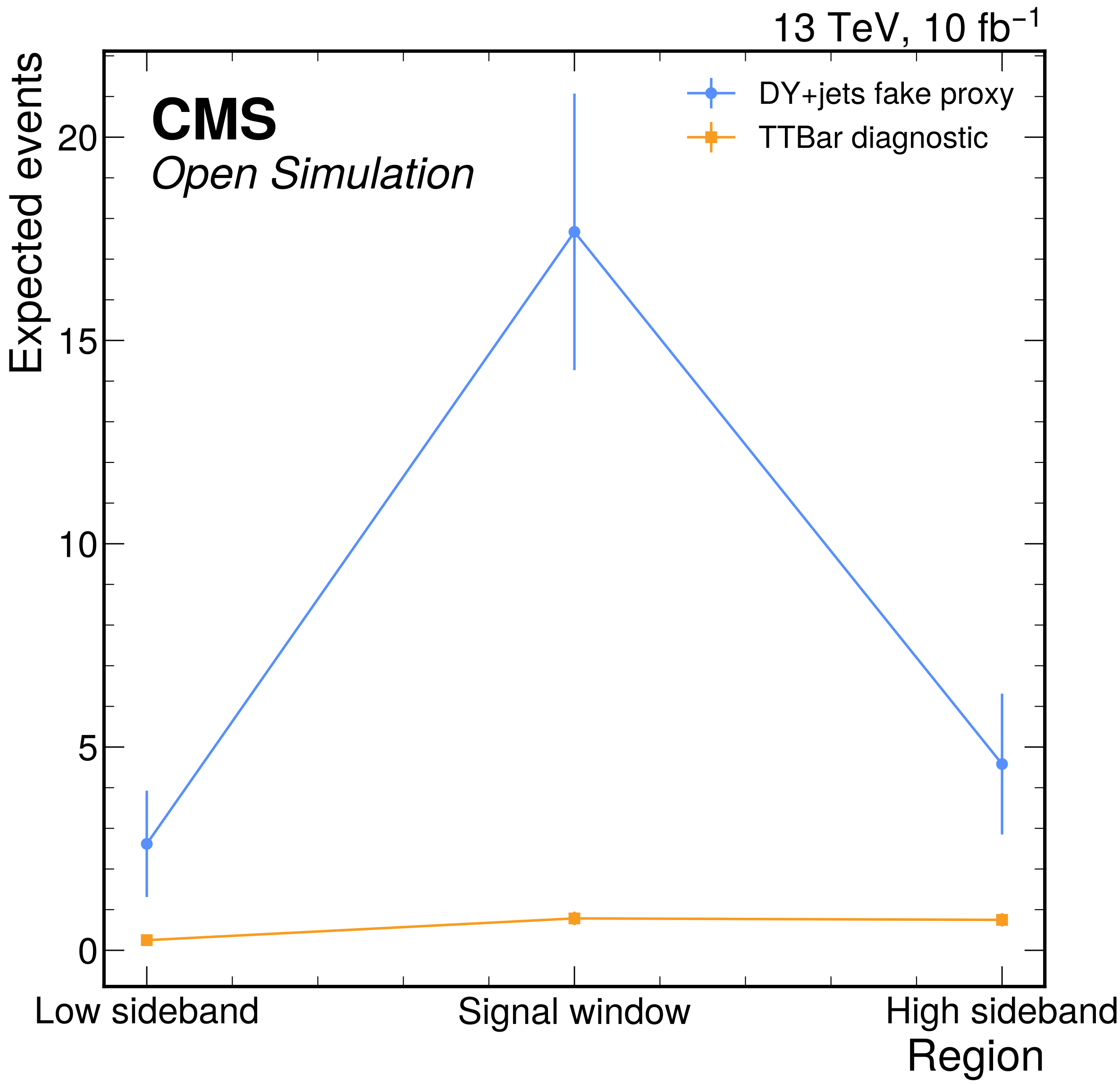

Figure 42: DY+jets fake proxy and TTBar diagnostic reducible-background checks across the predeclared sideband and signal regions. The TTBar diagnostic to DY+jets fake-proxy ratios are 0.16 in the high sideband, 0.10 in the low sideband, and 0.044 in the signal window, so TTBar is not promoted to the nominal fake model by the Phase 2 thresholds.

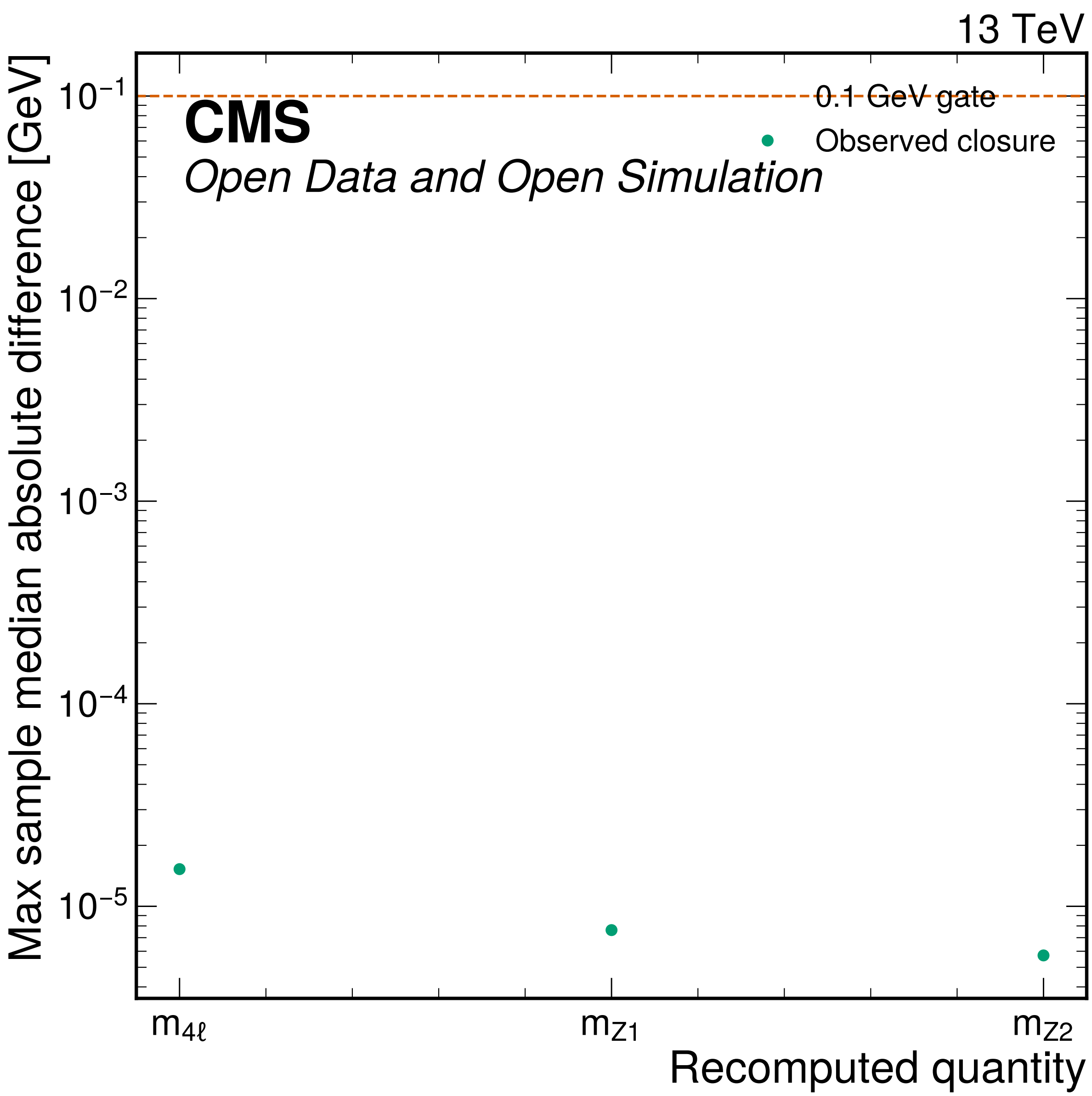

Figure 43: Angular reco closure summary showing the maximum per-sample median absolute mass difference for recomputed four-vector quantities. All medians are far below the 0.1 GeV closure gate and all angular physical-range checks have zero out-of-range entries.

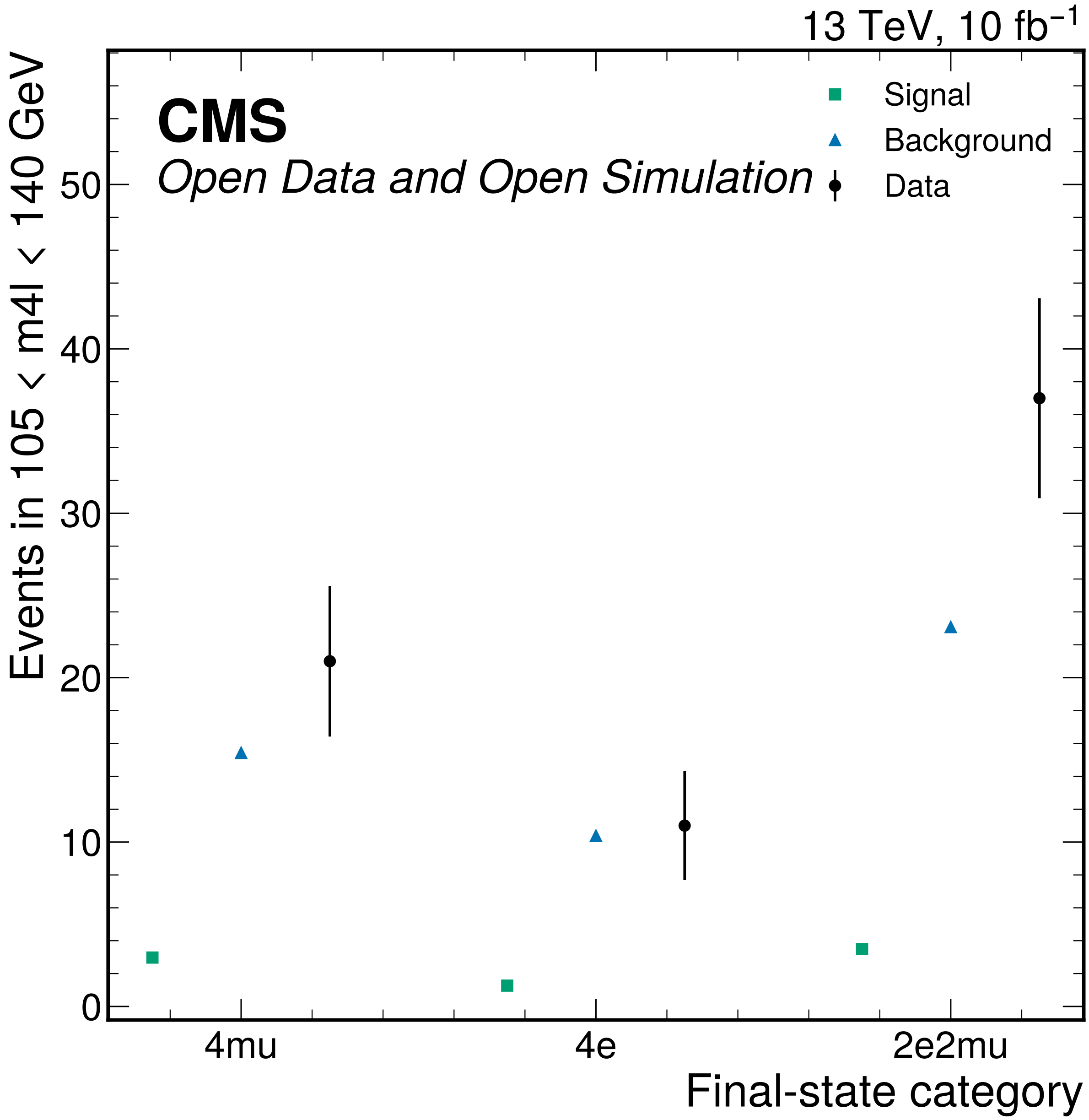

Figure 44: S1 final-state category viability summary for the fit window. The categories are retained only as a conditional Phase 4 handoff: 17/18 final-state bins have S+B below five expected events, while S2 classifier categories fail low-stat viability.

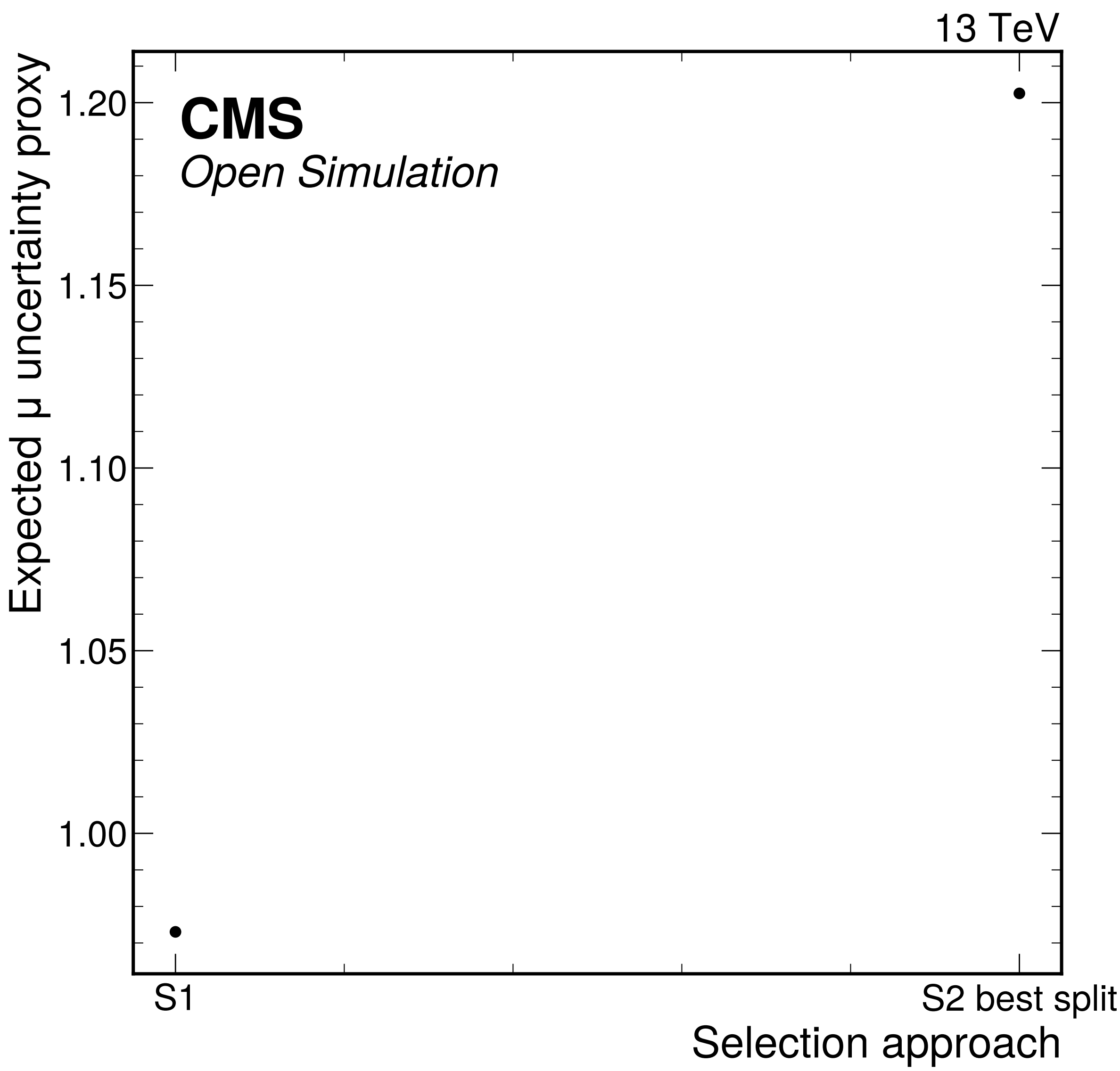

Figure 45: S1 and S2 approach comparison using the common Asimov counting precision proxy. The nominal choice is S1_reference_like_final_state_categories because S2 did not pass all input, score-shape, low-stat, overtraining, or >10% improvement gates; final-state binning is a conditional low-count Phase 4 handoff.

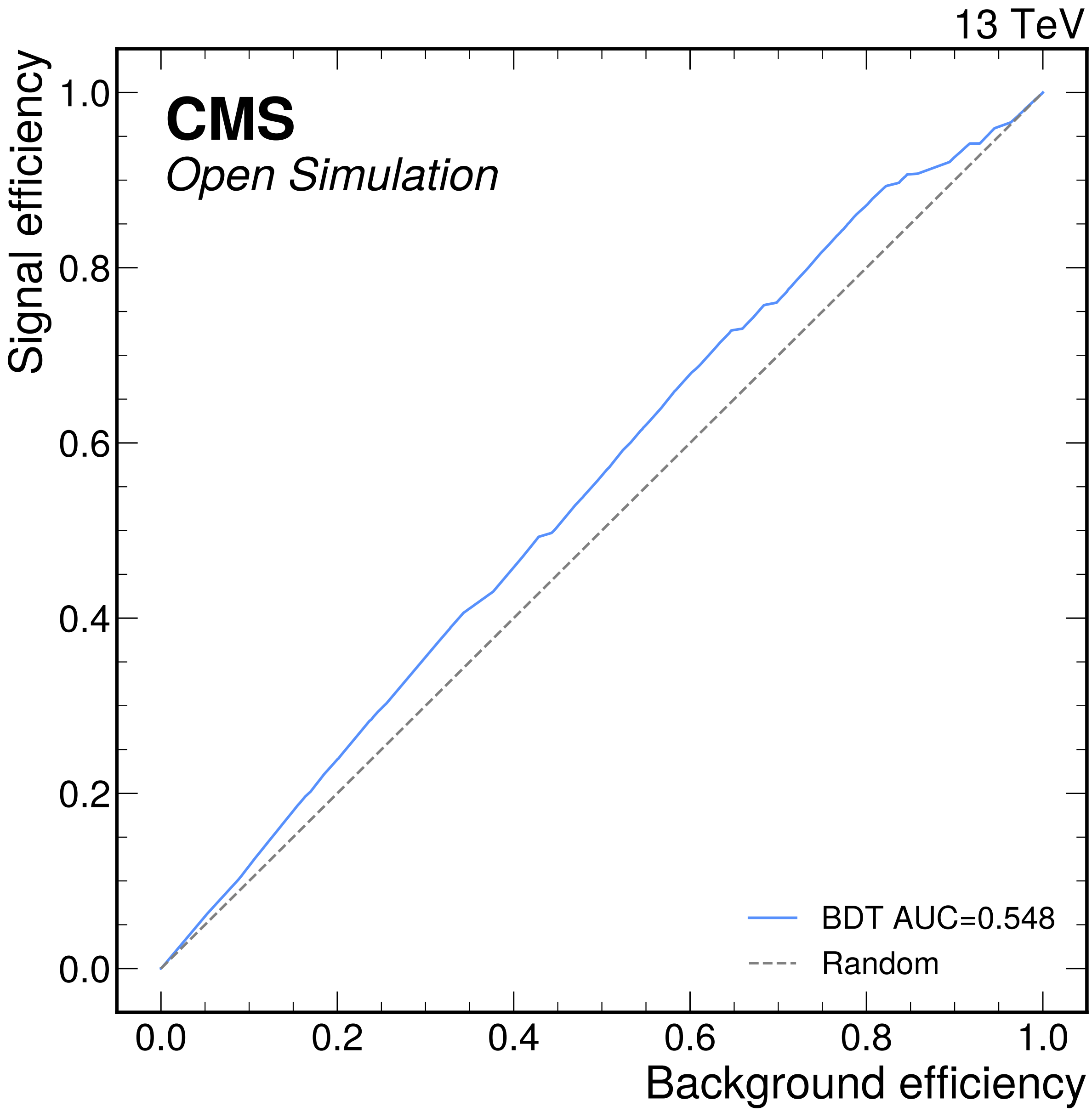

Figure 46: BDT classifier ROC curve for the S2 attempt. The weak separation and failed category-viability gates prevent promotion to nominal Phase 4 categories.

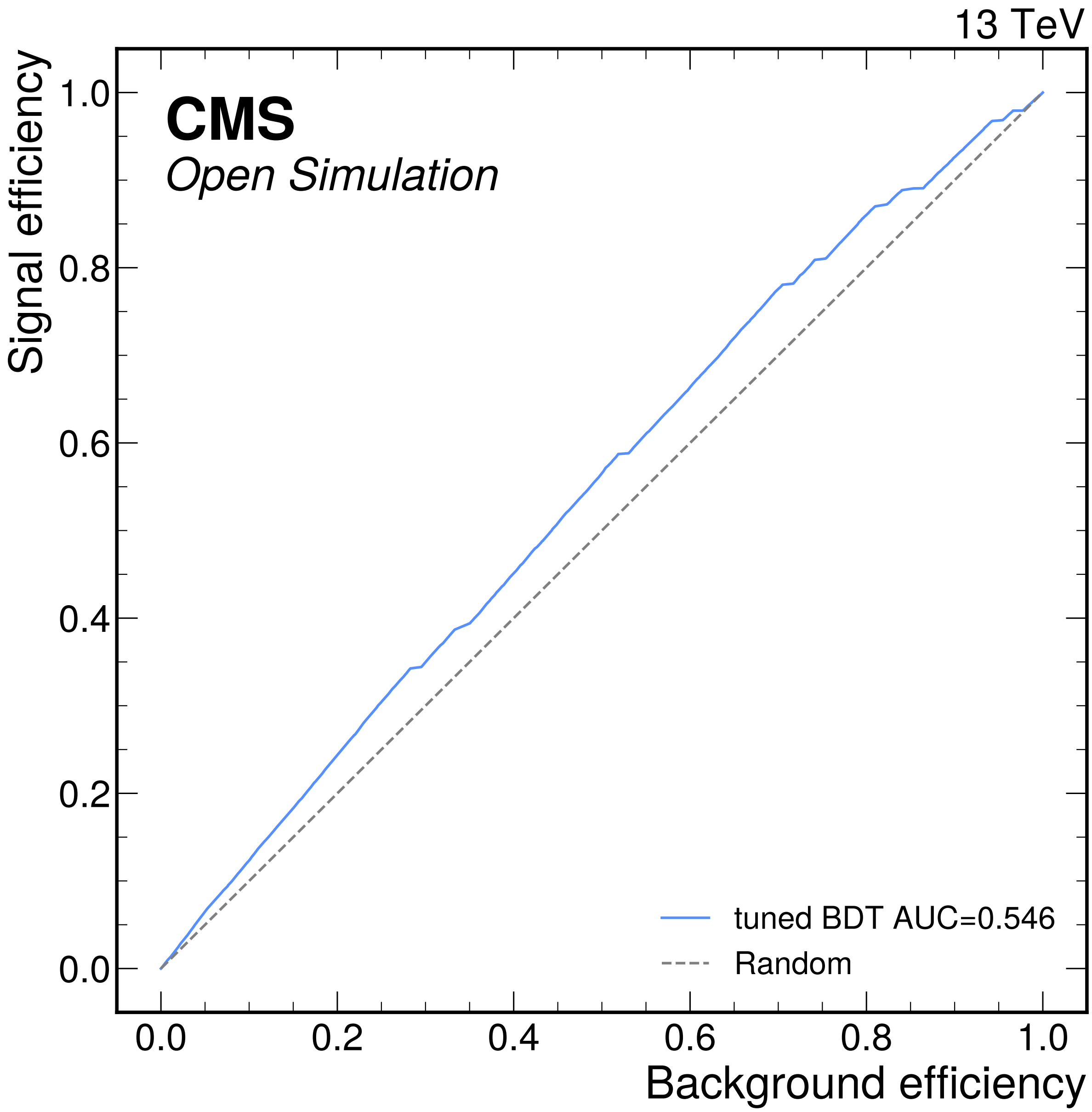

Figure 47: tuned BDT classifier ROC curve for the S2 attempt. The weak separation and failed category-viability gates prevent promotion to nominal Phase 4 categories.

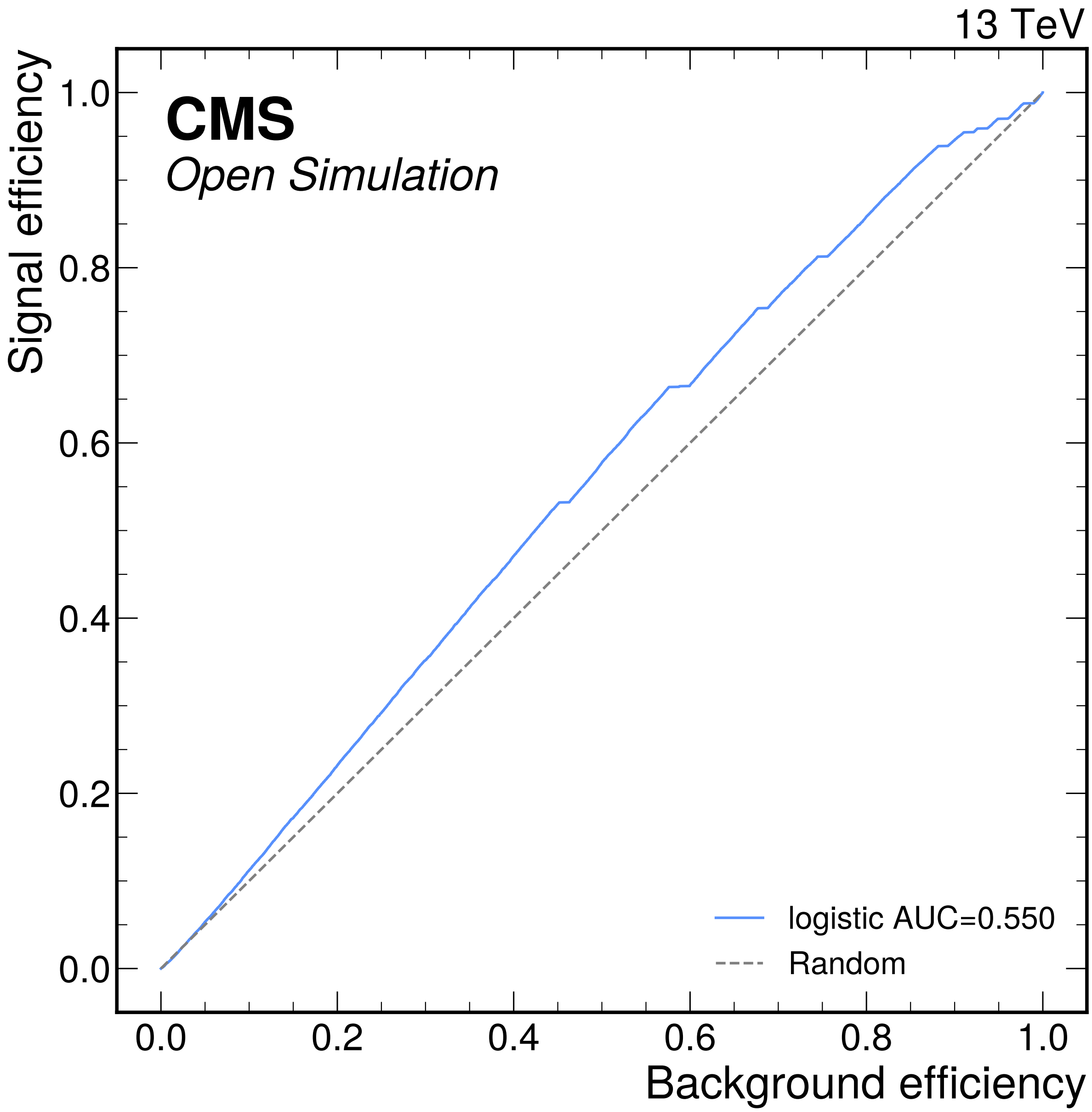

Figure 48: logistic classifier ROC curve for the S2 attempt. The weak separation and failed category-viability gates prevent promotion to nominal Phase 4 categories.

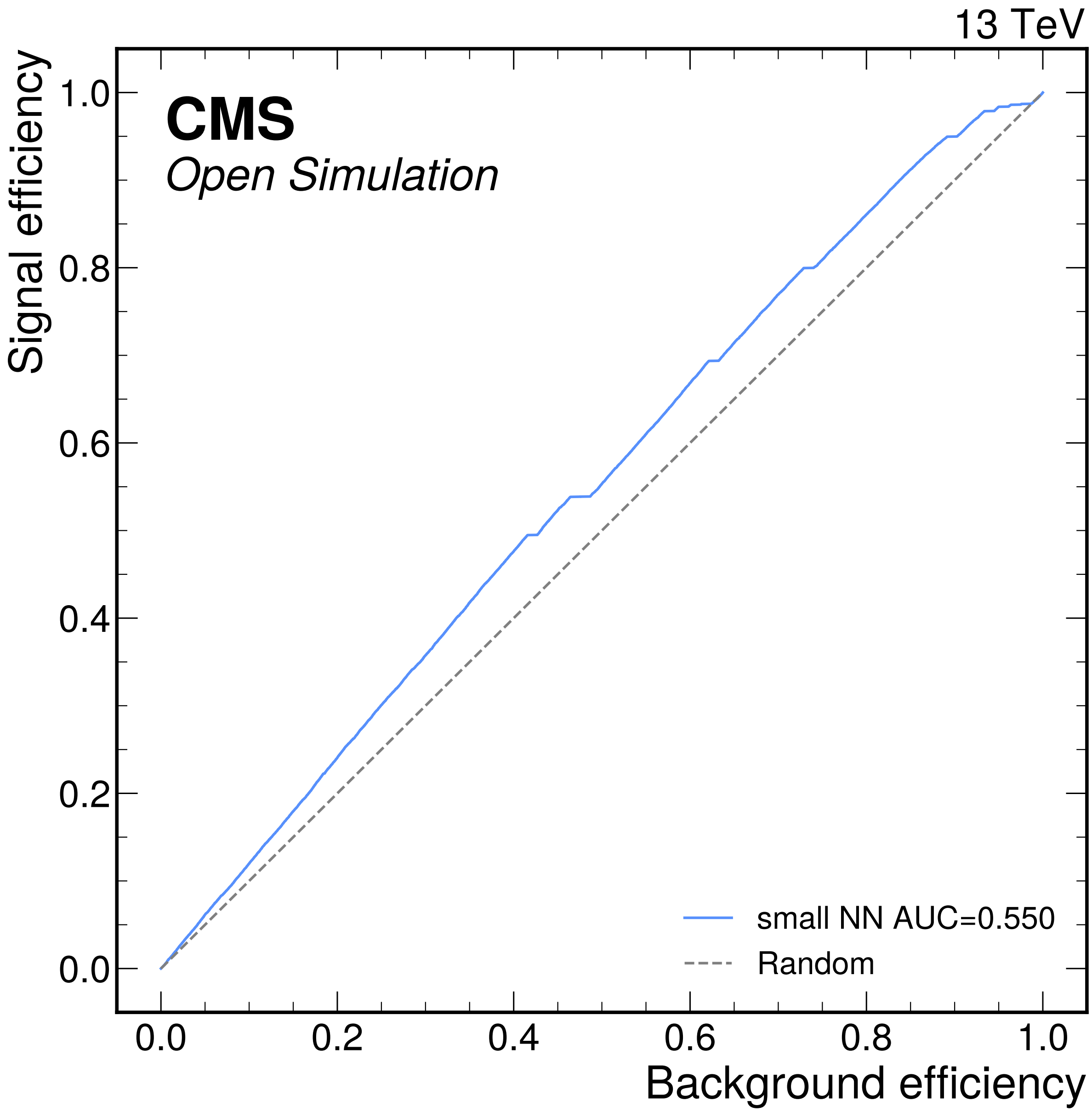

Figure 49: small NN classifier ROC curve for the S2 attempt. The weak separation and failed category-viability gates prevent promotion to nominal Phase 4 categories.

References

[1] CMS Collaboration. Observation of a new boson at a mass of 125 gev with the cms experiment at the lhc. *Physics Letters B*, 716:30–61, 2012.

[2] ATLAS Collaboration. Observation of a new particle in the search for the standard model higgs boson with the atlas detector at the lhc. *Physics Letters B*, 716:1–29, 2012.

[3] CMS Collaboration. Measurements of properties of the higgs boson decaying into the four-lepton final state in pp collisions at sqrt(s) = 13 tev. *Journal of High Energy Physics*, 2017(11):047, 2017.

[4] CMS Collaboration. Measurements of production cross sections of the higgs boson in the four-lepton final state in proton-proton collisions at sqrt(s) = 13 tev. *European Physical Journal C*, 81:488, 2021.

[5] S. Navas et al. Review of particle physics. *Physical Review D*, 110:030001, 2024.

[6] CMS Collaboration. The cms experiment at the cern lhc. *Journal of Instrumentation*, 3:S08004, 2008.

[7] CMS Collaboration. Cms open data portal, 2026.

[8] S. Agostinelli et al. Geant4: A simulation toolkit. *Nuclear Instruments and Methods in Physics Research A*, 506:250–303, 2003.

[9] Torbjorn Sjostrand, Stephen Mrenna, and Peter Z. Skands. A brief introduction to pythia 8.1. *Computer Physics Communications*, 178:852–867, 2008.

[10] Paolo Nason. A new method for combining nlo qcd with shower monte carlo algorithms. *Journal of High Energy Physics*, 2004(11):040, 2004.

[11] Tianqi Chen and Carlos Guestrin. Xgboost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 785–794, 2016.

[12] Fabian Pedregosa, Gael Varoquaux, Alexandre Gramfort, Vincent Michel, Bertrand Thirion, Olivier Grisel, Mathieu Blondel, Peter Prettenhofer, Ron Weiss, Vincent Dubourg, Jake Vanderplas, Alexandre Passos, David Cournapeau, Matthieu Brucher, Matthieu Perrot, and Edouard Duchesnay. Scikit-learn: Machine learning in python. *Journal of Machine Learning Research*, 12:2825–2830, 2011.

[13] Kyle Cranmer, George Lewis, Lorenzo Moneta, Akira Shibata, and Wouter Verkerke. Histfactory: A tool for creating statistical models for use with roofit and roostats. Technical Report CERN-OPEN-2012-016, CERN, 2012.

[14] Lukas Heinrich, Matthew Feickert, Giordon Stark, and Kyle Cranmer. pyhf: pure-python implementation of histfactory statistical models. *Journal of Open Source Software*, 6(58):2823, 2021.

[15] Glen Cowan, Kyle Cranmer, Eilam Gross, and Ofer Vitells. Asymptotic formulae for likelihood-based tests of new physics. *European Physical Journal C*, 71:1554, 2011.

[16] F. James and M. Roos. Minuit: A system for function minimization and analysis of the parameter errors and correlations. *Computer Physics Communications*, 10:343–367, 1975.